A Framework for Stochastic Dynamics, Coarse-Graining, and Learned Probability Flows

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Introduction

The starting point of both statistical physics and generative artificial intelligence is the relationship between the evolution of an underlying state and the induced evolution of observables and probability distributions. To describe this relationship, we distinguish three fundamental objects: the state x of a system, an observable f defined on its state space, and a probability measure μ describing the statistical distribution of its states. Once the underlying dynamics is specified, it induces two complementary descriptions: the evolution of observables and the transport of probability measures.

We begin by establishing the algebraic, measurable, and differentiable structures of the underlying state space and the corresponding observable space. We then specify deterministic and stochastic dynamics and investigate how they induce evolution on observables and probability measures. This provides the mathematical foundation for studying coarse-graining, where microscopic descriptions are mapped onto lower-dimensional collective variables, and conditional expectation provides a natural way to formulate reduced observables and effective dynamics.

The central question throughout this development is how these successive transformations are related and under what conditions they preserve the relevant dynamical and statistical structures. In particular, we examine the compatibility between flow composition, the pushforward of probability measures, conditional expectation, and the approximation of infinitesimal generators. These questions connect the mathematical description of microscopic dynamics with the construction of effective models and their numerical or data-driven approximations.

Original problem source and attribution. The point of departure is the supplied assessment portfolio [Assessment], primarily Project 9, together with the Hamiltonian application of Project 10 and selected questions from Projects 8 and 11–14. Projects 2–3 motivate the provisional neural PDE extension. Core questions develop the theoretical framework; secondary tasks retain their statements and original solution source, with detailed answers temporarily deferred.

All numerical labels of the form "Project n, item m" refer to the *Student Assessment Exercise* problem portfolio supplied with this project [Assessment] (original PDF retained locally and excluded from the repository). A source marker appears at the relevant result, and the full question-to-section concordance is given in the Source Index (Table 1).

Source Index and Direct Links

Each blue project entry links to its retained statement, problem, or answer. Sections 1.1–1.5 are included in full at the section level; individually deferred solutions remain explicitly marked. The new Projects 2–3 entries distinguish a proved sampling counterexample from an open operator-learning research task. *Core* identifies a question central to the theoretical framework; *Concise* identifies an auxiliary question with its necessary conclusions and a shortened argument; *Pending* identifies a task whose solution or proof has been commented out and whose entry leads to an explicit completion notice when its statement is included. A superscript $\star$ marks an entry with a solution, proof, or extension still to be completed, including every entry whose proof reads *Proof to be completed.* and the partially deferred extensions in Projects 9.11 and 13.1. For included entries, the page number is also a clickable link to that location. Project and question numbers refer to the *original assessment PDF* [Assessment], not to this manuscript. An unnumbered project task is listed as "Main task." The separate entry for Project 9, item 4 (numerics) leads to its discrete-time analysis.

Shaded *Problem* and *Exercise* panels identify task statements. Orange *Later thought* panels contain optional pending ideas, not additional solved source problems. Theorem, proposition, and lemma statements use the same blue panels; their proofs follow outside the panels. Corollaries are displayed without an enclosing panel.

Table 1: Selected assessment questions and their locations.

Original source	**Page**	**Placement in the unified theory**
Project 2 – Main task	105	Concise: Proved finite-sampling PINN counterexample
Project 3 – Main task*	107	Pending: Committor operator learning and source ambiguities
Project 8 – Main task	94	Concise: Boundary diagnosis and corrected no-flux Markov scheme
Project 8 – 1*	95	Pending: Printed ghost-cell stencil: positivity and mass failure
Project 9 – 1	11	Core: Affine ODE and variation of constants
Project 9 – 2	21	Core: Composition, inverse, and time-reversal of flows
Project 9 – 3	22	Core: Variational matrix and noncommuting-matrix correction

Source Index and Direct Links (continued)

Original source	Page	Placement in the unified theory
Project 9 – 4	22	Core: Liouville–Jacobi determinant theorem
Project 9 – 4 (numerics)	27	Concise: Euler Jacobian and scalar convergence validation
Project 9 – 5	39	Concise: Volume transport and Reynolds-type identity
Project 9 – 6⋆	40	Pending: Minimum-energy path and optimal velocity
Project 9 – 7	45	Core: Pushforward dynamics and continuity equation
Project 9 – 8	45	Core: Density change-of-variables formula
Project 9 – 9	50	Concise: Material derivative and transport theorem
Project 9 – 10	56	Core: Differential-entropy evolution
Project 9 – 11⋆	58	Concise: Score evolution; Hessian and long-time extensions deferred
Project 9 – 12	60	Concise: Relative-entropy evolution for two flows
Project 9 – 13	92	Concise: First variation of transported KL divergence
Project 9 – 14⋆	93	Pending: Transport from a Gaussian to a target law
Project 9 – 15⋆	93	Pending: Measure-valued kinetic optimal-control problem
Project 10 – 1	52	Concise: Hamilton equations from the mechanical Hamiltonian
Project 10 – 2	53	Concise: Energy conservation along trajectories
Project 10 – 3⋆	54	Pending: Time reversal and momentum reflection
Project 10 – 4	53	Concise: Symplecticity and preservation of phase volume
Project 11 – 1	59	Core: Nonnegativity of relative entropy
Project 11 – 2⋆	62	Pending: KL asymmetry and unequal-variance Gaussians
Project 11 – 3⋆	62	Pending: Truncated Gaussian: KL curve and quadratic constrained minimum
Project 11 – 4	66	Concise: Joint and separate convexity of relative entropy
Project 11 – 5	69	Core: Gibbs variational identity and exponential tilt
Project 11 – 6⋆	72	Core: Donsker–Varadhan dual representation
Project 11 – 7⋆	61	Core: Chain rule for relative entropy
Project 11 – 8⋆	58	Pending: Stein identity and its operator
Project 11 – 9⋆	79	Pending: f-divergence and NWJ form under attainment hypotheses
Project 11 – 10⋆	79	Pending: Exponential Fenchel dual and finite-state saddle proof
Project 11 – 11⋆	80	Pending: Shift optimization recovering Donsker–Varadhan
Project 11 – 12⋆	80	Pending: Logistic classification and density-ratio optimality

Source Index and Direct Links (continued)

Original source	Page	Placement in the unified theory
Project 11 – 13⋆	81	Pending: Reward-guided measure optimization and mean-field tilt
Project 12 – 1	83	Concise: Gaussian conditional distribution
Project 12 – 2⋆	84	Pending: Noisy conditioning and level-set approximation
Project 12 – 3	84	Core: Coarea formula and effective free energy
Project 12 – 4	85	Core: Conditional mean-force formula
Project 13 – 1⋆	86	Concise: Total covariance and Gaussian reconstruction
Project 13 – 2	87	Core: Itô equation for a collective variable
Project 13 – 3	88	Core: Markovian matching of time marginals
Project 14 – 1	89	Concise: Gaussian stochastic-interpolant statistics
Project 14 – 2	90	Core: Conditional interpolant velocity field
Project 14 – 3	91	Core: Conditional-score identity

Chapter 1

From Algebraic Structures to Probability Dynamics and Learned Flows

The organizing chain of this chapter is

$$\text{microscopic dynamics} \longrightarrow \text{probability evolution} \longrightarrow \text{coarse-graining}$$

$$\longrightarrow \text{effective dynamics} \longrightarrow \text{learned dynamics and numerical approximation.}$$

At every stage a state, an observable on states, and a probability measure remain distinct objects. Section 1.1 defines the spaces and dual pairings before dynamics is imposed. Section 1.2 constructs state flows and their induced evolution, then uses geometric and entropy identities to expose structure worth preserving. Section 1.3 passes from a microscopic law to conditional laws and distinguishes exact projection from a closed effective process. Section 1.4 formulates learning and numerical approximation of these objects; Section 1.5 tests the distinctions against exact OU formulas and retained numerical evidence. Chapter 2 develops a provisional extension to neural PDE solvers. Its mathematical results, executed integration checks, planned experiments, and open questions are identified separately. Earlier deferred proofs remain deferred; restoring their sections does not turn those notices into completed results.

1.1 State Algebra, Probability Spaces, and Observables

1.1.1 States, measurable structure, and observables

Consider N classical particles in three spatial dimensions. A microscopic configuration, including positions and momenta, is represented by

$$x = (q_1, \ldots, q_N, p_1, \ldots, p_N) \in V = \mathbb{R}^{6N}.$$

Here N is the particle count and $q_i, p_i \in \mathbb{R}^3$ are particle-indexed vectors, not time-indexed states. The equations of motion determine the trajectory of a given configuration. Physi-

cal predictions, however, usually concern functions of that configuration. For example, a Hamiltonian $H(x)$ assigns an energy to a state, while

$$f_\varphi(x) = \sum_{i=1}^{N} \varphi(q_i)$$

measures a spatially weighted particle number for a prescribed probe φ. Such functions are *observables*. If the preparation of the system does not specify a unique initial configuration, a probability law instead describes how likely different configurations are. The three objects we must therefore distinguish are a *state* x, an *observable* f, and a *probability law* μ.

The distinction is more than terminology. A dynamical rule moves states; the same rule changes the values of observables and transports probability laws. We will construct these induced actions rather than postulate three unrelated evolution equations. This is useful both in statistical physics, where microscopic trajectories give rise to ensemble predictions, and in machine learning, where learned flows act on samples but are evaluated by statistics of their resulting distributions.

States and linear structure. A *state space* is the set of admissible physical states, equipped with the structures required by the question under study. Throughout the finite-dimensional foundation we use a single symbol $V = \mathbb{R}^d$: it is both the physical state space and the real vector space on which linear operations are defined. In the particle example, $d = 6N$. Other systems need not have a vector-space state space; a fixed-length spin, for instance, lies on the sphere $S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$. When a general measurable space or a nonlinear state space is needed later, we introduce it explicitly as E rather than use E as another name for V.

Definition 1.1 (Linear endomorphisms)

Let $V = \mathbb{R}^d$ be the real vector space of states, with its usual addition and scalar multiplication. A map $A : V \to V$ is a *linear endomorphism* if

$$A(ax + by) = aA(x) + bA(y) \quad (x, y \in V,\ a, b \in \mathbb{R}).$$

The notation

$$\mathrm{End}(V) = \mathrm{Hom}_{\mathbb{R}}(V, V)$$

denotes the space of all real-linear maps from V to itself. With

- pointwise addition $(A + B)(x) = A(x) + B(x)$;
- scalar multiplication $(cA)(x) = cA(x)$;
- multiplication given by composition $AB = A \circ B$,

this space is a **unital associative algebra** over $\mathbb{R}$. Its unit is the identity map $I_V(x) = x$. Choosing a basis identifies this algebra with $M_d(\mathbb{R})$, the algebra of real

$d \times d$ matrices: addition and scalar multiplication correspond to the respective matrix operations, and composition corresponds to matrix multiplication.

Linear endomorphisms will reappear as linear ODE velocity fields and as derivatives of nonlinear flow maps. Their introduction does not constrain the dynamics to be linear. A general state transformation $F : V \to V$ need not belong to $\mathrm{End}(V)$. For example, $F(x) = x^3$ is a well-defined transformation of $\mathbb{R}$ but is not linear. To describe the effect of arbitrary state transformations, we next turn to functions on V.

Measurable structure and observable algebra. A physical observable assigns a numerical value to each state of a system. However, when the initial state is not known exactly, we must also be able to describe events involving possible states. For example, if $H : V \to \mathbb{R}$ denotes the energy of a state, the condition $H(x) \leq h$ determines the subset

$$\{x \in V : H(x) \leq h\}.$$

To assign probabilities to such events, we must first specify a collection of subsets of the state space on which probabilities can be consistently defined. This motivates the introduction of a measurable structure.

Definition 1.2 (σ-algebra and measurable space)

On the chosen state set V, a family of subsets $\mathcal{S} \subseteq 2^V$ is called a *σ-algebra* on V if it satisfies:

1. $V \in \mathcal{S}$;
2. If $A \in \mathcal{S}$, then $V \setminus A \in \mathcal{S}$;
3. If $(A_n)_{n\geq 1}$ is a sequence of sets in $\mathcal{S}$, then

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}.$$

The elements of $\mathcal{S}$ are called *measurable sets*. The pair $(V, \mathcal{S})$ is called a *measurable space*.

These conditions ensure that measurable sets remain measurable under complements and countable unions. They also imply that $\varnothing \in \mathcal{S}$ and that $\mathcal{S}$ is closed under countable intersections, since

$$\bigcap_{n=1}^{\infty} A_n = V \setminus \left(\bigcup_{n=1}^{\infty} (V \setminus A_n) \right).$$

In probability theory, measurable subsets of the state space represent events to which probabilities may be assigned. The measurable structure specifies which events are admissible;

it does not itself assign probabilities.

For the continuous dynamics considered here, we also need a notion of neighborhoods: when do states approach one another, and what does it mean for an observable or a state transformation to be continuous? A *topology* supplies the neighborhood structure underlying these notions of convergence and continuity. This structure is used to describe continuous trajectories and flow maps; later, bounded continuous observables also serve as tests for weak convergence of probability laws.

On $V = \mathbb{R}^d$, we use the topology induced by the Euclidean distance. Generating a σ-algebra from the open sets then connects this topological structure to measurable events, so that probability laws can be studied together with continuous observables and dynamics.

Definition 1.3 (Euclidean open sets and Borel σ-algebra)

Let $V = \mathbb{R}^d$ with its Euclidean distance. For $x \in V$ and $r > 0$, the *open ball* centered at x with radius r is

$$B_r(x) := \{y \in V : \|y - x\| < r\}.$$

The collection of *open subsets* of V is

$$\mathcal{O}(V) := \{O \subseteq V : \forall x \in O,\ \exists r > 0 :\ B_r(x) \subseteq O\}\,.$$

This collection is the *Euclidean topology* on V. The *Borel σ-algebra* on V is

$$\mathcal{B}(V) := \sigma(\mathcal{O}(V)),$$

where $\sigma(\mathcal{O}(V))$ denotes the smallest σ-algebra containing every open subset of V. The pair $(V, \mathcal{B}(V))$ is called the *Borel measurable space*.

The Borel σ-algebra contains all open and closed sets, as well as the sets obtained from them through countable unions, countable intersections, and complements. It provides a natural measurable structure compatible with the topology already introduced on the state space.

Having specified which subsets of V are measurable, we must next determine which functions preserve this structure.

Definition 1.4 (Measurable maps and observables)

Let $(V, \mathcal{S})$ and $(Y, \mathcal{G})$ be measurable spaces. A map $F : V \to Y$ is called *measurable* if $\forall B \in \mathcal{G}$, $F^{-1}(B) \in \mathcal{S}$, where

$$F^{-1}(B) := \{x \in V : F(x) \in B\}$$

denotes the inverse image of B under F. A real-valued observable on V is a Borel

measurable function

$$f : V \to \mathbb{R},$$

where both V and $\mathbb{R}$ are equipped with their Borel σ-algebras.

Measurability ensures that conditions imposed on observable values determine measurable events in the state space. For example, $\forall a \in \mathbb{R}$,

$$\{x \in V : f(x) \leq a\} = f^{-1}((-\infty, a]) \in \mathcal{B}(V).$$

It also guarantees compatibility with state transformations. If $F : V \to V$ and $f : V \to \mathbb{R}$ are Borel measurable, then their composition is Borel measurable, since

$$(f \circ F)^{-1}(B) = F^{-1}(f^{-1}(B)) \in \mathcal{B}(V)$$

for every $B \in \mathcal{B}(\mathbb{R})$. Thus, composing an observable with a measurable state transformation produces another measurable observable. In particular, every continuous state transformation is Borel measurable.

We now introduce a convenient algebraic space of observables. Although physical observables need not be bounded, bounded observables provide a natural class that is closed under the basic pointwise operations.

Definition 1.5 (Algebra of observables)

Let $V = \mathbb{R}^d$ with its Borel σ-algebra. Define

$$\begin{aligned}\mathcal{A} &:= B_b(V) \\ &= \left\{ f : V \to \mathbb{R} : \big(\forall B \in \mathcal{B}(\mathbb{R}), f^{-1}(B) \in \mathcal{B}(V)\big) \wedge \big(\sup_{x \in V} |f(x)| < \infty\big) \right\}.\end{aligned}$$

For $f, g \in \mathcal{A}$ and $a \in \mathbb{R}$, define the pointwise operations

$$\begin{aligned}(f + g)(x) &= f(x) + g(x), \\ (af)(x) &= af(x), \\ (fg)(x) &= f(x)g(x).\end{aligned}$$

These operations preserve boundedness and Borel measurability. Consequently, $\mathcal{A}$ is a *commutative unital algebra* over $\mathbb{R}$, with multiplicative identity

$$\mathbf{1}(x) = 1, \quad \forall x \in V.$$

The multiplication in $\mathcal{A}$ is pointwise multiplication of functions, whereas multiplication in $\mathrm{End}(V)$ is composition of linear maps. These are distinct algebraic operations on different mathematical objects.

The boundedness restriction in $\mathcal{A}$ is a mathematical convenience, not a physical require-

ment. Important observables, such as a Hamiltonian $H(x)$ or $\|x\|^2$, may be unbounded and therefore need not belong to $\mathcal{A}$. Such observables remain measurable functions and can be used in statistical expressions whenever the required integrability conditions are satisfied.

The space $\mathcal{A}$ also admits the supremum norm

$$\|f\|_\infty := \sup_{x \in V} |f(x)|.$$

The completeness of this normed algebra will be used when discussing operator domains in the next section.

1.1.2 Probability laws and integration

A microscopic system is characterized by a state belonging to a physical state space V. However, the actual microscopic state need not be completely determined by the information available to us. In some situations, the system occupies a definite but unknown state, and a probabilistic description represents our uncertainty about its value. In other situations, a preparation protocol may produce different microscopic states across repeated realizations, even when the same macroscopic conditions are imposed. The resulting variability can also be described probabilistically. These two situations have different physical interpretations: the former concerns incomplete knowledge of a state, whereas the latter concerns the statistical variability of a preparation procedure. Nevertheless, both can be represented mathematically by probability measures on the physical state space.

To formulate this description precisely, we distinguish the sample space Ω, whose elements represent possible outcomes of an experiment or an uncertainty model, from the physical state space V, whose elements represent the possible microscopic configurations of the system. A random state associates each outcome in the sample space with a corresponding physical state. By assigning probabilities to events in the sample space, we can consequently determine the probabilities of measurable events in V. Thus, the Borel measurable structure $(V, \mathcal{B}(V))$ introduced earlier provides the natural setting for describing statistical properties of microscopic states, while the sample space provides a general representation of the underlying uncertainty.

Definition 1.6 (Probability space, random state, and law)

A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where $\mathcal{F}$ is a σ-algebra on Ω and $\mathbb{P}$ is a countably additive map $\mathbb{P} : \mathcal{F} \to [0, 1]$ with $\mathbb{P}(\Omega) = 1$. A V-valued random state is a measurable map

$$X : (\Omega, \mathcal{F}) \longrightarrow (V, \mathcal{B}(V)).$$

Its law $\mu = \mathrm{Law}(X)$ is the Borel probability measure on V defined by $\mu(B) = \mathbb{P}(X \in B)$. Write $\mathcal{P}(V)$ for the set of all Borel probability measures on V.

For every $f \in \mathcal{A}$, the expectation of $f(X)$ depends only on its law:

$$\mathbb{E}[f(X)] = \int_V f(x)\, \mu(\mathrm{d}x) =: \langle f, \mu \rangle.$$

The bracket is an *integration pairing*. It is linear in the observable and extends to a bilinear pairing if probabilities are replaced by finite signed measures. One should not identify $\mathcal{P}(V)$ itself with a vector space: it is a convex subset of the space of signed measures. Similarly, a probability law is not necessarily a density. Only when μ is absolutely continuous with respect to Lebesgue measure do we write $\mu(\mathrm{d}x) = \rho(x)\, \mathrm{d}x$.

1.1.3 Transformations, duality, and evolution

A Borel measurable state transformation $F : V \to V$ induces two actions. An observable f is evaluated after the transformation as $f \circ F$; a law μ assigns the transformed event B the probability $\mu(F^{-1}(B))$. The integration pairing relates these actions. Neither linearity nor invertibility of F is needed.

Definition 1.7 (Composition operator and pushforward measure)

For a Borel measurable $F : V \to V$, its composition (Koopman) operator is the linear map $\mathcal{K}_F : \mathcal{A} \to \mathcal{A}$, defined by

$$(\mathcal{K}_F f)(x) := f(F(x)), \quad f \in \mathcal{A},$$

and its pushforward map $\mathsf{T}_F : \mathcal{P}(V) \to \mathcal{P}(V)$ by

$$\mathsf{T}_F \mu := F_{\#}\mu, \quad (F_{\#}\mu)(B) := \mu(F^{-1}(B)), \quad B \in \mathcal{B}(V).$$

For a measurable map $\xi : V \to \mathcal{Z}$ into another measurable state space $\mathcal{Z}$, the same definition gives $\xi_{\#}\mu \in \mathcal{P}(\mathcal{Z})$; here $\mathcal{P}(\mathcal{Z})$ uses its specified sigma-algebra.

Theorem 1.8 (Observable–measure duality)**.** *On $(V, \mathcal{B}(V))$, let $F : V \to V$ be Borel measurable. Then $\mathcal{K}_F$ is a positive, unital algebra homomorphism on $B_b(V)$ and $\|\mathcal{K}_F f\|_\infty \le \|f\|_\infty$. For every $f \in B_b(V)$ and $\mu \in \mathcal{P}(V)$,*

$$\boxed{\langle \mathcal{K}_F f, \mu \rangle = \langle f, \mathsf{T}_F \mu \rangle.} \tag{1.1}$$

Moreover, for measurable $F, G : V \to V$,

$$\mathcal{K}_F \mathcal{K}_G = \mathcal{K}_{G \circ F}, \quad \mathsf{T}_G \mathsf{T}_F = \mathsf{T}_{G \circ F}. \tag{1.2}$$

Proof. Linearity and preservation of products follow by evaluating at x:

$$\mathcal{K}_F(af + bg)(x) = af(F(x)) + bg(F(x)),$$
$$\mathcal{K}_F(fg)(x) = f(F(x))g(F(x)).$$

Positivity, preservation of the constant function $\mathbf{1}$, and the supremum-norm bound follow from the same definition. For an indicator $f = \mathbf{1}_B$, the defining identity of the pushforward reads

$$\int_V \mathbf{1}_B(y)\,(F_\#\mu)(\,\mathrm{d}y) = \mu(F^{-1}(B)) = \int_V \mathbf{1}_B(F(x))\,\mu(\,\mathrm{d}x).$$

Linearity proves this for simple functions. Monotone approximation and the monotone convergence theorem extend it to nonnegative measurable functions, and taking positive and negative parts proves (1.1) for bounded real observables. Finally, $\mathcal{K}_F\mathcal{K}_G f = f \circ G \circ F$ and $(G_\# F_\# \mu)(B) = \mu(F^{-1}(G^{-1}(B)))$, proving (1.2). □

Thus a nonlinear state map still induces a linear operator on observables, dual to its action on laws. No probability density is required.

1.1.3.1 Families of state transformations

An evolution is a compatible family of state transformations, not merely one map. Suppose maps $\Phi_{s,t} : V \to V$ are defined for $s \le t$ and satisfy

$$\Phi_{s,s} = \mathrm{id}_V, \quad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t} \quad (s \le r \le t).$$

Define $\mathcal{K}_{s,t} := \mathcal{K}_{\Phi_{s,t}} : \mathcal{A} \to \mathcal{A}$ and $\mathsf{T}_{s,t} := \mathsf{T}_{\Phi_{s,t}} : \mathcal{P}(V) \to \mathcal{P}(V)$. The preceding theorem immediately yields

$$\begin{aligned} \mathcal{K}_{s,r}\mathcal{K}_{r,t} = \mathcal{K}_{s,t}, \quad \mathsf{T}_{r,t}\mathsf{T}_{s,r} = \mathsf{T}_{s,t}, \\ \langle \mathcal{K}_{s,t} f, \mu_s \rangle = \langle f, \mathsf{T}_{s,t}\mu_s \rangle. \end{aligned} \tag{1.3}$$

In (1.3), the order of the composition operators is reversed because an observable at time t is pulled back to the initial state, whereas a probability law is pushed forward in chronological order. The next section will *construct* such $\Phi_{s,t}$ by solving an ODE, rather than assume their existence. In particular, the existence interval and invertibility of a flow require hypotheses on the underlying vector field.

A deterministic flow does not require a probability space: $\Phi_{s,t}(x)$ is defined for a specified initial state. If one subsequently introduces a random initial condition X_s with law μ_s, the random state $X_t = \Phi_{s,t}(X_s)$ has law $\mathsf{T}_{s,t}\mu_s$. Thus probability enters to describe an *ensemble of initial conditions*, not to define the underlying deterministic equation.

The remainder follows the construction from trajectories to transported laws (Project 9), its Hamiltonian application (Project 10), entropy and duality (Project 11), conditional reduction (Projects 12–13), and learned or numerical evolution (Projects 14 and 8). The Source Index records the individual questions; each result states its own assumptions.

1.2 Dynamical Equations, Flow Maps, and Probability Transport

This section develops a common framework for describing **state-space dynamics** and **the evolution of probability laws**. Taking flows generated by ordinary differential equations (ODEs) as the starting point, we study how the motion of individual states induces the transport of observables, measures, and densities. We then illustrate the geometric structure through Hamiltonian dynamics and turn to entropy identities and variational principles. The discussion is organized around the following topics and questions:

- *State flows and volume deformation.* Construct the flow $\Phi_{s,t}$ from an ODE with velocity field v_t and study its dependence on initial conditions.
  - How does the flow derivative $M_{s,t}(x)$ describe local deformation?
  - How does its determinant $J_{s,t}(x)$ determine changes in volume and transported densities?
- *Evolution of observables and probability laws.* Study the observable evolution operator $\mathcal{K}_{s,t}$, the law transport operator $\mathsf{T}_{s,t}$, and the generator $\mathcal{L}_t$.
  - How does the generator $\mathcal{L}_t$ relate to the velocity field v_t and the flow $\Phi_{s,t}$?
  - How does the continuity equation $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$ describe the evolution of densities?
- *Hamiltonian dynamics.* Illustrate the framework with the Hamiltonian $H(q, p)$ and its flow ϕ_t.
  - How does Hamiltonian structure lead to energy conservation and symplecticity?
  - What does volume preservation imply for probability transport?
- *Entropy and variational principles.* Develop transport identities for differential entropy $S(\rho)$ and relative entropy $D_{\mathrm{KL}}(\mu\|\nu)$, then connect convex duality and the log-partition functional $\Lambda_Q(g)$ to variational principles and free energy $F(\rho)$.
  - How do entropy functionals evolve under transport?
  - How do the variational representations of relative entropy and the free-energy gap characterize equilibrium and deviations from it?

1.2.1 State flows and geometric properties

Consider a moving fluid whose motion is described by a time-dependent velocity field $v_t(x)$. At each time t and position x, the vector $v_t(x)$ specifies the instantaneous velocity

of the fluid at that location. A fluid element moving with the surrounding fluid follows the local velocity field. If x_t denotes its position at time t, its motion is therefore governed by

$$\dot{x}_t = v_t(x_t).$$

Assume that v is sufficiently smooth in time and space. Given an initial time s and an initial position x, we seek a trajectory x_t that describes the subsequent motion of this fluid element.

However, understanding fluid motion involves more than determining individual trajectories. We are also interested in how the motion depends on the initial position. In particular, two fluid elements starting from nearby positions may move apart or approach each other. Their relative motion determines the local deformation of the fluid, including stretching, rotation, and expansion or contraction. These effects are closely related to the evolution of infinitesimal volume elements and, through mass conservation, the redistribution of fluid density.

This motivates a systematic study of the initial-value problem: how does a given initial state determine a unique trajectory, and how does that trajectory depend on its initial condition? We begin by formulating this problem precisely.

1.2.1.1 ODE initial-value problems

Let $V = \mathbb{R}^d$, where $d \geq 1$ is an integer, be the Euclidean state space, equipped with its Euclidean norm. The time-dependent velocity field specifies the fluid velocity $v(t, x)$ at each time t and position x. We write it as the map

$$v : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d, \quad (t, x) \mapsto v_t(x),$$

and assume that it is smooth and satisfies $v_t \in C_0(\mathbb{R}^d; \mathbb{R}^d)$, for all $t \in \mathbb{R}$. Given initial data $(s, x) \in \mathbb{R} \times \mathbb{R}^d$, the corresponding initial-value problem is to find a trajectory $x_t \in C^1(I, V)$ satisfying

$$\begin{cases} \dot{x}_t = v_t(x_t), \quad t \in [s, +\infty), \\ x_s = x. \end{cases} \tag{1.4}$$

We next state an existence and uniqueness theorem for the initial-value problem.

Theorem 1.9 (Picard–Lindelöf Theorem)**.** *Consider the initial-value problem*

$$\begin{cases} \dot{x}(t) = f(t, x(t)), \\ x(t_0) = x_0, \end{cases}$$

where $f : \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}^d$. Suppose that:

1. *f is continuous in (t, x).*
2. *f is locally Lipschitz continuous with respect to x, locally uniformly in t. That is, for every compact interval $J \subset \mathbb{R}$ and every $R > 0$, there exists $L_{J,R} < \infty$ such*

that

$$\|f(t,y) - f(t,z)\| \le L_{J,R}\|y - z\|, \quad t \in J, \quad y, z \in \overline{B}_R(0).$$

Then, for every initial condition $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^d$, there exists an open interval containing t_0 on which the initial-value problem admits a unique C^1 solution. Moreover, any two solutions with the same initial data agree on the intersection of their intervals of definition.

The proof is given in Appendix A.1. The theorem provides the existence and uniqueness of the trajectory, which allows us to define the corresponding flow map in the following discussion.

1.2.1.2 A linear inhomogeneous initial-value problem

Example 1.10 (Solution of a time-dependent linear inhomogeneous ODE)**.**

Source exercise: Project 9, item 1. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $A : I \to \mathrm{End}(V)$ and $B : I \to V$ be continuous functions. Consider the time-dependent affine vector field

$$v_t(y) = A_t y + B_t, \quad y \in V.$$

For a prescribed initial time $s \in I$ and initial state $x \in V$, consider the initial-value problem

$$\begin{cases} \dot{x}_t = A_t x_t + B_t, \\ x_s = x. \end{cases}$$

Prove that this initial-value problem admits a unique solution x_t on I, and derive an explicit representation of the solution in terms of the given functions A_t, B_t, and the initial state x.[a]

[a] In particular, construct the evolution operator associated with the homogeneous system and use it to obtain the solution of the inhomogeneous equation.

Solution.

We now establish the conditions under which this ODE admits a globally defined solution on I and prove its uniqueness.

1. **Existence.**

   On every compact subinterval $J \subset I$ containing s, continuity bounds A and B. Hence, the affine vector field $v_t(x) = A_t x + B_t$ is globally Lipschitz in the state variable and has at most linear growth, uniformly on J. By Theorem 1.9, a unique solution exists on each such interval.

2. **Global extension and uniqueness.**

   Since any two such intervals containing s overlap, uniqueness guarantees that their solutions agree on the overlap. Consequently, these solutions can be consistently joined into a single trajectory defined on all of I. The same uniqueness argument ensures that this trajectory is unique.

Since the unknown x_t describes the evolution of a state vector over time, the object to be determined is a trajectory rather than a single element of V. We therefore introduce the trajectory space

$$\mathcal{X} = C^1(I, V), \quad \mathcal{Y} = C(I, V).$$

Here, $\mathcal{X}$ consists of continuously differentiable V-valued trajectories, while $\mathcal{Y}$ consists of continuous V-valued functions. Both inherit their vector-space structures from V, with addition and scalar multiplication defined pointwise.

To express the differential equation in this functional framework, we introduce the

- differentiation operator:

$$\partial_t : \mathcal{X} \to \mathcal{Y}, \quad (\partial_t \gamma)(t) = \dot{\gamma}(t),$$

- operator induced by the time-dependent family A_t:

$$\mathsf{M}_A : \mathcal{X} \to \mathcal{Y}, \quad (\mathsf{M}_A \gamma)(t) = A_t \gamma(t).$$

Both are linear maps, and the given continuous forcing term satisfies $B \in \mathcal{Y}$. Defining $\mathsf{D}_A := \partial_t - \mathsf{M}_A$, the differential equation can be written as an equation for the entire trajectory:

$$\begin{cases} \mathsf{D}_A \gamma = B, \\ \gamma(s) = x. \end{cases}$$

Here, $\mathsf{D}_A \gamma = B$ means that

$$\dot{\gamma}(t) - A_t \gamma(t) = B_t$$

for every $t \in I$. Thus, the operator equation is simply the trajectory-space formulation of the original ODE, supplemented by its initial-value constraint.

1. The homogeneous equation. To solve the inhomogeneous equation, we first consider the associated homogeneous equation[1]:

$$\begin{cases} \mathsf{D}_A \eta = 0, \\ \eta(s) = v. \end{cases}$$

[1] Equivalently, $\dot{\eta}(t) - A_t \eta(t) = 0,\ \eta(s) = v$, where $v \in V$ is arbitrary.

Allowing the initial vector v to vary, all solutions of the homogeneous differential equation form the kernel of D_A:

$$\mathcal{H} := \ker \mathsf{D}_A = \{\eta \in \mathcal{X} : \mathsf{D}_A \eta = 0\}.$$

Since D_A is linear, $\mathcal{H}$ is a linear subspace of the trajectory space $\mathcal{X}$. To understand the structure of $\mathcal{H}$, we associate each homogeneous trajectory with its state at a fixed time. By uniqueness of the initial-value problem, this state uniquely determines the entire trajectory. Thus, we can identify the trajectory space $\mathcal{H}$ with the finite-dimensional state space V, without directly constructing a basis for $\mathcal{H}$.

1. **Evaluation at an arbitrary time t.** For each $t \in I$, define

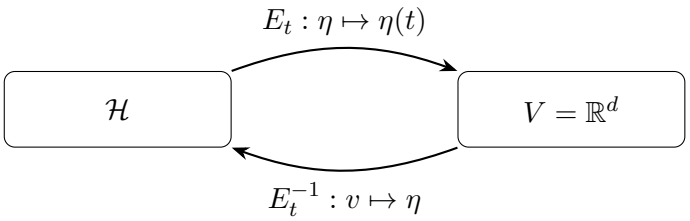
   $$E_t : \mathcal{H} \to V, \quad E_t(\eta) = \eta(t).$$

   By pointwise linearity, E_t is linear. Moreover, existence and uniqueness of the homogeneous initial-value problem at time t ensure that E_t is bijective. Hence, E_t is a linear isomorphism for every $t \in I$.

2. **The inverse solution map and the dimension of the solution space.** For each $t \in I$, define the solution map as the inverse of evaluation:

   $$S_t := E_t^{-1} : V \to \mathcal{H}.$$

   The output $S_t(v)$ is the unique complete homogeneous trajectory whose value at time t is v.



The inverse identities are $E_t \circ S_t = I_V$ and $S_t \circ E_t = I_{\mathcal{H}}$. Since E_t is a linear isomorphism, its inverse S_t is linear as well:

$$S_t(av + bw) = aS_t(v) + bS_t(w), \quad v, w \in V.$$

This also expresses the superposition principle for homogeneous solutions. In particular,

$$\mathcal{H} \cong V, \quad \dim \mathcal{H} = d.$$

Although the ambient trajectory space $\mathcal{X}$ is generally infinite-dimensional, its homogeneous solution subspace has exactly d degrees of freedom, parameterized by the initial vector.

For a fixed initial time $s \in I$, first use S_s to associate the initial state with its unique homogeneous trajectory, and then use E_t to evaluate that trajectory at time t:

$$V \xrightarrow{S_s} \mathcal{H} \xrightarrow{E_t} V.$$

This composition defines the homogeneous evolution operator

$$U(t,s) := E_t \circ S_s \in \mathrm{End}(V).$$

Thus every homogeneous trajectory $\eta \in \mathcal{H}$ satisfies

$$\begin{cases} \eta(t) = U(t,s)\eta(s), \\ \eta(s) = v. \end{cases}$$

In particular, $U(s,s) = I_V$.

For fixed s and any $v \in V$, the trajectory $\eta(t) : t \mapsto U(t,s)v$ is precisely the homogeneous solution whose value at time s is v. It therefore satisfies

$$\partial_1 U(t,s) = A_t U(t,s), \quad U(s,s) = I_V. \tag{1.5}$$

Here ∂_1 denotes differentiation with respect to the first time argument. The equation is an identity in $\mathrm{End}(V)$: it follows because the corresponding vector identity holds for every $v \in V$. Thus $U(t,s)$ describes the finite-time evolution generated by the instantaneous linear dynamics A_t.

We now establish the basic algebraic properties of the evolution operator $U(t,s)$.

- **Composition law.**

  For any $v \in V$, the trajectories $\tau \mapsto U(\tau,r)U(r,s)v$ and $\tau \mapsto U(\tau,s)v$ solve the same homogeneous equation and have the same value at $\tau = r$. By uniqueness,

  $$U(t,r)U(r,s) = U(t,s), \quad r,s,t \in I.$$

- **Invertibility.**

  Taking the intermediate time to be either endpoint in the composition law gives

  $$U(t,s)U(s,t) = U(t,t) = I_V,$$
  $$U(s,t)U(t,s) = U(s,s) = I_V.$$

  Hence,

  $$U(t,s)^{-1} = U(s,t), \quad U(t,s) \in GL(V).$$

Thus, the homogeneous evolution is an invertible linear transformation. Further details on the construction and algebraic properties of U are given in Appendices A.3 and A.4. Its time-ordered representation for noncommuting A_t is developed in Appendix A.5.

2. The inhomogeneous equation. Now consider the inhomogeneous equation

$$\mathsf{D}_A\gamma = B.$$

We already know that the homogeneous solution space is $\mathcal{H} = \ker \mathsf{D}_A$. Suppose that we have found one particular solution γ_{p} satisfying $\mathsf{D}_A\gamma_{\mathrm{p}} = B$. If γ is any other solution, then

$$\mathsf{D}_A(\gamma - \gamma_{\mathrm{p}}) = \mathsf{D}_A\gamma - \mathsf{D}_A\gamma_{\mathrm{p}} = B - B = 0,$$

so $\gamma - \gamma_{\mathrm{p}} \in \mathcal{H}$. Conversely, for every $\eta \in \mathcal{H}$, linearity gives

$$\mathsf{D}_A(\gamma_{\mathrm{p}} + \eta) = \mathsf{D}_A\gamma_{\mathrm{p}} + \mathsf{D}_A\eta = B.$$

Therefore all solutions of the inhomogeneous equation form

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H}.$$

This is an affine subspace of $\mathcal{X}$, obtained by translating the homogeneous solution space $\mathcal{H}$ by γ_{p}. Having determined this algebraic structure, it remains to construct one particular solution.

Constructing the particular solution.

To simplify the construction, choose the particular solution with zero initial value:

$$\begin{cases} \dot{\gamma}_{\mathrm{p}}(t) = A_t\gamma_{\mathrm{p}}(t) + B_t, \\ \gamma_{\mathrm{p}}(s) = 0. \end{cases}$$

Its existence and uniqueness follow from Theorem 1.9. The linear term $A_t\gamma_{\mathrm{p}}(t)$ is governed by the same operator A_t as in the homogeneous system, whose evolution is described by $U(t,s)$; B_t is the additional forcing.

To construct this particular solution, we use the homogeneous evolution operator established above. For each fixed pair of times, $U(t,s)$ is a linear map from the entire state space V to itself. Its inverse $U(s,t)$ is therefore also defined on all of V. Since $\gamma_{\mathrm{p}}(t) \in V$, we may apply $U(s,t)$ to $\gamma_{\mathrm{p}}(t)$, even though γ_{p} itself solves the inhomogeneous equation.

We introduce the auxiliary trajectory $c(t)$ by

$$c(t) = U(s,t)\gamma_{\mathrm{p}}(t). \tag{1.6}$$

The auxiliary trajectory $c \in \mathcal{X}$ represents the evolution of the particular solution in a reference frame that removes the homogeneous dynamics generated by A_t. Its time dependence is therefore governed solely by the external driving B_t, transformed by the inverse homogeneous evolution operator $U(s,t)$. Equivalently, by the inverse identity,

$$\gamma_{\mathrm{p}}(t) = U(t,s)c(t).$$

This is a time-dependent change of representation relative to the homogeneous evolution; it does not assume that γ_{p} itself follows the homogeneous dynamics.

To determine the evolution of $c(t)$, we first note that for fixed s, the map $t \mapsto U(t,s)$ is continuously differentiable and takes values in $GL(V)$. Since inversion is a smooth operation on $GL(V)$, its inverse $t \mapsto U(s,t)$ is also continuously differentiable. We can therefore differentiate the inverse identity $U(s,t)U(t,s) = I_V$ with respect to t. Using (1.5), we obtain

$$0 = \frac{\partial}{\partial t}\big(U(s,t)U(t,s)\big) = \partial_2 U(s,t)\, U(t,s) + U(s,t)\partial_1 U(t,s).$$

Right-multiplying by $U(s,t)$ and using $U(t,s)U(s,t) = I_V$ gives

$$\partial_2 U(s,t) = -U(s,t)A_t.$$

The product rule applied to (1.6), together with $\dot{\gamma}_{\mathrm{p}}(t) = A_t\gamma_{\mathrm{p}}(t) + B_t$, now yields

$$\begin{aligned}\dot{c}(t) &= -U(s,t)A_t\gamma_{\mathrm{p}}(t) + U(s,t)\dot{\gamma}_{\mathrm{p}}(t)\\ &= -U(s,t)A_t\gamma_{\mathrm{p}}(t) + U(s,t)\big(A_t\gamma_{\mathrm{p}}(t) + B_t\big)\\ &= U(s,t)B_t.\end{aligned}$$

The homogeneous terms cancel exactly, leaving only the forcing term in the transformed coordinates. Since $\gamma_{\mathrm{p}}(s) = 0$ and $U(s,s) = I_V$, we have $c(s) = 0$. Integrating from s to t gives

$$c(t) = \int_s^t U(s,\tau)B_\tau \, \mathrm{d}\tau.$$

Using $U(t,s)U(s,\tau) = U(t,\tau)$ to return to the original coordinates gives

$$\begin{aligned}\gamma_{\mathrm{p}}(t) &= U(t,s)\int_s^t U(s,\tau)B_\tau \, \mathrm{d}\tau\\ &= \int_s^t U(t,s)U(s,\tau)B_\tau \, \mathrm{d}\tau\\ &= \int_s^t U(t,\tau)B_\tau \, \mathrm{d}\tau.\end{aligned}$$

Hence the zero-initial-value particular solution is

$$\gamma_{\mathrm{p}}(t) = \int_s^t U(t,\tau)B_\tau \, \mathrm{d}\tau.$$

The homogeneous trajectory with initial state x is $\eta_x(t) := U(t,s)x$. Adding it to the particular solution gives

$$x_t = \eta_x(t) + \gamma_{\mathrm{p}}(t).$$

Therefore,

$$\boxed{x_t = U(t,s)x + \int_s^t U(t,\tau)B_\tau \, \mathrm{d}\tau.} \tag{1.7}$$

This is the variation-of-constants formula.

The solution has a natural algebraic interpretation. For fixed $s, t \in I$, its dependence on the initial state is given by

$$x \longmapsto U(t,s)x + \gamma_{\mathrm{p}}(t).$$

The chosen γ_{p} depends on the reference time s through $\gamma_{\mathrm{p}}(s) = 0$, but is independent of x. Since $U(t,s)$ is invertible, this is an invertible affine transformation of the state space.

The homogeneous equation generates linear transformations of V, while the forcing introduces a time-dependent translation. Correspondingly, the homogeneous solutions form a vector space, whereas the solutions of the inhomogeneous equation with fixed forcing form an affine space modeled on that homogeneous solution space. We can make this geometric structure more explicit within the trajectory space. Extend evaluation from $\mathcal{H}$ to the ambient trajectory space by defining

$$\widetilde{E}_s : \mathcal{X} \to V, \quad \widetilde{E}_s(\gamma) = \gamma(s),$$

whose restriction to $\mathcal{H}$ is E_s. Define its kernel by

$$\mathcal{K}_s := \ker \widetilde{E}_s = \{\gamma \in \mathcal{X} : \gamma(s) = 0\}.$$

Thus, $\mathcal{K}_s$ consists of all trajectories that vanish at the initial time. For any $\gamma \in \mathcal{X}$, the trajectory

$$\eta(t) = U(t,s)\gamma(s).$$

is homogeneous and has the same initial value as γ. Therefore, $\gamma - \eta \in \mathcal{K}_s$. Moreover, a homogeneous trajectory that vanishes at s must vanish identically. Hence

$$\mathcal{H} \cap \mathcal{K}_s = \{0\}.$$

We consequently obtain the direct-sum decomposition $\mathcal{X} = \mathcal{H} \oplus \mathcal{K}_s$.

Recall the chosen particular solution γ_{p}, with $\gamma_{\mathrm{p}}(s) = 0$, and the homogeneous trajectory $\eta_x(t) = U(t,s)x$ associated with the prescribed initial state x.The inhomogeneous solution set is

$$\mathcal{S}_B = \gamma_{\mathrm{p}} + \mathcal{H},$$

while the set of all trajectories satisfying the initial-value constraint is

$$\mathcal{X}_{s,x} := \{\gamma \in \mathcal{X} : \gamma(s) = x\} = \eta_x + \mathcal{K}_s.$$

The trajectory $\gamma_{\mathrm{p}} + \eta_x$ belongs to both sets: it solves the inhomogeneous equation and has initial value x. To see that it is the only element of the intersection, suppose

$$\gamma = \gamma_{\mathrm{p}} + \eta = \eta_x + k,$$

where $\eta \in \mathcal{H}$ and $k \in \mathcal{K}_s$. Then

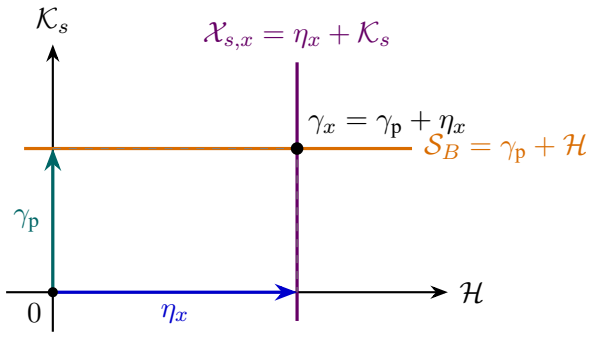
$$\eta - \eta_x = k - \gamma_{\mathrm{p}}.$$

The left-hand side belongs to $\mathcal{H}$, while the right-hand side belongs to $\mathcal{K}_s$, because $\gamma_{\mathrm{p}}(s) = 0$. Since $\mathcal{H} \cap \mathcal{K}_s = \{0\}$, both sides must vanish. Thus $\eta = \eta_x$ and $\gamma = \gamma_{\mathrm{p}} + \eta_x$. Therefore the intersection is exactly

$$\mathcal{S}_B \cap \mathcal{X}_{s,x} = \{\gamma_{\mathrm{p}} + \eta_x\}.$$

Denote this selected trajectory by $\gamma_x = \gamma_{\mathrm{p}} + \eta_x$, so that $\gamma_x(t) = x_t$.

The following diagram illustrates this geometry in a two-dimensional schematic slice of the trajectory space. The horizontal and vertical directions represent the complementary subspaces $\mathcal{H}$ and $\mathcal{K}_s$, respectively. They are not assumed to be orthogonal.



The homogeneous solution space determines the directions of the inhomogeneous solution set, while the particular solution determines its affine displacement. The initial-value constraint selects exactly one trajectory from this affine family. ♦

Example 1.11 (A concrete special case: Newton's law of cooling)**.** We now illustrate the preceding construction with a simple physical example. Consider a body whose temperature T_t relaxes toward a constant ambient temperature T_{env}. Under the lumped-temperature approximation and a constant heat-transfer coefficient, Newton's law of cooling takes the form

$$\dot{T}_t = -\lambda\big(T_t - T_{\mathrm{env}}\big), \quad T_0 = T_0,$$

where $\lambda > 0$ is the relaxation rate and $t \geq 0$.

Rearranging the equation,

$$\dot{T}_t = -\lambda T_t + \lambda T_{\mathrm{env}}.$$

This is a special case of the original problem with

$$V = \mathbb{R}, \quad A_t = -\lambda, \quad B_t = \lambda T_{\mathrm{env}},$$

where we identify $\mathrm{End}(\mathbb{R})$ with $\mathbb{R}$ through scalar multiplication.

Since A_t is constant, the evolution operator is simply

$$U(t,s) = e^{-\lambda(t-s)}.$$

The homogeneous contribution associated with theinitial temperature is

$$\eta_{T_0}(t) = U(t,0)T_0 = T_0 e^{-\lambda t}.$$

For this illustration the reference time is $s = 0$. The zero-initial-value particular solution is

$$\begin{aligned}\gamma_{\mathrm{p}}(t) &= \int_0^t U(t,\tau)B \, \mathrm{d}\tau \\ &= \lambda T_{\mathrm{env}} \int_0^t e^{-\lambda(t-\tau)} \, \mathrm{d}\tau \\ &= T_{\mathrm{env}} \left(1 - e^{-\lambda t}\right).\end{aligned}$$

Adding the homogeneous and particular contributions, we obtain

$$T_t = T_0 e^{-\lambda t} + T_{\mathrm{env}} \left(1 - e^{-\lambda t}\right).$$

Equivalently,

$$T_t = T_{\mathrm{env}} + (T_0 - T_{\mathrm{env}})e^{-\lambda t}.$$

For a concrete illustration, choose

$$T_0 = 80\,^\circ\mathrm{C}, \quad T_{\mathrm{env}} = 20\,^\circ\mathrm{C}, \quad \lambda = 0.5\,\mathrm{min}^{-1}.$$

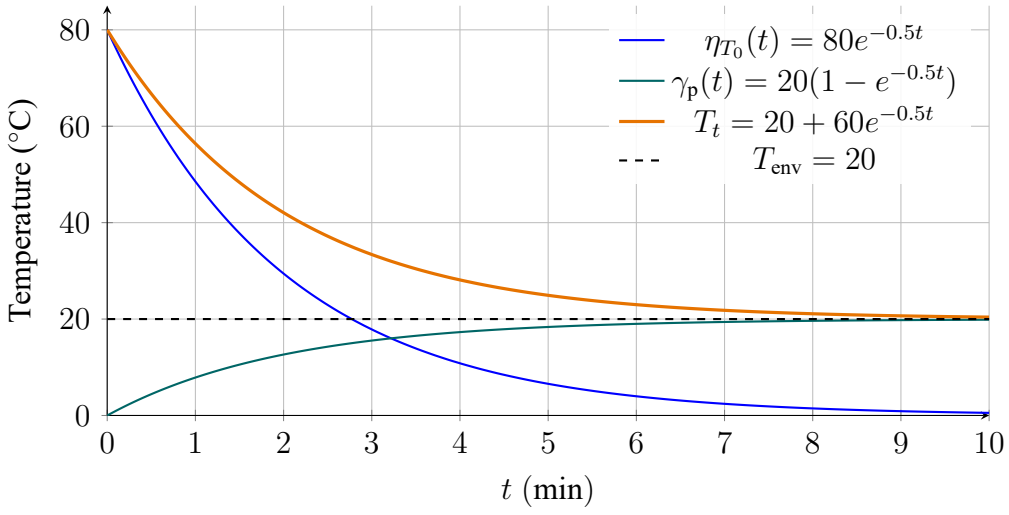
Then

$$\eta_{T_0}(t) = 80e^{-0.5t}, \quad \gamma_{\mathrm{p}}(t) = 20(1 - e^{-0.5t}),$$

and the complete trajectory is

$$T_t = 20 + 60e^{-0.5t}.$$

The following plot displays the homogeneous contribution, the zero-initial-value particular solution, and their sum. The horizontal dashed line indicates the ambient equilibrium temperature.



The homogeneous contribution describes the decay of the initial-state contribution, whereas the particular solution describes the accumulated effect of the ambient-temperature forcing. Their sum gives the actual temperature trajectory, which approaches T_{env} as $t \to +\infty$.

This example demonstrates that the affine-shift construction is directly applicable to physical relaxation processes. The decomposition is not merely a formal manipulation: it separates the contribution of the initial state from the accumulated response to external driving.

1.2.1.3 Two-time flows and variational matrices

In the homogeneous linear case, the evolution operator induces the flow map $\Phi_{s,t}(x) = U(t,s)x$. We now extend this construction to a general time-dependent velocity field $v_t(x)$, for which the flow is not necessarily linear in the initial state. Under the standing assumptions of Section 1.2.1.1, the initial-value problem admits a unique global solution for every initial time s and initial state $x \in V$. This follows from the Picard–Lindelöf theorem together with the growth control ensuring global existence. Denoting this solution by $x_t^{s,x}$, we define the associated two-time flow by

$$\Phi_{s,t} : V \to V, \quad x \mapsto \Phi_{s,t}(x) := x_t^{s,x}.$$

Figure 1.1 illustrates how the flow transports both an individual state and a material region.

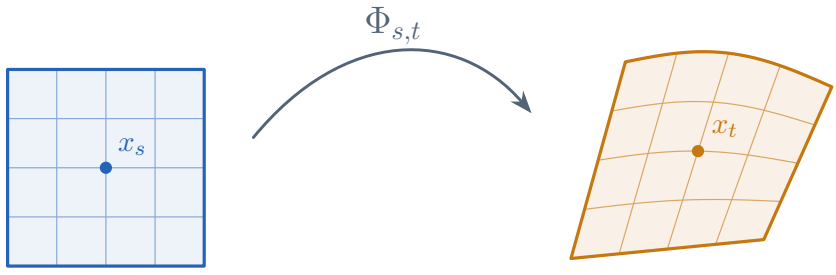


Figure 1.1: Transport of a material region by the two-time flow $\Phi_{s,t}$. The initial region Ω_s (left) and the marked state x_s are mapped to Ω_t (right) and x_t. The grid shows the deformation under the same smooth map; the area or volume need not be preserved.

The following theorem establishes the existence and basic structural properties of this two-time flow.

Theorem 1.12 (Existence, composition, and reversibility of the flow)**.**

Source exercise: Project 9, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Let $V = \mathbb{R}^d$. Assume that the velocity field $v : \mathbb{R} \times V \to V$ is jointly continuous, continuously differentiable in space with jointly continuous derivative, and globally Lipschitz in space uniformly on compact time intervals. Denote by $x_t^{s,x}$ the unique global solution of (1.4) with $x_s^{s,x} = x$, and define the two-time flow

$$\Phi_{s,t} : V \to V, \quad x \mapsto \Phi_{s,t}(x) := x_t^{s,x}.$$

Then the following statements hold:

1. ***Well-defined flow.*** *By Theorem 1.9 and the linear growth bound implied by the assumptions, the two-time flow $\Phi_{s,t} : V \to V$ is well-defined for every pair of times, including $t < s$.*

2. ***Composition and invertibility.*** *Without any ordering restriction on r, s, t, the flow satisfies*

$$\Phi_{s,s} = \mathrm{id}_V, \quad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t}, \quad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$

   Moreover, each $\Phi_{s,t} : V \to V$ is a C^1 diffeomorphism.

3. ***Autonomous flows.*** *If, in addition, $v_t = v$ is autonomous[a] and its solutions extend to all real times, then $\phi_t := \Phi_{0,t}$ defines a one-parameter group satisfying*

$$\phi_{t+r} = \phi_t \circ \phi_r, \quad \phi_{-t} = \phi_t^{-1}.$$

4. ***Time reversal.*** *For any solution x_t on $[0, T]$, the time-reversed trajectory*

$$y_t = x_{T-t}, \quad 0 \le t \le T,$$

   satisfies $\dot{y}_t = -v_{T-t}(y_t)$, $y_0 = x_T$.

[a] An autonomous dynamical system is one whose vector field has no explicit time dependence, i.e., $v_t(x) = v(x)$. Consequently, its evolution depends only on the elapsed time $t - s$, rather than on the initial time s itself.

Proof. (Proof in Appendix A.2.) □

Having established the existence, uniqueness, and algebraic properties of the flow, we now examine how its evolution depends on the initial state. In particular, the derivative of the flow with respect to the initial condition describes the evolution of infinitesimal perturbations, while its determinant characterizes the corresponding local volume change. These quantities satisfy the following evolution equations.

Proposition 1.13 (Variational equation and Jacobian evolution)**.**

Source exercise: Project 9, item 3–4. *Student Assessment Exercise* [Assessment] (adapted).

For the flow $\Phi_{s,t}$ under the standing assumptions, prove the following properties:

- ***Jacobian matrix.*** *The Jacobian matrix of the flow,*

$$M_{s,t}(x) := D_x\Phi_{s,t}(x) \in GL(V),$$

satisfies

$$\begin{cases} \partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x))M_{s,t}(x), \\ M_{s,s}(x) = I_d. \end{cases} \tag{1.8}$$

- ***Jacobian determinant.*** *The determinant of the Jacobian matrix satisfies*

$$J_{s,t}(x) := \det M_{s,t}(x) = \exp\left(\int_s^t \nabla \cdot v_r(\Phi_{s,r}(x))\ \mathrm{d}r\right). \tag{1.9}$$

Proof. Fix an initial time s and state x. By the definition of the flow and (1.4), we have

$$\partial_t\Phi_{s,t}(x) = v_t(\Phi_{s,t}(x)), \quad \Phi_{s,s}(x) = x. \tag{1.10}$$

By the continuous differentiability of the vector field in the state variable, the flow is differentiable with respect to its initial state. Differentiating the flow equation (1.10) with respect to x and applying the chain rule gives

$$\partial_t D_x\Phi_{s,t}(x) = Dv_t(\Phi_{s,t}(x))D_x\Phi_{s,t}(x).$$

By the definition $M_{s,t}(x) := D_x\Phi_{s,t}(x)$, we obtain

$$\partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x))M_{s,t}(x).$$

Since $\Phi_{s,s} = \mathrm{id}_V$, its derivative is the identity operator:

$$M_{s,s}(x) = D_x\Phi_{s,s}(x) = D_x\,\mathrm{id}_V(x) = I_d.$$

Thus, we obtain the matrix initial-value problem (1.8)

$$\begin{cases} \partial_t M_{s,t}(x) = Dv_t(\Phi_{s,t}(x))M_{s,t}(x), \\ M_{s,s}(x) = I_d. \end{cases}$$

Algebraic evolution in $\mathrm{End}(V)$**.** For a fixed initial state x, define

$$A_t := Dv_t(\Phi_{s,t}(x)) \in \mathrm{End}(V), \quad M_t := D_x\Phi_{s,t}(x) \in \mathrm{GL}(V).$$

The variational equation can then be written as an evolution equation in the associative algebra $\mathrm{End}(V)$:

$$\dot{M}_t = A_t \circ M_t, \quad M_s = I_V,$$

where multiplication is given by composition of linear operators. Its equivalent integral equation is

$$M_t = I_V + \int_s^t A_r \circ M_r \, \mathrm{d}r.$$

Successive substitution yields the Peano–Baker series in $\mathrm{End}(V)$:

$$M_t = I_V + \int_s^t A_{r_1} \, \mathrm{d}r_1 + \int_s^t \int_s^{r_1} A_{r_1} \circ A_{r_2} \, \mathrm{d}r_2 \, \mathrm{d}r_1 + \cdots .$$

More generally, its n-th order term is

$$\int_{s \le r_n \le \cdots \le r_1 \le t} A_{r_1} \circ \cdots \circ A_{r_n} \, \mathrm{d}r_n \cdots \mathrm{d}r_1.$$

Under the continuity of A_t on compact time intervals, this series converges absolutely in $\mathrm{End}(V)$.

The operator products retain their chronological order. Since $\mathrm{End}(V)$ is generally non-commutative, operators at different times need not commute:

$$A_{r_1} \circ A_{r_2} \neq A_{r_2} \circ A_{r_1}.$$

Consequently, the evolution generally cannot be represented by an ordinary exponential. Instead, it is expressed through the time-ordered exponential in $\mathrm{End}(V)$:

$$M_{s,t}(x) = \mathcal{T} \exp_{\mathrm{End}(V)} \left(\int_s^t A_r \, \mathrm{d}r \right).$$

If the operators commute pairwise at different times, namely

$$A_t \circ A_r = A_r \circ A_t, \quad s \le r, t,$$

the time ordering is unnecessary, and the evolution reduces to the ordinary exponential in the associative algebra:

$$M_{s,t}(x) = \exp_{\mathrm{End}(V)} \left(\int_s^t A_r \, \mathrm{d}r \right).$$

Furthermore, differentiating the flow composition identity

$$\Phi_{s,t} = \Phi_{r,t} \circ \Phi_{s,r}$$

with respect to the initial state gives

$$M_{s,t}(x) = M_{r,t}(\Phi_{s,r}(x)) \circ M_{s,r}(x).$$

Thus, the induced linear evolution inherits the composition structure of the original flow, known as the cocycle property.

Jacobian determinant and volume evolution. The flow composition and inverse identities imply

$$\Phi_{t,s} \circ \Phi_{s,t} = \mathrm{id}_V \, .$$

Differentiating with respect to the initial state x and applying the chain rule gives

$$D\Phi_{t,s}(\Phi_{s,t}(x)) \circ M_{s,t}(x) = I_V.$$

Since $M_{s,t}(x) \in \mathrm{End}(V)$ and V is finite-dimensional, the existence of a left inverse implies that $M_{s,t}(x)$ is invertible. Thus, $M_{s,t}(x) \in \mathrm{GL}(V)$. Define the scalar Jacobian

$$J_{s,t}(x) := \det M_{s,t}(x).$$

For a differentiable curve of invertible linear operators, Jacobi's formula gives

$$\partial_t J_{s,t}(x) = J_{s,t}(x) \, \mathrm{tr} \left(M_{s,t}(x)^{-1} \circ \partial_t M_{s,t}(x) \right) .$$

Using the variational equation established above,

$$\partial_t M_{s,t}(x) = A_t \circ M_{s,t}(x),$$

we obtain

$$\begin{aligned} \partial_t J_{s,t}(x) &= J_{s,t}(x) \, \mathrm{tr} \left(M_{s,t}(x)^{-1} \circ A_t \circ M_{s,t}(x) \right) \\ &= J_{s,t}(x) \, \mathrm{tr}(A_t). \end{aligned}$$

The second equality follows from the cyclic invariance of the trace. Since

$$A_t = Dv_t(\Phi_{s,t}(x)), \quad \mathrm{tr}(Dv_t(y)) = \nabla \cdot v_t(y),$$

the Jacobian satisfies the scalar initial-value problem

$$\begin{cases} \partial_t J_{s,t}(x) = \nabla \cdot v_t(\Phi_{s,t}(x)) J_{s,t}(x), \\ J_{s,s}(x) = 1. \end{cases}$$

Solving this scalar linear equation yields

$$J_{s,t}(x) = \exp \left(\int_s^t \nabla \cdot v_r(\Phi_{s,r}(x)) \, \mathrm{d}r \right) > 0.$$

This establishes the Liouville–Jacobi formula. In particular, the determinant evolution is governed by an ordinary scalar exponential, even when the operators A_t at different times do not commute. □

Figure 1.2 summarizes the local volume interpretation of the determinant formula (1.9).

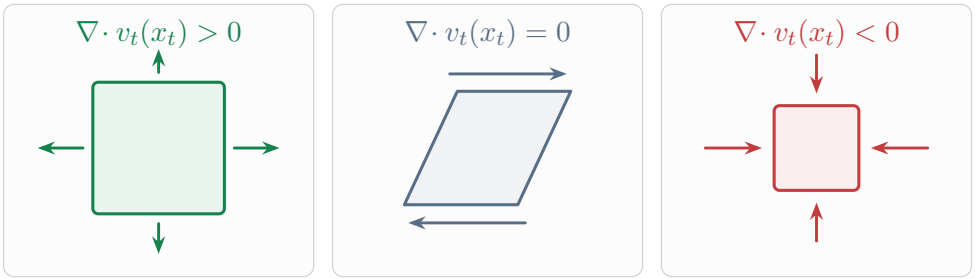


Figure 1.2: Divergence and instantaneous local volume change along $x_t = \Phi_{s,t}(x_s)$. From left to right, outward arrows illustrate expansion, shear illustrates zero instantaneous volume change, and inward arrows illustrate contraction. The Jacobian determinant gives the local volume ratio (area ratio in two dimensions); its value over a finite time interval depends on the integral of divergence along the trajectory.

Example 1.14. Consider the autonomous system

$$\begin{cases} \dot{x}_t = 0.2x_t, \\ \dot{y}_t = 0.1y_t + 0.3x_t^2, \end{cases} \qquad (x_{t=0}, y_{t=0}) = (x_0, y_0).$$

The second equation couples the y-motion nonlinearly to the x-coordinate. The velocity field is smooth but is not globally Lipschitz on $\mathbb{R}^2$ because of its quadratic term. The explicit solution below nevertheless exists for every real time and defines a smooth complete flow. Thus this example verifies the identity of Proposition 1.13 for this explicitly constructed flow, without asserting the global Lipschitz hypothesis used to guarantee existence in the preceding results.

The first equation gives $x_t = e^{0.2t}x_0$. Substitution into the second equation and multiplication by the integrating factor $e^{-0.1t}$ yield

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(e^{-0.1t}y_t\right) = 0.3x_0^2 e^{0.3t}, \quad e^{-0.1t}y_t - y_0 = x_0^2(e^{0.3t} - 1).$$

Consequently $y_t = e^{0.1t}y_0 + x_0^2(e^{0.4t} - e^{0.1t})$, and the flow map is

$$\Phi_{0,t}(x_0, y_0) = \begin{pmatrix} e^{0.2t}x_0 \\ e^{0.1t}y_0 + x_0^2(e^{0.4t} - e^{0.1t}) \end{pmatrix}.$$

Using the notation of Proposition 1.13 with $s = 0$, differentiation with respect to the initial state (x_0, y_0) gives

$$M_{0,t}(x_0, y_0) = D_{(x_0,y_0)}\Phi_{0,t}(x_0, y_0) = \begin{pmatrix} e^{0.2t} & 0 \\ 2x_0(e^{0.4t} - e^{0.1t}) & e^{0.1t} \end{pmatrix}.$$

Its determinant is therefore

$$J_{0,t}(x_0, y_0) = \det M_{0,t}(x_0, y_0) = e^{0.3t} > 0.$$

For the velocity field and its divergence, we have

$$v(x,y) = \begin{pmatrix} 0.2x \\ 0.1y + 0.3x^2 \end{pmatrix}, \quad \nabla \cdot v = \frac{\partial(0.2x)}{\partial x} + \frac{\partial(0.1y + 0.3x^2)}{\partial y} = 0.3.$$

The divergence integral therefore gives

$$J_{0,t}(x_0, y_0) = \exp\left(\int_0^t \nabla \cdot v(\Phi_{0,r}(x_0, y_0))\, \mathrm{d}r\right) = e^{0.3t},$$

in exact agreement with the direct determinant calculation.

Take the initial rectangle $R = [-1, 1] \times [-0.7, 0.7]$, whose area is $|R| = 2.8$. Since the flow is invertible and its Jacobian determinant is positive, the change-of-variables formula gives

$$|\Phi_{0,t}(R)| = \int_R J_{0,t}(x_0, y_0)\, \mathrm{d}x_0\, \mathrm{d}y_0 = 2.8e^{0.3t}.$$

In particular, at $t = 1.5$,

$$J_{0,1.5}(x_0, y_0) = e^{0.45} \approx 1.568312, \quad |\Phi_{0,1.5}(R)| \approx 4.391274.$$

The area has therefore increased by approximately 56.83%. Figure 1.3 visualizes this evolution from $t = 0$ to $t = 1.5$.

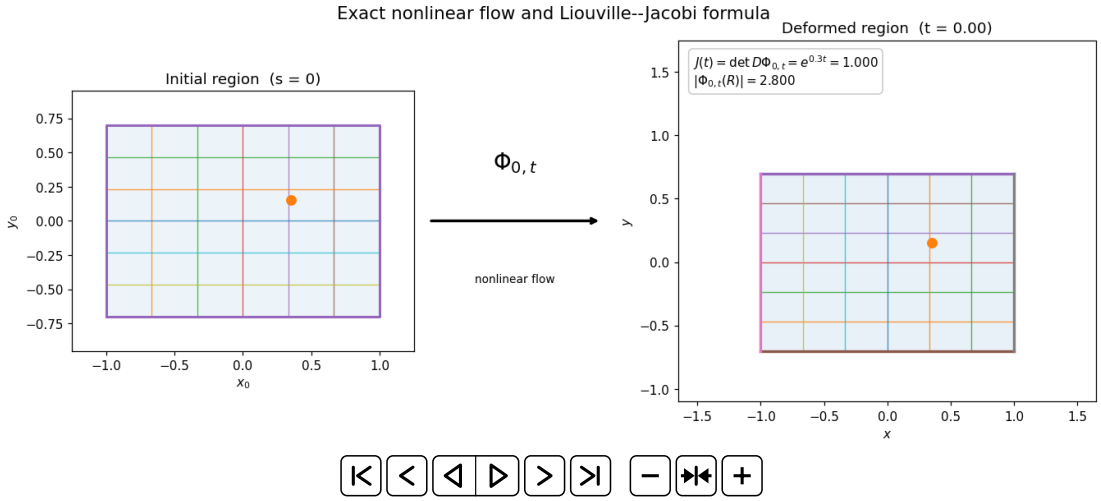


Figure 1.3: Nonlinear deformation of the initial rectangle under $\Phi_{0,t}$. The Jacobian determinant is $J_{0,t}(x_0, y_0) = e^{0.3t}$, in agreement with the Liouville–Jacobi formula. The first frame is the static preview.

The initially rectangular grid stretches and bends under the flow: the nonlinear term $0.3x^2$ changes its shape, while the divergence determines the instantaneous relative rate of local area change. Because $\nabla \cdot v = 0.3$ is independent of position, every initial point has the same local area multiplier $e^{0.3t}$, even though the deformation depends nonlinearly on position. Shape change and area change are distinct features of the flow.

Numerical flow Jacobians (Euler and higher-order solvers)

Source exercise: Project 9, item 4. *Student Assessment Exercise* [Assessment] (adapted).

Problem — Discrete Jacobian and convergence

Consider the differential equation[a]

$$\dot{x}_t = v_t(x_t).$$

Define

$$\widehat{x}_{k+1} = V_k(\widehat{x}_k), \quad \widehat{x}_0 = x_0, \quad V_k(x) = x + hv_{t_k}(x).$$

(a) **Single-step Jacobian.** Prove that

$$DV_k(x) = I_d + hDv_{t_k}(x)$$
$$\det DV_k(x) = 1 + h\nabla \cdot v_{t_k}(x) + O(h^2).$$

On this unit interval $h = 1/K$ and $t_k = kh$.

(b) **Discrete flow Jacobian.** Define $\widehat{\Phi}_{h,K} = V_{K-1} \circ \cdots \circ V_0$ and $\widehat{J}_{h,K}(x_0) = \det D\widehat{\Phi}_{h,K}(x_0)$. Show that

$$\widehat{J}_{h,K}(x_0) = \prod_{k=0}^{K-1} \det\big(I_d + hDv_{t_k}(\widehat{x}_k)\big).$$

(c) **Global convergence order.** With $J_1(x_0) := J_{0,1}(x_0) = \det D\Phi_{0,1}(x_0)$, use Proposition 1.13 to prove

$$E(h) = |\widehat{J}_{h,K}(x_0) - J_1(x_0)| = O(h).$$

Account separately for accumulated single-step errors, trajectory errors, and quadrature errors. Write $e_k := \widehat{x}_k - x_{t_k}$ for the state error.

(d) **Higher-order solvers.** What if you use a better ODE solver? Validate this convergence order numerically using an illustrative example.

[a] on time interval $[0, 1]$, with initial state $x_0 \in V = \mathbb{R}^d$. Assume v is C^2 on a neighbourhood of $[0, 1] \times B$, where B is a compact convex set containing the exact trajectory and all sufficiently fine Euler trajectories. All derivative bounds below are uniform on this set; part (d) requires the additional regularity appropriate to the solver.

Proof.

(a) Single-step Jacobian. Recall that the forward Euler map is

$$V_k(x) = x + hv_{t_k}(x).$$

To compute its derivative with respect to the state variable x, we first write the velocity field in component form:

$$v_{t_k}(x) = \begin{pmatrix} v_{t_k}^1(x) \\ v_{t_k}^2(x) \\ \vdots \\ v_{t_k}^d(x) \end{pmatrix}.$$

Consequently, the Euler map can be expressed componentwise as

$$V_k(x) = \begin{pmatrix} x^1 + h v_{t_k}^1(x) \\ x^2 + h v_{t_k}^2(x) \\ \vdots \\ x^d + h v_{t_k}^d(x) \end{pmatrix}.$$

The Jacobian matrix is obtained by differentiating each component of V_k with respect to each coordinate of x. Its (i, j)-th entry is

$$\frac{\partial V_k^i(x)}{\partial x^j} = \delta_{ij} + h \frac{\partial v_{t_k}^i(x)}{\partial x^j},$$

where δ_{ij} is the Kronecker delta, arising from $\partial x^i / \partial x^j = \delta_{ij}$. Therefore, the Jacobian matrix takes the form

$$DV_k(x) = \begin{pmatrix} 1 + h\frac{\partial v_{t_k}^1}{\partial x^1} & h\frac{\partial v_{t_k}^1}{\partial x^2} & \cdots & h\frac{\partial v_{t_k}^1}{\partial x^d} \\ h\frac{\partial v_{t_k}^2}{\partial x^1} & 1 + h\frac{\partial v_{t_k}^2}{\partial x^2} & \cdots & h\frac{\partial v_{t_k}^2}{\partial x^d} \\ \vdots & \vdots & \ddots & \vdots \\ h\frac{\partial v_{t_k}^d}{\partial x^1} & h\frac{\partial v_{t_k}^d}{\partial x^2} & \cdots & 1 + h\frac{\partial v_{t_k}^d}{\partial x^d} \end{pmatrix}.$$

Recognizing the identity matrix and the Jacobian of the velocity field, we obtain

$$DV_k(x) = I_d + h Dv_{t_k}(x).$$

To derive the determinant expansion, let $A = Dv_{t_k}(x) \in \mathrm{End}(V)$ have columns $a_1, \ldots, a_d$. By the multilinearity of the determinant,

$$\begin{aligned} \det(I_d + hA) &= \det(e_1 + h a_1, \ldots, e_d + h a_d) \\ &= 1 + h \sum_{j=1}^{d} \det(e_1, \ldots, a_j, \ldots, e_d) + R_A^{\det}(h) \\ &= 1 + h \,\mathrm{tr}(A) + R_A^{\det}(h), \end{aligned}$$

where $R_A^{\det}(h)$ collects all terms containing at least two columns from hA. Substituting $A = Dv_{t_k}(x)$ and using $\mathrm{tr}(Dv_{t_k}(x)) = \nabla \cdot v_{t_k}(x)$, we obtain

$$\det DV_k(x) = 1 + h \nabla \cdot v_{t_k}(x) + R_k^{\det}(h, x),$$

where $R_k^{\det}(h,x) := R^{\det}_{Dv_{t_k}(x)}(h)$. For the logarithmic expansion, define

$$u_k(h,x) := h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h,x).$$

Then

$$\det DV_k(x) = 1 + u_k(h,x).$$

For sufficiently small h, the determinant $\det DV_k(x)$ is positive uniformly for $x \in B$ and $k = 0, \dots, K-1$, as established in Appendix A.6. Recall the determinant expansion

$$\det DV_k(x) = 1 + h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h,x),$$

where $R_k^{\det}(h,x) = O(h^2)$ uniformly. Consequently,

$$u_k(h,x) := \det DV_k(x) - 1 = O(h)$$

uniformly. We may therefore apply the scalar Taylor expansion $\log(1+u) = u + O(u^2)$ for sufficiently small h. More precisely, define

$$r_{\log}(u) := \log(1+u) - u.$$

The combined logarithmic remainder is then

$$r_k^{\log}(h,x) := R_k^{\det}(h,x) + r_{\log}(u_k(h,x)).$$

It follows that

$$\begin{aligned}\log\det DV_k(x) &= \log(1 + u_k(h,x)) \\ &= h\nabla \cdot v_{t_k}(x) + r_k^{\log}(h,x).\end{aligned}$$

Since $u_k(h,x) = O(h)$ and $r_{\log}(u) = O(u^2)$, both contributions to the combined remainder are of order h^2. In particular, there exists a constant $C > 0$, independent of h, k, and x, such that

$$\sup_{x \in B} |r_k^{\log}(h,x)| \le Ch^2.$$

(b) Discrete flow Jacobian. Recall that the forward Euler scheme is defined by

$$\widehat{x}_{k+1} = V_k(\widehat{x}_k), \quad V_k(x) = x + hv_{t_k}(x).$$

Starting from the initial state x_0, successive applications of the single-step maps give

$$\widehat{x}_K = V_{K-1} \circ \cdots \circ V_1 \circ V_0(x_0).$$

$$x_0 = \widehat{x}_0 \xrightarrow{V_0} \widehat{x}_1 \xrightarrow{V_1} \cdots \xrightarrow{V_{K-2}} \widehat{x}_{K-1} \xrightarrow{V_{K-1}} \widehat{x}_K.$$

Thus, the discrete flow map is

$$\widehat{\Phi}_{h,K} = V_{K-1} \circ \cdots \circ V_1 \circ V_0.$$

To determine how the final state depends on the initial state, we differentiate this composition with respect to x_0.

By the chain rule, for the first two steps,

$$D(V_1 \circ V_0)(x_0) = DV_1(\widehat{x}_1)DV_0(x_0),$$

where $\widehat{x}_1 = V_0(x_0)$. Applying the chain rule successively yields

$$D\widehat{\Phi}_{h,K}(x_0) = DV_{K-1}(\widehat{x}_{K-1}) \cdots DV_1(\widehat{x}_1)DV_0(x_0).$$

The order of matrix multiplication is essential, since Jacobian matrices at different steps need not commute.

By definition, the Jacobian determinant of the discrete flow is

$$\widehat{J}_{h,K}(x_0) := \det D\widehat{\Phi}_{h,K}(x_0).$$

Using the multiplicative property of determinants, $\det(AB) = \det(A)\det(B)$, we obtain

$$\widehat{J}_{h,K}(x_0) = \prod_{k=0}^{K-1} \det DV_k(\widehat{x}_k).$$

From part (a), the Jacobian matrix of each single-step map is

$$DV_k(\widehat{x}_k) = I_d + hDv_{t_k}(\widehat{x}_k).$$

Substituting this expression gives

$$\widehat{J}_{h,K}(x_0) = \prod_{k=0}^{K-1} \det DV_k(\widehat{x}_k) = \prod_{k=0}^{K-1} \det\big(I_d + hDv_{t_k}(\widehat{x}_k)\big). \tag{1.11}$$

Hence, the Jacobian determinant of the entire discrete flow equals the product of the Jacobian determinants of its individual Euler steps.

(c) Global convergence order. We estimate the global error between the discrete and exact Jacobian determinants. Since both determinants are positive for sufficiently small h, define the logarithmic error

$$\delta(h) := \log \widehat{J}_{h,K}(x_0) - \log J_1(x_0).$$

Our goal is to prove that $|\delta(h)| = O(h)$ and then transfer this estimate to the Jacobian determinants themselves.

By the single-step expansion in part (a) and the product formula (1.11) for the discrete Jacobian,

$$\log \widehat{J}_{h,K}(x_0) = h \sum_{k=0}^{K-1} \nabla \cdot v_{t_k}(\widehat{x}_k) + R_{\log}(h), \quad |R_{\log}(h)| \le Ch,$$

whereas the continuous-time formula gives

$$\log J_1(x_0) = \int_0^1 \nabla \cdot v_t(x_t)\ \mathrm{d}t.$$

To compare these expressions, decompose the logarithmic error as

$$\delta(h) = R_{\log}(h) + E_{\text{traj}}(h) + E_{\text{quad}}(h),$$

where

$$E_{\text{traj}}(h) := h \sum_{k=0}^{K-1} \left[\nabla \cdot v_{t_k}(\widehat{x}_k) - \nabla \cdot v_{t_k}(x_{t_k})\right],$$

$$E_{\text{quad}}(h) := h \sum_{k=0}^{K-1} \nabla \cdot v_{t_k}(x_{t_k}) - \int_0^1 \nabla \cdot v_t(x_t)\ \mathrm{d}t.$$

Thus, it suffices to estimate these three contributions.

1. **Trajectory error.** By the uniform spatial Lipschitz bound L_x for $\nabla \cdot v_t$ and the standard Euler state estimate

   $$\max_{0 \le k \le K} \|\widehat{x}_k - x_{t_k}\| \le C_E h,$$

   we obtain

   $$|E_{\text{traj}}(h)| \le hKL_xC_Eh = L_xC_Eh.$$

2. **Quadrature error.** Let $g(t) := \nabla \cdot v_t(x_t)$. If L_t bounds $|\partial_t \nabla \cdot v_t(x)|$ and V_* bounds $\|v_t(x)\|$, then

   $$|g(t) - g(s)| \le L_t|t-s| + L_x\|x_t - x_s\| \le (L_t + L_xV_*)|t-s|.$$

   Writing $L_g := L_t + L_xV_*$, the Lipschitz estimate gives

   $$|E_{\text{quad}}(h)| \le \sum_{k=0}^{K-1} \int_{t_k}^{t_{k+1}} |g(t_k) - g(t)|\ \mathrm{d}t \le \frac{L_g}{2} Kh^2 = \frac{L_g}{2} h.$$

   Together with $|R_{\log}(h)| \le Ch$, these estimates yield

   $$|\delta(h)| \le \left(C + L_xC_E + \frac{L_g}{2}\right) h = C_*h.$$

By the continuous-time Jacobian formula and the uniform bound $|\nabla \cdot v_t(x)| \le B_{\text{div}}$,

$$0 < J_1(x_0) \le e^{B_{\text{div}}}.$$

Since $\widehat{J}_{h,K}(x_0) = J_1(x_0)e^{\delta(h)}$, the mean value estimate $|e^z - 1| \le e^{|z|}|z|$ gives

$$|\widehat{J}_{h,K}(x_0) - J_1(x_0)| \le J_1(x_0)e^{|\delta(h)|}|\delta(h)|$$

Therefore,

$$|\widehat{J}_{h,K}(x_0) - J_1(x_0)| = O(h).$$

Thus, the discrete Jacobian determinant converges to the exact continuous-time Jacobian determinant with first-order global accuracy.

(d) Higher-order solvers. Consider the augmented initial-value problem

$$\begin{cases} \dot{x}_t = v_t(x_t), & x_{t=0} = x_0, \\ \dot{M}_t = Dv_t(x_t)M_t, & M_0 = I_d. \end{cases}$$

Assume sufficient smoothness and stability for the chosen method to have order p on this augmented system, with bounded exact and numerical solutions. Its matrix approximation then satisfies

$$\|\widehat{M}_{h,K} - M_1\| = O(h^p), \quad M_1 := M_{0,1}(x_0) = D\Phi_{0,1}(x_0).$$

To identify $\widehat{M}_{h,K}$ with $D\widehat{\Phi}_{h,K}(x_0)$, the variational update must be obtained by differentiating the same numerical state update. For Runge–Kutta methods, differentiating every stage produces exactly the corresponding Runge–Kutta method on the augmented system. The determinant is a polynomial in the matrix entries and is Lipschitz on bounded matrix sets, so

$$|\det \widehat{M}_{h,K} - \det M_1| \le C\|\widehat{M}_{h,K} - M_1\| = O(h^p).$$

Thus classical RK4 gives fourth-order Jacobian and determinant convergence when its stage derivatives are computed consistently and the required regularity holds. State accuracy alone does not establish derivative accuracy; combining an RK4 state step with the Euler factor $I_d + hDv_{t_k}(\widehat{x}_k)$ does not generally give a fourth-order Jacobian. □

Problem — Numerical validation

Consider the scalar initial-value problem

$$\dot{x}_t = x_t, \quad x_0 = 1, \quad t \in [0, 1],$$

whose exact flow is $\Phi_{0,t}(x) = e^t x$ and whose Jacobian determinant at $t = 1$ is $J_1 = e$. Numerically validate the predicted convergence orders of the Jacobian determinants of the numerical flow maps by performing the following experiments:

1. Construct the numerical flow maps using forward Euler and classical RK4, and compute their Jacobian determinants $\widehat{J}_{h,K} = \det D\widehat{\Phi}_{h,K}(x_0)$.
2. Choose $K = 4, 8, 16, 32, 64, 128$ and set $h = 1/K$. Compute the absolute errors
$$E(h) = |\widehat{J}_{h,K} - e|$$
for both methods.
3. Plot the errors against h on a log–log scale.
4. Estimate the observed convergence orders using consecutive mesh refinements and compare them with the theoretical predictions.

Solution.

For the scalar field $v(x) = x$, differentiation of the Euler and RK4 updates gives

$$\widehat{J}^{\text{Euler}}_{h,K} = (1+h)^K, \quad \widehat{J}^{\text{RK4}}_{h,K} = \left(1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}\right)^K.$$

Here $Kh = 1$ and the exact Jacobian is e. The leading errors are $(e/2)h$ and $(e/120)h^4$, respectively. The existing high-precision validation is retained below.

1. Construction of the numerical flow maps. We first construct the numerical flow maps and differentiate them with respect to the initial state. For the given equation $\dot{x}_t = x_t$, the exact flow is

$$\Phi_{0,t}(x) = e^t x,$$

and hence

$$J_1 = \det D\Phi_{0,1}(x_0) = e,$$

Let $t_k = kh$, where $h = 1/K$, and denote the numerical state at time t_k by X_k, starting from an arbitrary initial state $X_0 = x$.

- For the forward Euler method, the numerical update is
$$X_{k+1} = X_k + hX_k = (1+h)X_k.$$
Iterating over K time steps gives
$$\widehat{\Phi}^{\text{Euler}}_{h,K}(x) = (1+h)^K x.$$
Differentiating with respect to the initial state yields
$$\widehat{J}^{\text{Euler}}_{h,K} = \det D\widehat{\Phi}^{\text{Euler}}_{h,K}(x_0) = (1+h)^K.$$
- For the classical fourth-order Runge–Kutta method (RK4), the four intermediate

stages are

$$k_1 = X_k, \quad k_2 = X_k + \frac{h}{2}k_1, \quad k_3 = X_k + \frac{h}{2}k_2, \quad k_4 = X_k + hk_3.$$

Substituting the preceding stages successively gives

$$k_1 = X_k, \quad k_2 = \left(1 + \frac{h}{2}\right) X_k,$$
$$k_3 = \left(1 + \frac{h}{2} + \frac{h^2}{4}\right) X_k, \quad k_4 = \left(1 + h + \frac{h^2}{2} + \frac{h^3}{4}\right) X_k.$$

The RK4 update is

$$X_{k+1} = X_k + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

Substituting the four stages and collecting powers of h, we obtain

$$X_{k+1} = \left(1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}\right) X_k.$$

Define the one-step amplification factor

$$R_4(h) = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24}.$$

After K steps, the numerical flow map is

$$\widehat{\Phi}^{\text{RK4}}_{h,K}(x) = R_4(h)^K x.$$

Differentiating with respect to the initial state gives

$$\widehat{J}^{\text{RK4}}_{h,K} = \det D\widehat{\Phi}^{\text{RK4}}_{h,K}(x_0) = R_4(h)^K.$$

Thus, both discrete Jacobian determinants are obtained directly from the derivatives of their respective numerical flow maps.

2. Asymptotic error analysis. We now determine the leading-order errors of the two numerical Jacobians. Since $Kh = 1$, we have $K = h^{-1}$. For $h > 0$, the inequalities

$$1 + h < e^h, \quad R_4(h) < e^h$$

follow from the exponential series. Raising both inequalities to the power K gives

$$0 < \widehat{J}^{\text{Euler}}_{h,K} < e, \quad 0 < \widehat{J}^{\text{RK4}}_{h,K} < e.$$

Therefore, the absolute errors can be written as

$$E(h) = e - \widehat{J}_{h,K}$$

for both methods.

1. *Euler method.* Taking the logarithm of the discrete Jacobian determinant gives

$$\log \widehat{J}^{\text{Euler}}_{h,K} = \log(1+h)^K = \frac{1}{h}\log(1+h).$$

Using the Taylor expansion

$$\log(1+h) = h - \frac{h^2}{2} + \frac{h^3}{3} + O(h^4),$$

we obtain

$$\log \widehat{J}^{\text{Euler}}_{h,K} = 1 - \frac{h}{2} + \frac{h^2}{3} + O(h^3).$$

Exponentiating both sides and using $\exp(z) = 1 + z + O(z^2)$ as $z \to 0$, we find

$$\widehat{J}^{\text{Euler}}_{h,K} = e \exp\left(-\frac{h}{2} + \frac{h^2}{3} + O(h^3)\right) = e\left(1 - \frac{h}{2} + O(h^2)\right).$$

Subtracting from the exact Jacobian determinant yields

$$E_{\text{Euler}}(h) = \frac{e}{2}h + O(h^2).$$

Thus, the Euler approximation exhibits first-order convergence for the Jacobian determinant.

2. *Classical RK4 method.* Similarly, taking the logarithm gives

$$\log \widehat{J}^{\text{RK4}}_{h,K} = \frac{1}{h}\log R_4(h).$$

To determine its leading-order error, we compare the RK4 amplification factor with the Taylor expansion of the exact exponential:

$$\begin{aligned} e^h &= 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \frac{h^4}{24} + \frac{h^5}{120} + O(h^6) \\ &= R_4(h) + \frac{h^5}{120} + O(h^6). \end{aligned}$$

Consequently,

$$R_4(h) = e^h - \frac{h^5}{120} + O(h^6).$$

Factoring out e^h gives

$$R_4(h) = e^h\left(1 - \frac{h^5}{120} + O(h^6)\right).$$

Taking logarithms and using $\log(1+z) = z + O(z^2)$ as $z \to 0$, we obtain

$$\log R_4(h) = h + \log\left(1 - \frac{h^5}{120} + O(h^6)\right) = h - \frac{h^5}{120} + O(h^6).$$

Dividing by h yields

$$\log \widehat{J}^{\mathrm{RK4}}_{h,K} = 1 - \frac{h^4}{120} + O(h^5).$$

Exponentiating,

$$\widehat{J}^{\mathrm{RK4}}_{h,K} = e \exp\left(-\frac{h^4}{120} + O(h^5)\right) = e\left(1 - \frac{h^4}{120} + O(h^5)\right).$$

Subtracting from the exact Jacobian determinant gives

$$E_{\mathrm{RK4}}(h) = \frac{e}{120} h^4 + O(h^5).$$

Thus, the RK4 approximation exhibits fourth-order convergence for the Jacobian determinant.

The two asymptotic expansions predict that the limiting slopes of the corresponding log–log error curves are one for Euler and four for RK4.

3. Numerical validation. We now verify these predictions numerically. Choose

$$K = 4, 8, 16, 32, 64, 128, \quad h = \frac{1}{K}.$$

For each mesh, evaluate the numerical flow maps and their Jacobians derived above, and compute the absolute errors

$$E_{\mathrm{Euler}}(h) = \left|\widehat{J}^{\mathrm{Euler}}_{h,K} - e\right|, \quad E_{\mathrm{RK4}}(h) = \left|\widehat{J}^{\mathrm{RK4}}_{h,K} - e\right|.$$

To estimate the convergence order from consecutive meshes, suppose that the error has the asymptotic form

$$E(h) = Ch^p + O(h^{p+1}),$$

where $C > 0$ and p is the convergence order. Replacing h by $h/2$ gives

$$E(h/2) = C\left(\frac{h}{2}\right)^p + O(h^{p+1}).$$

Taking the ratio of the two expressions, we obtain

$$\frac{E(h)}{E(h/2)} = 2^p\big(1 + O(h)\big).$$

Taking logarithms and dividing by $\log 2$ motivates the observed-order estimate

$$p_{\text{obs}}(h) = \frac{\log(E(h)/E(h/2))}{\log 2}. \tag{1.12}$$

As $h \to 0$, this estimate approaches the theoretical order p.

We evaluate both numerical flow Jacobians and the reference value e using 80-digit decimal arithmetic. This precision is particularly important for RK4, whose errors become very small and would otherwise be affected by cancellation when subtracting nearly equal quantities. Table 1.1 reports the resulting errors and observed orders. Each reported order compares its row with the next finer mesh; no order is reported for the final row. Figure 1.4 plots both error curves against h on logarithmic axes. For comparison, we also include the leading-order error predictions derived above:

$$E^{\text{lead}}_{\text{Euler}}(h) = \frac{e}{2}h, \quad E^{\text{lead}}_{\text{RK4}}(h) = \frac{e}{120}h^4.$$

Table 1.1: Scalar flow-Jacobian errors and observed orders from (1.12). Errors are rounded from the 80-digit calculation.

K	h	E_{Euler}	E_{RK4}	p_{Euler}	p_{RK4}
4	0.25	2.768756×10^{-1}	7.188926×10^{-5}	0.8605	3.8504
8	0.125	1.524973×10^{-1}	4.984042×10^{-6}	0.9244	3.9250
16	0.0625	8.035333×10^{-2}	3.281185×10^{-7}	0.9605	3.9625
32	0.03125	4.129170×10^{-2}	2.104785×10^{-8}	0.9798	3.9812
64	0.015625	2.093688×10^{-2}	1.332722×10^{-9}	0.9898	3.9906
128	0.0078125	1.054281×10^{-2}	8.383902×10^{-11}	–	–

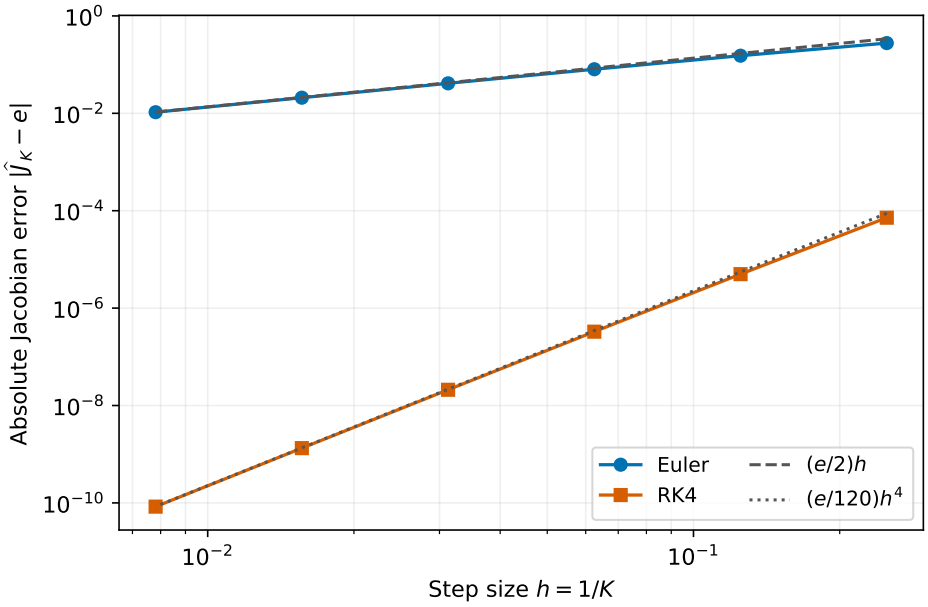


Figure 1.4: Scalar flow-Jacobian errors for Euler and RK4 on six successively refined meshes. The dashed and dotted lines show the leading terms $(e/2)h$ and $(e/120)h^4$, respectively.

The observed orders approach one for Euler and four for RK4, in agreement with the theoretical predictions. Moreover, the numerical error curves approach their respective leading-order reference curves as the time step decreases.

The reproducible calculation is `python experiments/scalar_jacobian.py`; the full-precision values are saved in `results/benchmarks/scalar_jacobian_convergence.csv`. Only the plotting coordinates are converted to ordinary floating-point numbers. ♦

1.2.1.4 Volume transport and minimum action

Let $\Omega_0 \subset \mathbb{R}^d$ be a bounded Lebesgue-measurable region, and let Φ_t denote the flow map generated by the ODE (1.4), with $\Phi_0 = \mathrm{id}$. The deformed domain at time t is defined by

$$\Omega_t := \Phi_t(\Omega_0) = \{\Phi_t(x_0) : x_0 \in \Omega_0\}.$$

We define its volume by

$$A_t := \mathrm{vol}(\Omega_t) := \lambda^d(\Omega_t) = \int_{\mathbb{R}^d} \mathbf{1}_{\Omega_t}(x)\,\mathrm{d}x,$$

where λ^d denotes the d-dimensional Lebesgue measure and $\mathbf{1}_{\Omega_t}$ is the indicator function of Ω_t. Assume that Φ_t is a smooth diffeomorphism. By the change-of-variables formula,

$$A_t = \int_{\Omega_0} |\det D\Phi_t(x_0)|\ \mathrm{d}x_0.$$

For a flow generated by an ODE with $\Phi_0 = \mathrm{id}$, the Jacobian determinant remains positive, so the absolute value can be omitted. Moreover, differentiating the volume with respect to time gives (1.13), whose measurable-region formulation is (1.14):

$$\dot{A}_t = \int_{\Omega_t} \nabla \cdot v_t(x)\, \mathrm{d}x. \tag{1.13}$$

Corollary 1.15 (Volume transport and divergence)**.**

Source exercise: Project 9, item 5. *Student Assessment Exercise* [Assessment] (adapted).

Let $\mathcal{D}_0 \subset \mathbb{R}^d$ be a bounded Lebesgue-measurable region, and let $\Phi_{0,t}$ denote the flow map generated by (1.4)*, with $\Phi_{0,0} = \mathrm{id}$. Define the deformed domain by*

$$\mathcal{D}_t := \Phi_{0,t}(\mathcal{D}_0) = \{\Phi_{0,t}(x_0) : x_0 \in \mathcal{D}_0\}.$$

Its volume is defined as the d-dimensional Lebesgue measure

$$|\mathcal{D}_t| := \lambda^d(\mathcal{D}_t) = \int_{\mathbb{R}^d} \mathbf{1}_{\mathcal{D}_t}(y)\, \mathrm{d}y,$$

where λ^d denotes the d-dimensional Lebesgue measure and $\mathbf{1}_{\mathcal{D}_t}$ is the indicator function of $\mathcal{D}_t$. Assume that $\Phi_{0,t}$ is a smooth diffeomorphism and that differentiation under the integral is legitimate. Then the volume evolves according to

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_t} \nabla \cdot v_t(y)\, \mathrm{d}y. \tag{1.14}$$

Proof. Let

$$M_{0,t}(x_0) := D\Phi_{0,t}(x_0), \quad J_{0,t}(x_0) := \det M_{0,t}(x_0).$$

By the change-of-variables formula,

$$|\mathcal{D}_t| = \int_{\mathcal{D}_0} J_{0,t}(x_0)\, \mathrm{d}x_0,$$

where $J_{0,t}(x_0) > 0$ because $\Phi_{0,t}$ is a flow starting from the identity map.

Differentiating under the integral gives

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_0} \frac{\partial}{\partial t} J_{0,t}(x_0)\, \mathrm{d}x_0.$$

The variational equation for the flow yields

$$\frac{\partial}{\partial t} M_{0,t}(x_0) = Dv_t(\Phi_{0,t}(x_0)) M_{0,t}(x_0).$$

Applying the Jacobi formula for the derivative of a determinant, we obtain

$$\begin{aligned}\frac{\partial}{\partial t}J_{0,t}(x_0) &= J_{0,t}(x_0)\,\mathrm{tr}\left(M_{0,t}(x_0)^{-1}\frac{\partial M_{0,t}(x_0)}{\partial t}\right)\\ &= J_{0,t}(x_0)\,\mathrm{tr}\left(Dv_t(\Phi_{0,t}(x_0))\right)\\ &= J_{0,t}(x_0)\nabla\cdot v_t(\Phi_{0,t}(x_0)).\end{aligned}$$

Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_0} J_{0,t}(x_0)\nabla\cdot v_t(\Phi_{0,t}(x_0))\,\mathrm{d}x_0.$$

Finally, applying the change-of-variables formula $y = \Phi_{0,t}(x_0)$, we arrive at

$$\frac{\mathrm{d}}{\mathrm{d}t}|\mathcal{D}_t| = \int_{\mathcal{D}_t} \nabla\cdot v_t(y)\,\mathrm{d}y,$$

which completes the proof. □

Example 1.16 (Minimum weighted kinetic action between two states)**.**

Source exercise: Project 9, item 6. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $w : [0,1] \to (0,\infty)$ be continuous, and fix two points $y, z \in \mathbb{R}^d$. Consider a time-dependent velocity field $v_t : \mathbb{R}^d \to \mathbb{R}^d$ generating a flow map $\Phi_{0,t}$ through

$$\frac{\partial}{\partial t}\Phi_{0,t}(x) = v_t(\Phi_{0,t}(x)), \quad \Phi_{0,0}(x) = x.$$

Find an optimal velocity field v_t^* that minimizes

$$I[v] = \int_0^1 w_t\|v_t(\Phi_{0,t}(y))\|_2^2\,\mathrm{d}t,$$

subject to the endpoint constraint $\Phi_{0,1}(y) = z$. Determine the corresponding optimal flow map $\Phi_{0,t}^*(x)$ for $t \in [0,1]$.

To be completed.

1.2.2 Observable evolution and probability transport

In statistical physics, the dynamics of a physical system governs not only the evolution of individual trajectories but also the evolution of probability distributions over the state space, describing the statistical ensemble of its possible states. To formulate such ensembles mathematically, we first introduce a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let X_t be a measurable random variable taking values in the state space $(E, \mathcal{S})$, where $\mathcal{S}$ is the Borel σ-algebra. Thus,

$$X_t : \Omega \to E, \quad \omega \mapsto X_t(\omega).$$

For each sample point $\omega \in \Omega$, the value $X_t(\omega)$ specifies the state of the system at time t corresponding to that realization.

Trajectories space and Probability space In the classical dynamical setting, we may identify the sample space with a trajectory space, whose elements are entire state trajectories. More precisely, for a flow generated by $\dot{x}_t = v_t(x_t)$, define the trajectories space:

$$\mathcal{X} := \{\gamma \in C([0,T]; E) \mid \gamma(t) = \Phi_{0,t}(\gamma(0)), \quad t \in [0,T]\}.$$

Each $\gamma \in \mathcal{X}$ represents a possible trajectory of the system, and the random variable X_t is the corresponding evaluation map:

$$X_t : \mathcal{X} \to E, \quad \gamma \mapsto \gamma(t) = x_t.$$

We equip Ω with the σ-algebra generated by these evaluation maps:

$$\mathcal{F} := \sigma\left(\{X_t^{-1}(A) \;\middle|\; t \in [0,T],\ A \in \mathcal{S}\}\right).$$

Thus, the probability measure $\mathbb{P}$ assigns probabilities to different trajectories, while X_t extracts the state of a trajectory at a specified time,

$$\mathbb{P} : \mathcal{F} \to [0,1], \quad A \mapsto \mathbb{P}(A), \quad A \in \mathcal{F}.$$

Pushforward measures $\mathbb{P}$ is the trajectory-space probability measure. The law at time s is the pushforward $\mu_s = (X_s)_{\#}\mathbb{P}$, so that for every $A \in \mathcal{S}$,

$$\mu_s(A) = \mathbb{P}\big(X_s^{-1}(A)\big) = \mathbb{P}\big(\{\gamma \in \Omega : \gamma(s) \in A\}\big).$$

This operation merely transfers a measure from trajectory space to state space; it does not by itself describe temporal evolution. Two distinct operations are involved.

- **Evaluation map.** The map X_t extracts the state at time t and pushes the trajectory law forward to a law on the state space,

$$\mu_t = (X_t)_{\#}\mathbb{P} \in \mathcal{P}(E, \mathcal{S}).$$

- **Dynamical flow.** The deterministic flow $\Phi_{s,t} : E \to E$, which moves states along $x_t = \Phi_{s,t}(x_s)$, transports a measure on the state space from time s to time t,

$$(\Phi_{s,t})_{\#} : \mathcal{P}(E, \mathcal{S}) \to \mathcal{P}(E, \mathcal{S}), \quad \mu \mapsto (\Phi_{s,t})_{\#}\mu.$$

The two operations are compatible: because every trajectory follows the flow, the evaluation maps satisfy, for $0 \le s \le t \le T$,

$$X_t = \Phi_{s,t} \circ X_s,$$

i.e. the following diagram commutes:

$$\begin{array}{ccc} \mathcal{X} & \xrightarrow{X_s} & E \\ & \searrow^{X_t} & \downarrow{\scriptstyle \Phi_{s,t}} \\ & & E \end{array}$$

Consequently both routes give the same law:

$$\mu_t = (X_t)_{\#}\mathbb{P} = (\Phi_{s,t} \circ X_s)_{\#}\mathbb{P} = (\Phi_{s,t})_{\#}(X_s)_{\#}\mathbb{P} = (\Phi_{s,t})_{\#}\mu_s,$$

or, equivalently, for every $A \in \mathcal{S}$,

$$\mu_t(A) = \mu_s\big(\Phi_{s,t}^{-1}(A)\big).$$

The relationship between these objects and their evolution is summarized by

$$\underbrace{(\mathcal{X}, \mathcal{F}, \mathbb{P})}_{\text{trajectory space}} \xrightarrow{X_s} \underbrace{(E, \mathcal{S}, \mu_s)}_{\text{state space, law at time } s} \xrightarrow{\Phi_{s,t}} \underbrace{(E, \mathcal{S}, \mu_t)}_{\text{state space, law at time } t}.$$

This description corresponds to the classical Schrödinger picture, in which probability measures evolve under the dynamics while observables are kept fixed. Rather than following individual trajectories, we study how their statistical distribution is transported through the state space.

When the probability measures are absolutely continuous with respect to Lebesgue measure on $\mathbb{R}^d$, we may write

$$\mu_t(\,\mathrm{d}x) = \rho_t(x)\,\mathrm{d}x,$$

where ρ_t denotes the probability density at time t. Our objective is then to characterize the evolution of ρ_t in terms of the velocity field v_t. For a sufficiently regular invertible flow, the density can be expressed using the flow map and its Jacobian. Alternatively, by studying the action of μ_t on test functions, we can derive a differential equation for probability transport directly from v_t, without explicitly computing the flow map.

1.2.2.1 Generators and observable evolution

The observable–measure duality has a dynamical consequence that is central to statistical physics and Markov processes. Given a sufficiently smooth observable f, define the Koopman operator

$$(\mathcal{K}_{s,t}f)(x) = f(\Phi_{s,t}(x)).$$

It describes the evolution of observables induced by the state flow. Differentiating with respect to the terminal time and applying the chain rule gives

$$\partial_t(\mathcal{K}_{s,t}f) = \mathcal{K}_{s,t}\mathcal{L}_t f, \quad \mathcal{L}_t f = v_t \cdot \nabla f.$$

Here $\mathcal{L}_t$ is the infinitesimal generator induced by the velocity field. Differentiating instead with respect to the initial time, using the flow composition identity, yields the backward equation

$$\partial_s \mathcal{K}_{s,t} f = -\mathcal{L}_s(\mathcal{K}_{s,t} f).$$

The different operator orders reflect the distinction between terminal-time and initial-time differentiation.

The generator is a linear differential operator acting on a suitable domain of smooth observables. It is neither the velocity field itself nor the finite-time flow, but describes their induced infinitesimal action. More precisely, for sufficiently regular f,

$$\mathcal{L}_t f = \lim_{h \downarrow 0} \frac{\mathcal{K}_{t,t+h} f - f}{h}.$$

The existence of this limit depends on the chosen function space and topology; throughout, the differential expression is understood on the appropriate smooth test functions.

The same operator framework extends to stochastic dynamics. Let $(E, \mathcal{S})$ be a measurable state space, which need not have a vector space structure. A Markov transition kernel $\mathsf{K}_{s,t}(x, \mathrm{d}y)$ assigns a probability measure to each initial state and satisfies the Chapman–Kolmogorov composition rule. It induces an evolution operator on bounded measurable observables:

$$(\mathsf{P}_{s,t} f)(x) = \int_E f(y) \mathsf{K}_{s,t}(x, \mathrm{d}y).$$

The corresponding operators satisfy $\mathsf{P}_{s,r}\mathsf{P}_{r,t} = \mathsf{P}_{s,t},\ s \le r \le t$. Meanwhile, the probability law evolves according to

$$\mu_t(B) = \int_E \mathsf{K}_{s,t}(x, B) \mu_s(\mathrm{d}x).$$

Thus, observables and probability measures evolve in dual ways, related by the integral pairing

$$\langle f, \mu_t \rangle = \langle \mathsf{P}_{s,t} f, \mu_s \rangle.$$

A deterministic flow is recovered by taking

$$\mathsf{K}_{s,t}(x, \mathrm{d}y) = \delta_{\Phi_{s,t}(x)}(\mathrm{d}y),$$

for which $\mathsf{P}_{s,t} = \mathcal{K}_{s,t}$.

When the corresponding infinitesimal limit exists, the Markov evolution has a generator

$$\mathcal{L}_t f = \lim_{h \downarrow 0} \frac{\mathsf{P}_{t,t+h} f - f}{h}.$$

For suitable test functions and integrability conditions, the dual evolution then gives the weak forward equation

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle f, \mu_t \rangle = \langle \mathcal{L}_t f, \mu_t \rangle.$$

When a probability density exists, this weak equation can be expressed through the adjoint generator $\mathcal{L}_t^*$ as $\partial_t \rho_t = \mathcal{L}_t^* \rho_t$.

The form of the generator depends on the underlying dynamical mechanism. For example, consider a diffusion process

$$\mathrm{d}X_t = b_t(X_t)\mathrm{d}t + \sigma_t(X_t)\mathrm{d}W_t.$$

Its differential generator, obtained from Itô's formula, is

$$\mathcal{L}_t f = b_t \cdot \nabla f + \frac{1}{2}\,\mathrm{tr}\left(\sigma_t \sigma_t^\top \nabla^2 f\right).$$

For a finite-state jump process with transition rates $q_t(x, y)$, the generator instead takes the form

$$\mathcal{L}_t f(x) = \sum_{y \neq x} q_t(x, y)[f(y) - f(x)].$$

These examples illustrate how different microscopic mechanisms lead to different generator expressions while retaining the same underlying dual structure. The corresponding adjoint operators and probability evolution equations will be derived when the respective dynamics are introduced.

1.2.2.2 Probability transport and observable evolution

Density transport and the classical continuity equation. We first describe probability conservation through the geometry of the deterministic flow, taking $E = \mathbb{R}^d$. Recall the ODE $\dot{x}_t = v_t(x_t)$, that state dynamics

$$x_t = \Phi_{0,t}(x_0), \quad 0 \leq t \leq T.$$

Differentiability with respect to initial conditions requires additional spatial regularity; here we assume that v_t is C^1 in space with jointly continuous spatial derivatives. For the change of variables below, we further assume that each $\Phi_{0,t}$ is a C^1 diffeomorphism of $\mathbb{R}^d$ onto itself. This global invertibility requires the corresponding backward trajectories to exist and does not follow from forward existence and uniqueness alone.

Let $X_0 \sim \rho_0(x)\ \mathrm{d}x$,

$$X_t = \Phi_{0,t}(X_0), \quad \mu_t = (\Phi_{0,t})_{\#}\mu_0.$$

The pushforward relation established above conserves the probability assigned to a region transported by the flow. The following result expresses this conservation in terms of the local volume deformation.

Proposition 1.17 (Density along trajectories)**.**

Source exercise: Project 9, item 7–8. *Student Assessment Exercise* [Assessment] (adapted).

Under the preceding diffeomorphism assumptions, the transported law has a Lebesgue density ρ_t. Set

$$M_{0,t}(x_0) := D\Phi_{0,t}(x_0), \quad J_{0,t}(x_0) := \det M_{0,t}(x_0).$$

Then $J_{0,t}(x_0) > 0$, and a representative of ρ_t can be chosen so that, for almost every x_0,

$$\rho_t(\Phi_{0,t}(x_0)) = \rho_0(x_0)\big|J_{0,t}(x_0)\big|^{-1} = \rho_0(x_0)\exp\left(-\int_0^t \nabla\cdot v_s(\Phi_{0,s}(x_0))\,\mathrm{d}s\right).$$

If, in addition, this representative $\rho(t,x) = \rho_t(x)$ is C^1 in (t,x), then it satisfies the continuity equation

$$\partial_t\rho_t + \nabla\cdot(\rho_t v_t) = 0 \tag{1.15}$$

pointwise on $(0,T)\times\mathbb{R}^d$.

Proof. Let $A \in \mathcal{B}(\mathbb{R}^d)$ be a Borel set representing a region of the state space at time t. Its preimage under the flow map $\Phi_{0,t}$ is defined by

$$\Phi_{0,t}^{-1}(A) := \left\{\, x_0 \in \mathbb{R}^d : \Phi_{0,t}(x_0) \in A \,\right\},$$

i.e., the set of initial states at time $t = 0$ that are mapped into A at time t (Fig 1.5).

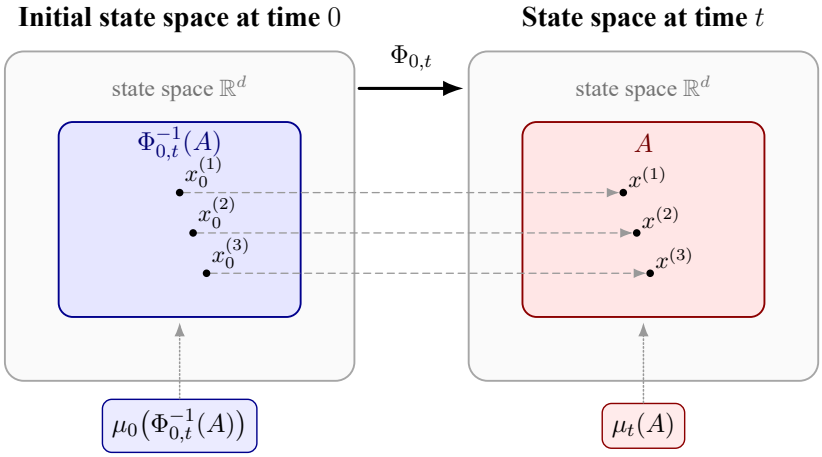


Figure 1.5: Measure transport under the flow $\Phi_{0,t}$: the mass of A at time t equals the mass of its preimage $\Phi_{0,t}^{-1}(A)$ at time 0.

Since $\Phi_{0,t}$ is continuous and A is Borel measurable, the preimage $\Phi_{0,t}^{-1}(A)$ is also a Borel set. The pushforward relation then reads

$$\mu_t(A) = \big((\Phi_{0,t})_{\#}\mu_0\big)(A) = \mu_0\big(\Phi_{0,t}^{-1}(A)\big), \quad A \in \mathcal{B}(\mathbb{R}^d).$$

Since μ_0 admits a Lebesgue density and $\Phi_{0,t}$ is a C^1 diffeomorphism, the change-of-variables formula implies that μ_t also admits a Lebesgue density.

$$\mu_t(\mathrm{d}x) = \rho_t(x)\,\mathrm{d}x, \quad \mu_0(\mathrm{d}x_0) = \rho_0(x_0)\,\mathrm{d}x_0,$$

we obtain

$$\mu_t(A) = \int_A \rho_t(x)\,\mathrm{d}x = \int_{\Phi_{0,t}^{-1}(A)} \rho_0(x_0)\,\mathrm{d}x_0.$$

Since we know the initial probability density ρ_0 and aim to determine the probability density ρ_t at time t, we need to express the integral in terms of the evolved state variable. Because $\Phi_{0,t}$ is a C^1 diffeomorphism, its inverse is also continuously differentiable. We can therefore rewrite the integral over the initial region $\Phi_{0,t}^{-1}(A)$ as an integral over the evolved region A by applying the change-of-variables formula with $x_0 = \Phi_{0,t}^{-1}(x)$:

$$\mathrm{d}x_0 = \left|\det D\Phi_{0,t}^{-1}(x)\right|\,\mathrm{d}x.$$

Therefore, we obtain

$$\mu_t(A) = \int_{\Phi_{0,t}^{-1}(A)} \rho_0(x_0)\,\mathrm{d}x_0 = \int_A \rho_0(\Phi_{0,t}^{-1}(x))\left|\det D\Phi_{0,t}^{-1}(x)\right|\,\mathrm{d}x.$$

The density at time t is

$$\rho_t(x) = \rho_0(\Phi_{0,t}^{-1}(x))\left|\det D\Phi_{0,t}^{-1}(x)\right|$$

for almost every $x \in \mathbb{R}^d$. Equivalently, using $x = \Phi_{0,t}(x_0)$ and the inverse Jacobian relation, we obtain

$$\rho_t(\Phi_{0,t}(x_0))\,|\det D\Phi_{0,t}(x_0)| = \rho_0(x_0)$$

for almost every $x_0 \in \mathbb{R}^d$.

By the Liouville–Jacobi formula (1.9), we have

$$J_{0,t}(x_0) = \exp\left(\int_0^t \nabla \cdot v_s(x_s)\,\mathrm{d}s\right).$$

Consequently, the density along trajectories satisfies

$$\rho_t(x_t) = \rho_0(x_0)\exp\left(-\int_0^t \nabla \cdot v_s(x_s)\,\mathrm{d}s\right).$$

This formula shows that spatial differentiability of ρ_0 is not required to establish differentiability along trajectories. Indeed, for almost every initial state x_0, the value $\rho_0(x_0)$ is finite and independent of time, while the time dependence of $\rho_t(x_t)$ is entirely determined by the exponential factor. Since the velocity field has jointly continuous spatial derivatives, this factor is continuously differentiable in time. Thus, for the representative of ρ_t defined by the

transport formula, $t \mapsto \rho_t(\Phi_{0,t}(x_0))$ is continuously differentiable for almost every x_0, even when ρ_0 is merely integrable.

If $\rho_0 \in L^1(\mathbb{R}^d)$ and the velocity field has jointly continuous spatial derivatives, then, provided the flow is a C^1 diffeomorphism, the transport formula defines a representative of ρ_t whose restriction to almost every trajectory is continuously differentiable in time. Indeed, along the trajectory $x_t = \Phi_{0,t}(x_0)$, the density transport formula gives

$$\widetilde{\rho}_t(x_t) J_{0,t}(x_0) = \rho_0(x_0).$$

For almost every initial state x_0, the right-hand side is finite and independent of time. Differentiating this conserved product and using the Liouville–Jacobi formula

$$\frac{\mathrm{d}}{\mathrm{d}t} J_{0,t}(x_0) = J_{0,t}(x_0) \nabla \cdot v_t(x_t),$$

we obtain

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}t} \left[\widetilde{\rho}_t(x_t) J_{0,t}(x_0) \right] \\ &= J_{0,t}(x_0) \frac{\mathrm{d}}{\mathrm{d}t} \widetilde{\rho}_t(x_t) + \widetilde{\rho}_t(x_t) \frac{\mathrm{d}}{\mathrm{d}t} J_{0,t}(x_0) \\ &= J_{0,t}(x_0) \left[\frac{\mathrm{d}}{\mathrm{d}t} \widetilde{\rho}_t(x_t) + \widetilde{\rho}_t(x_t) \nabla \cdot v_t(x_t) \right]. \end{aligned}$$

Since $J_{0,t}(x_0) > 0$, division by the Jacobian yields

$$\frac{\mathrm{d}}{\mathrm{d}t} \widetilde{\rho}_t(x_t) = -\widetilde{\rho}_t(x_t) \nabla \cdot v_t(x_t).$$

Thus, no spatial differentiability of the initial density is required to establish differentiability of the transported density along almost every trajectory.

If the transport representative $\widetilde{\rho}(t, x)$ is C^1 in (t, x), the chain rule and the trajectory identity give

$$\partial_t \widetilde{\rho}(t, x_t) + v_t(x_t) \cdot \nabla \widetilde{\rho}(t, x_t) = -\widetilde{\rho}(t, x_t) \nabla \cdot v_t(x_t).$$

Equivalently,

$$\left[\partial_t \widetilde{\rho} + \nabla \cdot (\widetilde{\rho} v_t) \right](t, x_t) = 0.$$

For each fixed t, the expression

$$x_0 \mapsto \left[\partial_t \widetilde{\rho} + \nabla \cdot (\widetilde{\rho} v_t) \right](t, \Phi_{0,t}(x_0))$$

is continuous and vanishes for almost every x_0. A continuous function that vanishes almost everywhere with respect to Lebesgue measure must vanish everywhere. Since $\Phi_{0,t}$ is onto, the identity therefore holds at every state $x \in \mathbb{R}^d$. Hence,

$$\partial_t \widetilde{\rho} + \nabla \cdot (\widetilde{\rho} v_t) = 0.$$

□

Notice

The conserved quantity is probability mass: for every Borel region A,

$$\mu_t(\Phi_{0,t}(A)) = \mu_0(A).$$

The density itself generally changes along trajectories. Local volume expansion, measured by an increasing $J_{0,t}$, decreases the density carried by the same material element; contraction increases it. It is the product $\rho_t(\Phi_{0,t}(x_0))J_{0,t}(x_0)$ that remains constant. Thus, conservation of probability mass must be distinguished from constancy of density.

Statistical observables and weak probability transport. The statistical consequences of microscopic motion are measured through ensemble averages. A measurable observable $f : E \to \mathbb{R}$, with $E = \mathbb{R}^d$ here, represents a physical property such as a position component, energy, or an order parameter. Whenever f is integrable under μ_t, its expectation is

$$\langle f \rangle_t := \langle f, \mu_t \rangle = \int_E f(x)\, \mu_t(\,\mathrm{d}x) = \mathbb{E}[f(X_t)].$$

The trajectory specifies the microscopic state, whereas these expectations describe statistical properties of the ensemble. The pushforward identity connects the two descriptions:

$$\langle f, \mu_t \rangle = \int_{\mathbb{R}^d} f(\Phi_{0,t}(x_0))\, \mu_0(\,\mathrm{d}x_0).$$

For a C^1 observable, differentiation along each realization gives

$$\frac{\mathrm{d}}{\mathrm{d}t} f(X_t) = v_t(X_t) \cdot \nabla f(X_t) = (\mathcal{L}_t f)(X_t), \quad \mathcal{L}_t f = v_t \cdot \nabla f.$$

This is the deterministic generator introduced above, now acting on a measured physical property. On every compact time interval, assume that $f(\Phi_{0,t}(x_0))$ and its time derivative are dominated by μ_0-integrable functions of x_0. We may then differentiate the preceding integral to obtain

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle f, \mu_t \rangle = \langle \mathcal{L}_t f, \mu_t \rangle. \tag{1.16}$$

Thus, (1.16) says that the rate of change of an ensemble average is the ensemble average of the microscopic rate of change of its observable. No differentiability of a density is needed for this identity.

Smooth observables as test functions. A physical observable need not be smooth or compactly supported. For a distributional formulation, however, it is useful to select $f \in C_c^\infty(\mathbb{R}^d)$. Such functions can represent smooth, spatially localized measurements and also

serve as mathematical test functions. These roles are related, but this choice of test space does not restrict all physical observables to that class.

For the following measure formulation, retain the given deterministic flow and assume the local flux integrability condition

$$\int_0^T \int_K |v_t(x)|\, \mu_t(\mathrm{d}x)\, \mathrm{d}t < \infty \quad \forall K \subset \mathbb{R}^d.$$

This holds, in particular, for a jointly continuous velocity field on $[0,T] \times \mathbb{R}^d$. The spatial differentiability and invertibility assumptions used for the density change of variables are not needed here.

Proposition 1.18 (Measure-valued continuity equation)**.** *The transported probability law $\mu_t = (\Phi_{0,t})_{\#}\mu_0$ satisfies*

$$\int_0^T \int_{\mathbb{R}^d} [\partial_t \psi(t,x) + v_t(x) \cdot \nabla_x \psi(t,x)]\, \mu_t(\mathrm{d}x)\, \mathrm{d}t = 0 \tag{1.17}$$

for every $\psi \in C_c^\infty((0,T) \times \mathbb{R}^d)$. Equivalently,

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0$$

in the distributional sense.

Proof. Fix $\psi \in C_c^\infty((0,T) \times \mathbb{R}^d)$ and define

$$F(t) := \int_{\mathbb{R}^d} \psi(t,x)\, \mu_t(\mathrm{d}x) = \int_{\mathbb{R}^d} \psi\big(t, \Phi_{0,t}(x)\big)\, \mu_0(\mathrm{d}x),$$

the second identity being the definition of the pushforward $\mu_t = (\Phi_{0,t})_{\#}\mu_0$.

Assume that, for μ_0-a.e. x, the trajectory $t \mapsto \Phi_{0,t}(x)$ is absolutely continuous with

$$\frac{\mathrm{d}}{\mathrm{d}t}\Phi_{0,t}(x) = v_t\big(\Phi_{0,t}(x)\big) \quad \text{for a.e. } t \in (0,T).$$

Then, for μ_0-a.e. x and a.e. t, the chain rule yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi\big(t, \Phi_{0,t}(x)\big) = \partial_t \psi\big(t, \Phi_{0,t}(x)\big) + \nabla_x \psi\big(t, \Phi_{0,t}(x)\big) \cdot v_t\big(\Phi_{0,t}(x)\big).$$

Integrating in x and applying the pushforward formula once more,

$$\begin{aligned} F'(t) &= \int_{\mathbb{R}^d} \Big[\partial_t \psi\big(t, \Phi_{0,t}(x)\big) + \nabla_x \psi\big(t, \Phi_{0,t}(x)\big) \cdot v_t\big(\Phi_{0,t}(x)\big)\Big]\, \mu_0(\mathrm{d}x) \\ &= \int_{\mathbb{R}^d} \Big[\partial_t \psi(t,x) + v_t(x) \cdot \nabla_x \psi(t,x)\Big]\, \mu_t(\mathrm{d}x). \end{aligned} \tag{1.18}$$

Next, integrate (1.18) over $t \in [0,T]$. Since $\psi \in C_c^\infty((0,T)\times\mathbb{R}^d)$, it vanishes in a neighbourhood of the hyperplanes $t=0$ and $t=T$; hence

$$F(0) = \int_{\mathbb{R}^d} \psi(0,x)\,\mu_0(\mathrm{d}x) = 0, \quad F(T) = \int_{\mathbb{R}^d} \psi(T,x)\,\mu_T(\mathrm{d}x) = 0.$$

Consequently, by the fundamental theorem of calculus,

$$\begin{aligned} 0 &= F(T) - F(0) \\ &= \int_0^T F'(t)\,\mathrm{d}t = \int_0^T \int_{\mathbb{R}^d} \Big[\partial_t\psi(t,x) + v_t(x)\cdot\nabla_x\psi(t,x)\Big]\mu_t(\mathrm{d}x)\,\mathrm{d}t, \end{aligned}$$

which is (1.17). To deduce the distributional form, note that for every $\psi \in C_c^\infty((0,T)\times\mathbb{R}^d)$,

$$\Big\langle \partial_t\mu_t + \nabla\cdot(v_t\mu_t),\, \psi \Big\rangle = -\int_0^T \int_{\mathbb{R}^d} \Big[\partial_t\psi(t,x) + v_t(x)\cdot\nabla_x\psi(t,x)\Big]\mu_t(\mathrm{d}x)\,\mathrm{d}t = 0.$$

Hence $\partial_t\mu_t + \nabla\cdot(v_t\mu_t) = 0$ in $\mathcal{D}'((0,T)\times\mathbb{R}^d)$; conversely, this distributional identity clearly implies (1.17). This proves the assertion. □

Time-dependent observables and dynamical pictures. An observable may also depend explicitly on time:

$$f_t : E \to \mathbb{R}, \quad \langle f_t\rangle_t := \langle f_t, \mu_t\rangle = \int_E f_t(x)\,\mu_t(\,\mathrm{d}x), \quad E = \mathbb{R}^d.$$

At a fixed state x, the measured property can change because an external parameter or a measurement protocol changes. This explicit variation is distinct from the change in $f_t(X_t)$ caused by the motion of the state. It is also distinct from assigning dynamical time dependence to observables through a change of representation, as in Koopman evolution. We first keep the physical state coordinates fixed and account for both contributions.

Theorem 1.19 (Transport formula)**.**

Source exercise: Project 9, item 9. *Student Assessment Exercise* [Assessment] (adapted).

Let $F(t,x) = f_t(x)$ be C^1 in (t,x). On each compact time interval assume that $F(t,\Phi_{0,t}(x_0))$ and its time derivative are dominated by integrable functions of x_0 under μ_0. Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[F(t,X_t)] = \mathbb{E}[(\partial_t + v_t\cdot\nabla)F(t,X_t)],$$

or, equivalently,

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle f_t,\mu_t\rangle = \langle \partial_t f_t + \mathcal{L}_t f_t, \mu_t\rangle. \tag{1.19}$$

Formula (1.19) adds the explicit time variation of the observable to its change along states.

Proof. Since $X_t = \Phi_{0,t}(X_0)$ and $X_0 \sim \mu_0$, we have

$$\mathbb{E}[F(t, X_t)] = \int_{\mathbb{R}^d} F(t, \Phi_{0,t}(x_0))\, \mu_0(\mathrm{d}x_0).$$

For a fixed initial state x_0, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t} F(t, \Phi_{0,t}(x_0)) = \partial_t F(t, \Phi_{0,t}(x_0)) + \nabla F(t, \Phi_{0,t}(x_0)) \cdot \frac{\mathrm{d}}{\mathrm{d}t} \Phi_{0,t}(x_0).$$

By the defining ODE of the flow,

$$\frac{\mathrm{d}}{\mathrm{d}t} \Phi_{0,t}(x_0) = v_t(\Phi_{0,t}(x_0)).$$

Consequently,

$$\frac{\mathrm{d}}{\mathrm{d}t} F(t, \Phi_{0,t}(x_0)) = \left[\partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)\right]_{x=\Phi_{0,t}(x_0)}.$$

By the assumed integrable domination, differentiation under the integral sign is justified. Hence,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[F(t, X_t)] &= \int_{\mathbb{R}^d} \frac{\mathrm{d}}{\mathrm{d}t} F(t, \Phi_{0,t}(x_0))\, \mu_0(\mathrm{d}x_0) \\ &= \int_{\mathbb{R}^d} \left[\partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)\right]_{x=\Phi_{0,t}(x_0)} \mu_0(\mathrm{d}x_0).\end{aligned}$$

Finally, using the pushforward relation

$$\mu_t = (\Phi_{0,t})_{\#}\mu_0,$$

we obtain, for every integrable function g,

$$\int_{\mathbb{R}^d} g(\Phi_{0,t}(x_0))\, \mu_0(\mathrm{d}x_0) = \int_{\mathbb{R}^d} g(x)\, \mu_t(\mathrm{d}x).$$

Applying this identity to $g(x) = \partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)$ yields

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[F(t, X_t)] &= \int_{\mathbb{R}^d} \left[\partial_t F(t, x) + v_t(x) \cdot \nabla F(t, x)\right] \mu_t(\mathrm{d}x) \\ &= \mathbb{E}\left[(\partial_t + v_t \cdot \nabla) F(t, X_t)\right].\end{aligned}$$

Since $F(t, x) = f_t(x)$ and $\mathcal{L}_t f_t = v_t \cdot \nabla f_t$, this is equivalent to

$$\frac{\mathrm{d}}{\mathrm{d}t} \langle f_t, \mu_t \rangle = \langle \partial_t f_t + \mathcal{L}_t f_t, \mu_t \rangle.$$

□

The term $\partial_t f_t$ measures explicit variation at a fixed state, while $\mathcal{L}_t f_t$ measures variation

caused by the state's motion. Their sum is the material derivative,

$$\frac{Df_t}{Dt} := \partial_t f_t + v_t \cdot \nabla f_t.$$

When a smooth density exists and boundary terms vanish, this distinction can also be read from

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} f_t \rho_t \, \mathrm{d}x = \int_{\mathbb{R}^d} \rho_t \partial_t f_t \, \mathrm{d}x + \int_{\mathbb{R}^d} f_t \partial_t \rho_t \, \mathrm{d}x.$$

Using (1.15), the second term equals $\int_{\mathbb{R}^d} \rho_t v_t \cdot \nabla f_t \, \mathrm{d}x$. It accounts for the changing distribution; the first term accounts for the changing observable. Compact spatial support, or sufficient decay and integrability, justifies this integration by parts.

1.2.3 Hamiltonian dynamics: a geometric application

Consider a conservative mechanical system with unit mass. Let $q, p \in \mathbb{R}^n$ denote the generalized position and momentum, respectively, and define the phase-space state $z = (q, p) \in \mathbb{R}^{2n}$. The Hamiltonian is

$$H(q,p) = \frac{1}{2}\|p\|^2 + \mathcal{V}(q), \quad \mathcal{V} \in C^2(\mathbb{R}^n).$$

Hamilton's equations yield

$$\dot{q}_t = \nabla_p H(q_t, p_t) = p_t, \quad \dot{p}_t = -\nabla_q H(q_t, p_t) = -\nabla \mathcal{V}(q_t).$$

Consequently, the dynamics in phase space are governed by the first-order ODE

$$\dot{z}_t = \begin{pmatrix} \dot{q}_t \\ \dot{p}_t \end{pmatrix} = \begin{pmatrix} p_t \\ -\nabla \mathcal{V}(q_t) \end{pmatrix}.$$

The coupling between position and momentum can be expressed more compactly by introducing the canonical matrix

$$\mathbb{J}_n = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Writing

$$\nabla_z H(z) = \begin{pmatrix} \nabla_q H(q,p) \\ \nabla_p H(q,p) \end{pmatrix},$$

Hamilton's equations take the compact form

$$\dot{z}_t = \mathbb{J}_n \nabla_z H(z_t). \tag{1.20}$$

The matrix $\mathbb{J}_n$ in (1.20) is skew-symmetric and invertible, and defines the canonical symplectic form on phase space:

$$\omega(u, w) = u^\top \mathbb{J}_n w, \quad u, w \in \mathbb{R}^{2n}.$$

In particular, for $u = (u_q, u_p)$ and $w = (w_q, w_p)$,

$$\omega(u, w) = u_q^\top w_p - u_p^\top w_q.$$

This structure characterizes the geometric coupling between position and momentum in Hamiltonian mechanics. We next examine the conservation properties of the resulting flow, including energy conservation and preservation of the symplectic form.

Theorem 1.20 (Energy conservation, symplecticity, and Liouville's theorem)**.**

Source exercise: Project 10, item 2–4. *Student Assessment Exercise* [Assessment] (adapted).

Suppose $\mathcal{V} \in C^2(\mathbb{R}^n)$ and the Hamiltonian flow ϕ_t exists on an interval containing $t = 0$. Let $M_t(z) = D\phi_t(z)$ denote its variational matrix. Then the following properties hold.

1. ***Energy conservation.*** *Along every trajectory,*

$$H(\phi_t(z)) = H(z).$$

2. ***Symplecticity.*** *The variational matrix preserves the canonical symplectic form:*

$$M_t(z)^\top \mathbb{J}_n M_t(z) = \mathbb{J}_n. \tag{1.21}$$

3. ***Liouville's theorem.*** *The Hamiltonian flow preserves phase-space volume:*

$$\det M_t(z) = 1.$$

Equivalently, the Hamiltonian vector field is divergence-free:

$$\nabla_z \cdot v(z) = 0.$$

Proof. Since $\mathbb{J}_n^\top = -\mathbb{J}_n$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t} H(z_t) = \nabla H(z_t)^\top \dot{z}_t = \nabla H(z_t)^\top \mathbb{J}_n \nabla H(z_t) = 0,$$

because $a^\top \mathbb{J}_n a = 0$ for every real vector a. The Hessian $\nabla^2 H(z_t)$ is symmetric. Differentiating $v(z) = \mathbb{J}_n \nabla H(z)$ yields $Dv(z_t) = \mathbb{J}_n \nabla^2 H(z_t)$. Thus the variational equation reads

$\dot{M}_t = \mathbb{J}_n \nabla^2 H(z_t) M_t$ with $M_0 = I_{2n}$. Differentiating $M_t^\top \mathbb{J}_n M_t$ and using $\mathbb{J}_n^\top = -\mathbb{J}_n$ and $\mathbb{J}_n^2 = -I_{2n}$ gives

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(M_t^\top \mathbb{J}_n M_t) &= M_t^\top \left[(\mathbb{J}_n \nabla^2 H(z_t))^\top \mathbb{J}_n + \mathbb{J}_n(\mathbb{J}_n \nabla^2 H(z_t))\right] M_t \\ &= M_t^\top \left[\nabla^2 H(z_t)(-\mathbb{J}_n)\mathbb{J}_n + \mathbb{J}_n^2 \nabla^2 H(z_t)\right] M_t \\ &= M_t^\top (\nabla^2 H(z_t) - \nabla^2 H(z_t)) M_t = 0.\end{aligned}$$

The initial matrix is $M_0^\top \mathbb{J}_n M_0 = \mathbb{J}_n$, which proves (1.21). Taking determinants gives $(\det M_t)^2 = 1$. Continuity in time and $\det M_0 = 1$ fix $\det M_t = 1$. Independently, $\nabla \cdot v = \operatorname{tr}(\mathbb{J}_n \nabla^2 H) = 0$ since the trace of a product of an antisymmetric and a symmetric matrix vanishes. Liouville's formula therefore also yields $\det M_t = 1$. □

Proposition 1.21 (Two meanings of reversing a Hamiltonian trajectory)**.** *For any solution $z_t = (q_t, p_t)$ on $[0, T]$, the simple reversed curve $y_t = z_{T-t}$ satisfies $\dot{y}_t = -\mathbb{J}_n \nabla H(y_t)$, which is Hamilton's equation with Hamiltonian $-H$, not generally with H. For the mechanical Hamiltonian above define the involution $R(q, p) = (q, -p)$. Then*

$$\widetilde{z}_t = R z_{T-t} = (q_{T-t}, -p_{T-t})$$

solves the original *Hamilton equations for H.*

Proof. *Proof to be completed.* □

Example 1.22 (Harmonic oscillator: exact solution and a symplectic scheme)**.**

Problem

Take $\mathcal{V}(q) = \frac{1}{2}\omega^2 q^2$, $\omega > 0$, in one dimension. The system $\dot{q} = p$, $\dot{p} = -\omega^2 q$ is linear with $A = \left(\begin{smallmatrix} 0 & 1 \\ -\omega^2 & 0 \end{smallmatrix}\right)$.

Solution.

Exponentiating the oscillator matrix gives the exact trajectory below. Therefore $q_t = q_0 \cos(\omega t) + (p_0/\omega)\sin(\omega t)$, and $p_t = p_0 \cos(\omega t) - \omega q_0 \sin(\omega t)$. For comparison, velocity Verlet uses numerical states $(\widehat{q}_k, \widehat{p}_k)$ in a *kick–drift–kick* step:

$$\begin{aligned}\widehat{p}_{k+1/2} &= \widehat{p}_k - \tfrac{h}{2}\omega^2 \widehat{q}_k, \\ \widehat{q}_{k+1} &= \widehat{q}_k + h\widehat{p}_{k+1/2}, \\ \widehat{p}_{k+1} &= \widehat{p}_{k+1/2} - \tfrac{h}{2}\omega^2 \widehat{q}_{k+1}.\end{aligned} \tag{1.22}$$

Each kick or drift is the exact Hamiltonian flow of a purely potential or kinetic subsystem, respectively; their composition is symplectic. For the harmonic oscillator the resulting matrix is

$$V_h = \begin{pmatrix} 1 - \omega^2 h^2/2 & h \\ -h\omega^2(1 - \omega^2 h^2/4) & 1 - \omega^2 h^2/2 \end{pmatrix}, \quad \det V_h = 1.$$

In the stable regime $0 < h\omega < 2$, it preserves exactly the modified quadratic energy

$$\widetilde{H}_h(q,p) = \frac{1}{2}p^2 + \frac{1}{2}\omega^2(1-\omega^2h^2/4)q^2. \tag{1.23}$$

This explicitly explains the bounded $O(h^2)$ true-energy error on bounded oscillatory orbits. Explicit Euler, whose step matrix $I+hA$ has determinant $1+\omega^2h^2 > 1$, is not symplectic and generically produces secular energy growth; the numerical comparison follows in Table 1.2. ♦

Executed Hamiltonian experiment (Project 10): symplectic versus Euler evolution.

In the harmonic example take $\omega = 1$, $q_0 = 1$, $p_0 = 0$, step size $h = 0.1$ and $K = 1000$ steps, giving $T = 100$. The exact energy is $H_0 = 0.5$; write $\widehat{H}_k := H(\widehat{q}_k, \widehat{p}_k)$ for the computed true energy. Executing the updates of (1.22) and explicit Euler produces the results in Table 1.2. The bounds derived from (1.23) predict a maximum Verlet energy excursion of $\omega^4h^2q_0^2/8 = 0.00125$ for this particular initial state.

Table 1.2: Harmonic-oscillator energy comparison at $T = 100$ with $h = 0.1$.

Method	Energy at $T = 100$	Maximum $\lvert\widehat{H}_k - H_0\rvert$
Exact evolution	0.500000	0
Explicit Euler	10479.577819	10479.077819
Velocity Verlet	0.499724	0.001250

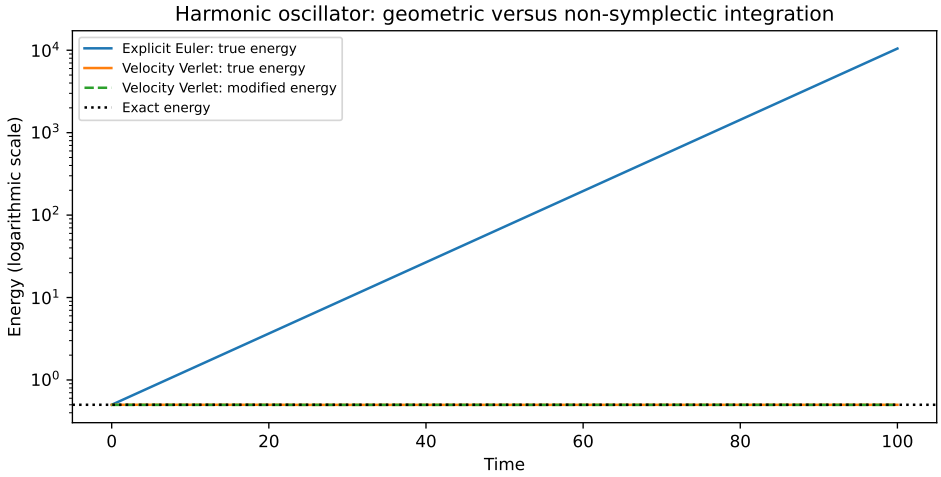


Figure 1.6: True and modified energies in the numerical Hamiltonian benchmark. The y-axis is logarithmic; the modified Verlet energy is constant up to roundoff, while Euler introduces an expanding phase-space map and secular energy growth.

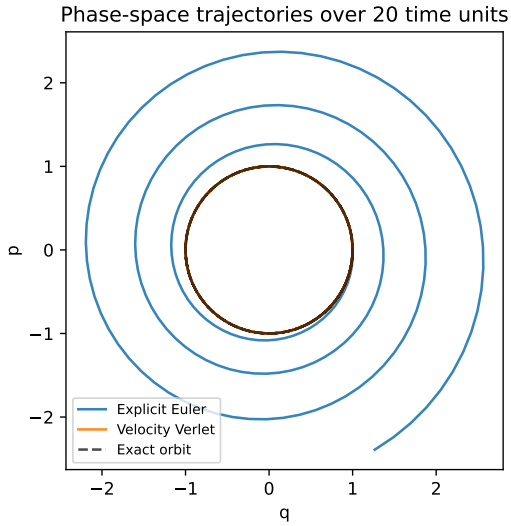


Figure 1.7: Exact, Euler, and Verlet trajectories in (q, p) over $0 \leq t \leq 20$. The differences arise from their discrete geometric structures, not from the continuous Hamiltonian system.

1.2.4 Entropy, variational principles, and free energy

1.2.4.1 Entropy and score under transport

For a smooth, strictly positive probability density ρ on $V = \mathbb{R}^d$, define its differential entropy by

$$S(\rho) = -\int_{\mathbb{R}^d} \rho(x) \log \rho(x) \, \mathrm{d}x,$$

provided that $\int_{\mathbb{R}^d} |\rho(x) \log \rho(x)| \, \mathrm{d}x < \infty$.

Theorem 1.23 (Evolution of differential entropy)**.**

Source exercise: Project 9, item 10. *Student Assessment Exercise* [Assessment] (adapted).

Let ρ_t be a strictly positive probability density that solves the continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$$

classically. Assume that differentiation under the integral is justified, all integrals below are finite, and the boundary terms arising from integration by parts vanish.
Define the differential entropy and the Stein score function by

$$S(\rho_t) = -\int_{\mathbb{R}^d} \rho_t(x) \log \rho_t(x) \, \mathrm{d}x, \quad s_t(x) := \nabla \log \rho_t(x).$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = \int_{\mathbb{R}^d} \rho_t(x)\nabla\cdot v_t(x)\,\mathrm{d}x = -\int_{\mathbb{R}^d} \rho_t(x)v_t(x)\cdot\nabla\log\rho_t(x)\,\mathrm{d}x$$
$$= \mathbb{E}[\nabla\cdot v_t(X_t)] = -\mathbb{E}[v_t(X_t)\cdot s_t(X_t)],$$

where X_t is a random variable with probability density ρ_t.

Proof. By the definition of differential entropy,

$$S(\rho_t) = -\int_{\mathbb{R}^d} \rho_t(x)\log\rho_t(x)\,\mathrm{d}x.$$

Differentiating with respect to time and applying the product rule, we obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = -\int_{\mathbb{R}^d} \big(1+\log\rho_t(x)\big)\partial_t\rho_t(x)\,\mathrm{d}x.$$

Using the continuity equation $\partial_t\rho_t + \nabla\cdot(\rho_t v_t) = 0$, we can rewrite this as

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = \int_{\mathbb{R}^d} \big(1+\log\rho_t(x)\big)\nabla\cdot(\rho_t(x)v_t(x))\,\mathrm{d}x.$$

Integration by parts, together with the assumed vanishing of boundary terms, gives

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = -\int_{\mathbb{R}^d} \rho_t(x)v_t(x)\cdot\nabla\big(1+\log\rho_t(x)\big)\,\mathrm{d}x$$
$$= -\int_{\mathbb{R}^d} \rho_t(x)v_t(x)\cdot\nabla\log\rho_t(x)\,\mathrm{d}x.$$

Since ρ_t is strictly positive, we have

$$\rho_t(x)\nabla\log\rho_t(x) = \nabla\rho_t(x).$$

Consequently, another integration by parts yields

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = -\int_{\mathbb{R}^d} v_t(x)\cdot\nabla\rho_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \rho_t(x)\nabla\cdot v_t(x)\,\mathrm{d}x.$$

Finally, since X_t has probability density ρ_t, the expectation identity

$$\mathbb{E}[g(X_t)] = \int_{\mathbb{R}^d} g(x)\rho_t(x)\,\mathrm{d}x$$

implies that

$$\frac{\mathrm{d}}{\mathrm{d}t}S(\rho_t) = \mathbb{E}[\nabla\cdot v_t(X_t)] = -\mathbb{E}[v_t(X_t)\cdot s_t(X_t)],$$

where $s_t(x) = \nabla\log\rho_t(x)$ is the Stein score function. □

The entropy rate is therefore the ensemble average of local volume expansion: positive mean divergence increases differential entropy, negative mean divergence decreases it, and

volume-preserving flows leave it constant. Deterministic transport has no universal entropy increase law.

Proposition 1.24 (Lagrangian score dynamics)**.**

Source exercise: Project 9, item 11. *Student Assessment Exercise* [Assessment] (adapted).

Assume v_t is twice continuously differentiable in space, the positive density has $\ell_t = \log \rho_t$ of class C^1 in time and C^2 in space, and the mixed derivatives used below are continuous. Let $X_t = \Phi_{0,t}(x)$ satisfy

$$\frac{\mathrm{d}}{\mathrm{d}t} X_t = v_t(X_t), \quad X_0 = x.$$

Set

$$s_t^\rho = \nabla \log \rho_t, \quad G_t(x) = s_t^\rho(\Phi_{0,t}(x)).$$

Then, along each trajectory,

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} s_t^\rho(X_t) &= \frac{\mathrm{d}}{\mathrm{d}t} G_t(x) \\ &= -(Dv_t(X_t))^\top G_t(x) - \nabla(\nabla \cdot v_t)(X_t), \end{aligned} \tag{1.24}$$

with the initial condition

$$G_0(x) = \nabla \log \rho_0(x).$$

In (1.24), Dv_t denotes the Jacobian matrix of v_t, and $\nabla \cdot v_t = \mathrm{tr}(Dv_t)$.

Proof. *Proof to be completed.* □

Proposition 1.25 (Stein's integration-by-parts identity)**.**

Source exercise: Project 11, item 8. *Student Assessment Exercise* [Assessment] (adapted).

Let ρ be positive and C^1 and let $s^\rho = \nabla \log \rho$. If $f \in C^1$, $\mathbb{E}_\rho[\|\nabla f\| + \|f s^\rho\|] < \infty$, and the boundary terms in the coordinate integrations by parts vanish, then

$$\mathbb{E}_\rho[\nabla f(X) + f(X) s^\rho(X)] = 0.$$

For $X \sim \mathcal{N}(0, I_d)$ this reads $\mathbb{E}[X f(X)] = \mathbb{E}[\nabla f(X)]$.

Proof. *Proof to be completed.* □

1.2.4.2 KL divergence and its basic identities

In the information-theoretic results below, $(E, \mathcal{S})$ denotes an explicitly chosen measurable space, which may be V with its Borel structure but need not be a vector space. Write $B_b(E)$ for bounded $\mathcal{S}$-measurable real functions and $\mathcal{P}(E)$ for probability measures on $(E, \mathcal{S})$; when a topology is specified, its Borel sigma-algebra is understood. For proba-

bilities $\mu \ll \nu$ define

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log\left(\frac{\mathrm{d}\mu}{\mathrm{d}\nu}\right) \mathrm{d}\mu,$$

and set it to $+\infty$ when $\mu \not\ll \nu$. The convention $0 \log 0 = 0$ is used. Its negative part is integrable, since $-r \log r \le 1/e$ for $0 \le r \le 1$ and ν is a probability law; thus the extended value is well defined. For densities we use $D_{\mathrm{KL}}(\rho\|\eta)$ only as shorthand for $D_{\mathrm{KL}}(\rho(x)\,\mathrm{d}x\|\eta(x)\,\mathrm{d}x)$.

Theorem 1.26 (Gibbs' inequality)**.**

Source exercise: Project 11, item 1. *Student Assessment Exercise* [Assessment] (adapted).

For probability measures μ, ν, $D_{\mathrm{KL}}(\mu\|\nu) \ge 0$, with equality if and only if $\mu = \nu$.

Proof.

- If $\mu \not\ll \nu$, then by definition $D_{\mathrm{KL}}(\mu\|\nu) = +\infty$, and the inequality is immediate.
- Otherwise, since $\mu \ll \nu$, the Radon–Nikodym theorem gives the density

$$r = \frac{\mathrm{d}\mu}{\mathrm{d}\nu}, \quad \int_E r \,\mathrm{d}\nu = 1.$$

By the change-of-measure formula,

$$D_{\mathrm{KL}}(\mu\|\nu) = \int_E r \log r \,\mathrm{d}\nu,$$

with the convention $0 \log 0 = 0$. Consider the function $f(r) = r \log r,\ r \ge 0$. Since

$$f''(r) = \frac{1}{r} > 0, \quad r > 0,$$

f is strictly convex on $[0, \infty)$, where $f(0) = 0$ is defined by continuity. Applying Jensen's inequality with respect to the probability measure ν, we obtain

$$D_{\mathrm{KL}}(\mu\|\nu) = \int_E f(r) \,\mathrm{d}\nu \ge f\left(\int_E r \,\mathrm{d}\nu\right) = f(1) = 0.$$

Therefore, $D_{\mathrm{KL}}(\mu\|\nu) \ge 0$.

Moreover, since f is strictly convex, equality in Jensen's inequality holds if and only if r is constant ν-almost everywhere. Together with $\int_E r \,\mathrm{d}\nu = 1$, this gives

$$r = 1, \quad \nu\text{-a.e.}$$

Consequently, for every $A \in \mathcal{S}$,

$$\mu(A) = \int_A r \,\mathrm{d}\nu = \int_A 1 \,\mathrm{d}\nu = \nu(A),$$

and hence $\mu = \nu$. Conversely, if $\mu = \nu$, then $\mathrm{d}\mu/\,\mathrm{d}\nu = 1$ ν-almost everywhere, so

$$D_{\mathrm{KL}}(\mu\|\nu) = \int_E \log 1 \; \mathrm{d}\mu = 0.$$

Thus,

$$D_{\mathrm{KL}}(\mu\|\nu) \geq 0, \quad D_{\mathrm{KL}}(\mu\|\nu) = 0 \iff \mu = \nu.$$

□

Proposition 1.27 (Relative entropy of two transported densities)**.**

Source exercise: Project 9, item 12. *Student Assessment Exercise* [Assessment] (adapted).

Let positive smooth ρ_t, η_t solve continuity equations with velocities v_t, w_t, respectively, and suppose all boundary terms vanish. Set $s_t^\rho = \nabla \log \rho_t$ and $s_t^\eta = \nabla \log \eta_t$. Then

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t\|\eta_t) = \int \rho_t (v_t - w_t) \cdot (s_t^\rho - s_t^\eta)\, \mathrm{d}x. \tag{1.25}$$

In particular, (1.25) *vanishes when the two densities are transported by the same velocity field.*

Proof. By the definition of the KL divergence,

$$D_{\mathrm{KL}}(\rho_t\|\eta_t) = \int_{\mathbb{R}^d} \rho_t \log \frac{\rho_t}{\eta_t}\, \mathrm{d}x.$$

Differentiating under the integral and applying the product and chain rules give

$$\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t\|\eta_t) = \int_{\mathbb{R}^d} \left[\partial_t \rho_t \left(\log \frac{\rho_t}{\eta_t} + 1 \right) - \frac{\rho_t}{\eta_t} \partial_t \eta_t \right] \mathrm{d}x.$$

Since ρ_t and η_t satisfy their respective continuity equations,

$$\partial_t \rho_t = -\nabla \cdot (\rho_t v_t), \qquad \partial_t \eta_t = -\nabla \cdot (\eta_t w_t),$$

we obtain

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t\|\eta_t) = & - \int_{\mathbb{R}^d} \nabla \cdot (\rho_t v_t) \left(\log \frac{\rho_t}{\eta_t} + 1 \right) \mathrm{d}x \\ & + \int_{\mathbb{R}^d} \frac{\rho_t}{\eta_t} \nabla \cdot (\eta_t w_t)\, \mathrm{d}x. \end{aligned}$$

We evaluate the two terms separately. By integration by parts and the vanishing of boundary terms, the first term becomes

$$\begin{aligned}
&-\int_{\mathbb{R}^d} \nabla \cdot (\rho_t v_t) \left(\log \frac{\rho_t}{\eta_t} + 1 \right) \mathrm{d}x \\
&\quad = \int_{\mathbb{R}^d} \rho_t v_t \cdot \nabla \log \frac{\rho_t}{\eta_t} \, \mathrm{d}x \\
&\quad = \int_{\mathbb{R}^d} \rho_t v_t \cdot (s_t^\rho - s_t^\eta) \, \mathrm{d}x.
\end{aligned}$$

Similarly, integration by parts gives

$$\begin{aligned}
&\int_{\mathbb{R}^d} \frac{\rho_t}{\eta_t} \nabla \cdot (\eta_t w_t) \, \mathrm{d}x \\
&\quad = -\int_{\mathbb{R}^d} \eta_t w_t \cdot \nabla \left(\frac{\rho_t}{\eta_t} \right) \mathrm{d}x \\
&\quad = -\int_{\mathbb{R}^d} \rho_t w_t \cdot \nabla \log \frac{\rho_t}{\eta_t} \, \mathrm{d}x \\
&\quad = -\int_{\mathbb{R}^d} \rho_t w_t \cdot (s_t^\rho - s_t^\eta) \, \mathrm{d}x,
\end{aligned}$$

where we have used

$$\nabla \left(\frac{\rho_t}{\eta_t} \right) = \frac{\rho_t}{\eta_t} \nabla \log \frac{\rho_t}{\eta_t}.$$

Combining the two terms yields

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t} D_{\mathrm{KL}}(\rho_t \| \eta_t) &= \int_{\mathbb{R}^d} \rho_t v_t \cdot (s_t^\rho - s_t^\eta) \, \mathrm{d}x \\
&\quad - \int_{\mathbb{R}^d} \rho_t w_t \cdot (s_t^\rho - s_t^\eta) \, \mathrm{d}x \\
&= \int_{\mathbb{R}^d} \rho_t (v_t - w_t) \cdot (s_t^\rho - s_t^\eta) \, \mathrm{d}x.
\end{aligned}$$

This completes the proof. □

Theorem 1.28 (KL disintegration and data processing)**.**

Source exercise: Project 11, item 7. *Student Assessment Exercise* [Assessment] (adapted).

Let E, F be standard Borel spaces, let μ, ν be probability measures on $E \times F$, and consider disintegrations

$$\begin{aligned}
\mu(\mathrm{d}x, \mathrm{d}y) &= \mu_X(\mathrm{d}x) \mu(\mathrm{d}y \mid x), \\
\nu(\mathrm{d}x, \mathrm{d}y) &= \nu_X(\mathrm{d}x) \nu(\mathrm{d}y \mid x).
\end{aligned}$$

Under $\mu \ll \nu$,

$$D_{\mathrm{KL}}(\mu\|\nu) = D_{\mathrm{KL}}(\mu_X\|\nu_X) + \int_E D_{\mathrm{KL}}\big(\mu(\cdot \mid x)\|\nu(\cdot \mid x)\big)\mu_X(\,\mathrm{d}x).$$

In particular, projection cannot increase KL divergence. For product laws,

$$D_{\mathrm{KL}}(\mu_X \otimes \mu_Y\|\nu_X \otimes \nu_Y) = D_{\mathrm{KL}}(\mu_X\|\nu_X) + D_{\mathrm{KL}}(\mu_Y\|\nu_Y),$$

with the same extended-real convention.

Proof. *Proof to be completed.* □

1.2.4.3 Convex geometry of relative entropy

The positivity of KL divergence does not make it a metric. Its direction specifies which law supplies the integration measure, and changing that law usually changes the penalty. This is important when an approximating model is trained by minimizing either $D_{\mathrm{KL}}(P\|Q)$ or $D_{\mathrm{KL}}(Q\|P)$.

Proposition 1.29 (Relative entropy is generally asymmetric)**.**

Source exercise: Project 11, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Let $P = \mathcal{N}(0,1)$ and $Q = \mathcal{N}(0,4)$ on $\mathbb{R}$. Then

$$D_{\mathrm{KL}}(P\|Q) = \log 2 - \frac{3}{8}, \quad D_{\mathrm{KL}}(Q\|P) = \frac{3}{2} - \log 2.$$

In particular, the two relative entropies are different, although both are finite and both vanish only when their arguments agree.

Proof. *Proof to be completed.* □

Example 1.30 (Truncated Gaussian: an exactly solvable KL optimization)**.**

Source exercise: Project 11, item 3. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $Z \sim \mathcal{N}(0,1)$ be a standard Gaussian random variable with probability density

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Denote its probability distribution by $\nu = \mathcal{N}(0,1)$. For each $b \in \mathbb{R}$, define the Gaussian upper-tail probability and the corresponding conditional distribution by

$$\bar{\Phi}(b) := \mathbb{P}(Z > b) = \int_b^\infty \varphi(x)\,\mathrm{d}x > 0, \quad \mu_b := \mathrm{Law}(Z \mid Z > b).$$

1. Determine the relative entropy $D_{\mathrm{KL}}(\mu_b\|\nu)$ explicitly as a function of b, and plot

its dependence on $b \in \mathbb{R}$.

2. Let $U : \mathbb{R} \to \mathbb{R}$ be the quadratic potential $U(x) = \frac{\lambda x^2}{2}$, $\lambda > 0$. Solve the optimization problem

$$\inf_{b \geq 0} \left\{ D_{\mathrm{KL}}(\mu_b \| \nu) + \int_{\mathbb{R}} U(x)\, \mu_b(\,\mathrm{d}x) \right\}.$$

Determine the global minimizer and the corresponding minimum value.

Solution.

1. Relative entropy of the truncated Gaussian. By the definition of conditional probability, for every Borel set $A \subseteq \mathbb{R}$,

$$\mu_b(A) = \frac{\nu(A \cap (b, \infty))}{\bar{\Phi}(b)} = \frac{1}{\bar{\Phi}(b)} \int_{A \cap (b,\infty)} \varphi(x)\ \mathrm{d}x.$$

Consequently, the Lebesgue density of μ_b is

$$\mu_b(\,\mathrm{d}x) = \frac{\varphi(x)}{\bar{\Phi}(b)} \mathbf{1}_{\{x > b\}}\ \mathrm{d}x.$$

Since $\nu(\,\mathrm{d}x) = \varphi(x)\ \mathrm{d}x$, the Radon–Nikodym derivative is

$$\frac{\mathrm{d}\mu_b}{\mathrm{d}\nu}(x) = \frac{\mathbf{1}_{\{x>b\}}}{\bar{\Phi}(b)}.$$

Using the definition of relative entropy, we obtain

$$\begin{aligned} D_{\mathrm{KL}}(\mu_b \| \nu) &= \int_{\mathbb{R}} \log\left(\frac{\mathrm{d}\mu_b}{\mathrm{d}\nu}(x) \right) \mu_b(\,\mathrm{d}x) \\ &= \int_{(b,\infty)} \log\left(\frac{1}{\bar{\Phi}(b)} \right) \mu_b(\,\mathrm{d}x) \\ &= -\log \bar{\Phi}(b) \int_{(b,\infty)} \mu_b(\,\mathrm{d}x) \\ &= -\log \bar{\Phi}(b). \end{aligned}$$

Therefore,

$$D_{\mathrm{KL}}(\mu_b \| \nu) = -\log \bar{\Phi}(b). \tag{1.26}$$

Differentiating with respect to b, and using $\bar{\Phi}'(b) = -\varphi(b)$, gives

$$\frac{\mathrm{d}}{\mathrm{d}b} D_{\mathrm{KL}}(\mu_b \| \nu) = \frac{\varphi(b)}{\bar{\Phi}(b)} > 0.$$

Hence, the relative entropy is strictly increasing in b. Furthermore,

$$\lim_{b\to-\infty} D_{\mathrm{KL}}(\mu_b\|\nu) = 0, \qquad \lim_{b\to+\infty} D_{\mathrm{KL}}(\mu_b\|\nu) = +\infty,$$

and $D_{\mathrm{KL}}(\mu_0\|\nu) = \log 2$.

The dependence of the relative entropy on b is illustrated in Figure 1.8.

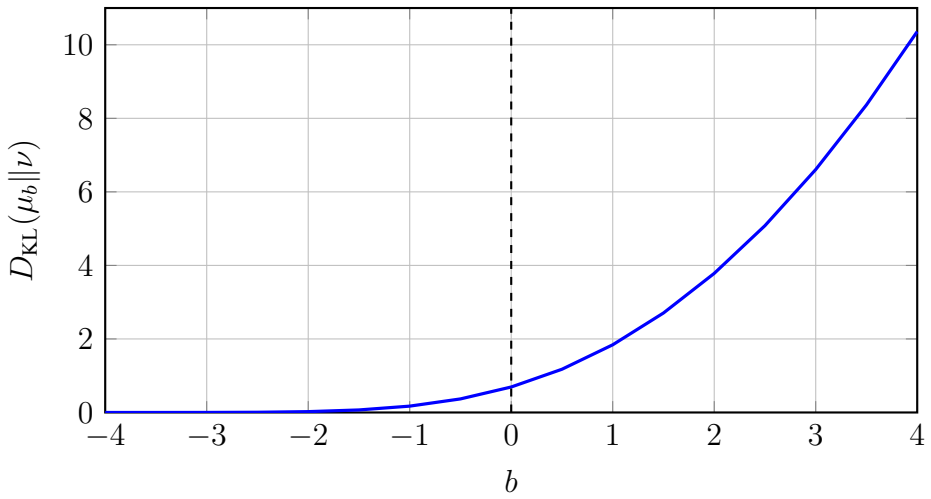


Figure 1.8: Relative entropy of the truncated Gaussian distribution as a function of b.

2. Optimization with a quadratic potential. Let $U(x) = \frac{\lambda x^2}{2}$, $\lambda > 0$. The potential energy averaged under μ_b is

$$\int_{\mathbb{R}} U(x)\,\mu_b(\,\mathrm{d}x) = \frac{\lambda}{2}\mathbb{E}[Z^2 \mid Z > b].$$

By the definition of conditional expectation,

$$\mathbb{E}[Z^2 \mid Z > b] = \frac{1}{\bar{\Phi}(b)} \int_b^\infty x^2\varphi(x)\ \mathrm{d}x.$$

Since $\varphi'(x) = -x\varphi(x)$, integration by parts yields

$$\begin{aligned}\int_b^\infty x^2\varphi(x)\ \mathrm{d}x &= -\int_b^\infty x\varphi'(x)\ \mathrm{d}x \\ &= -\left[x\varphi(x)\right]_b^\infty + \int_b^\infty \varphi(x)\ \mathrm{d}x \\ &= b\varphi(b) + \bar{\Phi}(b).\end{aligned}$$

Therefore,

$$\mathbb{E}[Z^2 \mid Z > b] = 1 + \frac{b\varphi(b)}{\bar{\Phi}(b)},$$

and hence

$$\int_{\mathbb{R}} U(x)\,\mu_b(\,\mathrm{d}x) = \frac{\lambda}{2}\left(1 + \frac{b\varphi(b)}{\bar{\Phi}(b)}\right). \tag{1.27}$$

Combining (1.26) and (1.27), the optimization problem becomes $\inf_{b\geq 0} F(b)$, where

$$F(b) = -\log\bar{\Phi}(b) + \frac{\lambda}{2}\left(1 + \frac{b\varphi(b)}{\bar{\Phi}(b)}\right). \tag{1.28}$$

To determine the global minimizer of (1.28), introduce the inverse Mills ratio

$$h(b) := \frac{\varphi(b)}{\bar{\Phi}(b)}.$$

Differentiating and using $\varphi'(b) = -b\varphi(b)$ and $\bar{\Phi}'(b) = -\varphi(b)$, we obtain

$$h'(b) = \frac{-b\varphi(b)\bar{\Phi}(b) + \varphi(b)^2}{\bar{\Phi}(b)^2} = h(b)\big(h(b) - b\big).$$

Thus, the objective function can be written as

$$F(b) = -\log\bar{\Phi}(b) + \frac{\lambda}{2}\big(1 + bh(b)\big).$$

Differentiating gives

$$\begin{aligned} F'(b) &= h(b) + \frac{\lambda}{2}\left(h(b) + bh'(b)\right) \\ &= h(b)\left[1 + \frac{\lambda}{2}\left(1 + b(h(b) - b)\right)\right]. \end{aligned}$$

It remains to establish the sign of $F'(b)$ on the admissible domain $b \geq 0$. For $b > 0$, since $x > b$ on the integration domain,

$$\bar{\Phi}(b) = \int_b^\infty \varphi(x)\ \mathrm{d}x < \frac{1}{b}\int_b^\infty x\varphi(x)\ \mathrm{d}x = \frac{\varphi(b)}{b},$$

where the last equality follows from $\varphi'(x) = -x\varphi(x)$. Consequently, $h(b) > b, \quad b > 0$. At $b = 0$, we have $h(0) > 0$. Therefore,

$$b(h(b) - b) \geq 0, \quad b \geq 0.$$

Since $h(b) > 0$ and $\lambda > 0$, it follows that

$$F'(b) = h(b)\left[1 + \frac{\lambda}{2}\left(1 + b(h(b) - b)\right)\right] > 0, \quad b \geq 0.$$

Thus, F is strictly increasing on $[0, \infty)$. Its unique global minimizer is therefore attained at the boundary point $b^* = 0$.

Finally, using $\bar{\Phi}(0) = \frac{1}{2}, \quad \varphi(0) = \frac{1}{\sqrt{2\pi}}$, we obtain

$$F(0) = -\log\frac{1}{2} + \frac{\lambda}{2}\left(1 + \frac{0 \cdot \varphi(0)}{\bar{\Phi}(0)}\right) = \log 2 + \frac{\lambda}{2}.$$

Hence, the optimization problem admits the unique global minimizer and minimum value

$$b^* = 0, \quad \inf_{b \geq 0} F(b) = \log 2 + \frac{\lambda}{2}.$$

♦

Theorem 1.31 (Convexity properties of relative entropy and its Bregman representation)**.**

Source exercise: Project 11, item 4. *Student Assessment Exercise* [Assessment] (adapted).

Let $(E, \mathcal{S})$ be a measurable space, and let $\mathcal{P}(E)$ denote the space of probability measures on it. The relative entropy $D_{\mathrm{KL}}(\gamma\|\theta)$ has the following properties:

1. ***Joint convexity.*** *$D_{\mathrm{KL}}(\gamma\|\theta)$ is jointly convex in the pair $(\gamma, \theta) \in \mathcal{P}(E) \times \mathcal{P}(E)$.*
2. ***Separate convexity.*** *For fixed θ, $D_{\mathrm{KL}}(\gamma\|\theta)$ is convex in γ. Similarly, for fixed γ, it is convex in θ.*
3. ***Bregman representation.*** *When expressed in terms of densities with respect to a common reference measure, the KL divergence is a Bregman divergence generated by the negative entropy functional.*

Solution.

Let $(E, \mathcal{S})$ be a measurable space and let $\gamma, \theta \in \mathcal{P}(E)$. Recall that the relative entropy (Kullback–Leibler divergence) is defined by

$$R(\gamma\|\theta) = \begin{cases} \displaystyle\int_E \log\left(\frac{\mathrm{d}\gamma}{\mathrm{d}\theta}\right) \mathrm{d}\gamma, & \gamma \ll \theta, \\ +\infty, & \text{otherwise.} \end{cases}$$

We first prove that R is jointly convex in (γ, θ). We then derive its separate convexity in each argument and discuss its relation to Bregman divergences.

1. Joint convexity. Let $\gamma_1, \gamma_2, \theta_1, \theta_2 \in \mathcal{P}(E)$ and let $\alpha \in (0, 1)$. Define the convex combinations

$$\gamma_\alpha = \alpha\gamma_1 + (1-\alpha)\gamma_2, \quad \theta_\alpha = \alpha\theta_1 + (1-\alpha)\theta_2.$$

We aim to show that

$$R(\gamma_\alpha\|\theta_\alpha) \leq \alpha R(\gamma_1\|\theta_1) + (1-\alpha)R(\gamma_2\|\theta_2). \tag{1.29}$$

To this end, introduce a common dominating measure

$$\lambda = \gamma_1 + \gamma_2 + \theta_1 + \theta_2.$$

By the Radon–Nikodym theorem, there exist nonnegative measurable densities

$$g_i = \frac{\mathrm{d}\gamma_i}{\mathrm{d}\lambda}, \quad h_i = \frac{\mathrm{d}\theta_i}{\mathrm{d}\lambda}, \quad i = 1, 2.$$

The densities of the convex combinations are therefore

$$g_\alpha = \alpha g_1 + (1-\alpha) g_2, \quad h_\alpha = \alpha h_1 + (1-\alpha) h_2.$$

Consequently,

$$R(\gamma_\alpha \| \theta_\alpha) = \int_E (\alpha g_1 + (1-\alpha) g_2) \times \log \left(\frac{\alpha g_1 + (1-\alpha) g_2}{\alpha h_1 + (1-\alpha) h_2} \right) \mathrm{d}\lambda.$$

We now use the log-sum inequality: for nonnegative numbers a_1, a_2, b_1, b_2,

$$(a_1 + a_2) \log \frac{a_1 + a_2}{b_1 + b_2} \leq a_1 \log \frac{a_1}{b_1} + a_2 \log \frac{a_2}{b_2},$$

with the usual extended-real conventions. Applying this inequality pointwise with

$$a_1 = \alpha g_1, \quad a_2 = (1-\alpha) g_2, \quad \text{and} \quad b_1 = \alpha h_1, \quad b_2 = (1-\alpha) h_2,$$

we obtain

$$(\alpha g_1 + (1-\alpha) g_2) \log \left(\frac{\alpha g_1 + (1-\alpha) g_2}{\alpha h_1 + (1-\alpha) h_2} \right) \quad \leq \quad \alpha g_1 \log \frac{g_1}{h_1} + (1-\alpha) g_2 \log \frac{g_2}{h_2}.$$

Integrating both sides with respect to λ gives

$$\begin{aligned} R(\gamma_\alpha \| \theta_\alpha) &\leq \alpha \int_E g_1 \log \frac{g_1}{h_1} \, \mathrm{d}\lambda + (1-\alpha) \int_E g_2 \log \frac{g_2}{h_2} \, \mathrm{d}\lambda \\ &= \alpha R(\gamma_1 \| \theta_1) + (1-\alpha) R(\gamma_2 \| \theta_2). \end{aligned}$$

This establishes the joint convexity of relative entropy in the pair (γ, θ).

2. Convexity in γ or θ.

- To establish convexity in the first argument, fix θ and choose $\theta_1 = \theta_2 = \theta$ in (1.29). We immediately obtain

$$R(\alpha \gamma_1 + (1-\alpha) \gamma_2 \| \theta) \leq \alpha R(\gamma_1 \| \theta) + (1-\alpha) R(\gamma_2 \| \theta). \tag{1.30}$$

  Therefore, (1.30) proves convexity in γ for fixed θ.

- Similarly, fix γ and choose $\gamma_1 = \gamma_2 = \gamma$ in (1.29). Then

$$R(\gamma\|\alpha\theta_1 + (1-\alpha)\theta_2) \le \alpha R(\gamma\|\theta_1) + (1-\alpha)R(\gamma\|\theta_2). \tag{1.31}$$

  Hence, (1.31) proves convexity in θ for fixed γ.

3. Relation to Bregman divergence. We next examine the connection between relative entropy and Bregman divergence. Let $\Phi : \mathbb{R}^n \to \mathbb{R}$ be a differentiable convex function. Its associated Bregman divergence is defined by

$$D_\Phi(p,q) = \Phi(p) - \Phi(q) - \langle \nabla\Phi(q), p-q\rangle. \tag{1.32}$$

Geometrically, this quantity measures the difference between the function value $\Phi(p)$ and the first-order Taylor approximation of Φ at q.

Consider the negative Shannon entropy

$$\Phi(p) = \sum_{i=1}^n p_i \log p_i,$$

defined on the interior of the probability simplex. Its gradient is

$$\nabla\Phi(q) = (1+\log q_1, \dots, 1+\log q_n).$$

Substituting into (1.32), we obtain

$$D_\Phi(p,q) = \sum_{i=1}^n p_i \log p_i - \sum_{i=1}^n q_i \log q_i - \sum_{i=1}^n (1+\log q_i)(p_i - q_i).$$

Since p and q are probability vectors, $\sum_{i=1}^n p_i = \sum_{i=1}^n q_i = 1$. The linear terms therefore cancel, yielding

$$D_\Phi(p,q) = \sum_{i=1}^n p_i \log \frac{p_i}{q_i} = R(p\|q). \tag{1.33}$$

Thus, (1.33) identifies KL as a Bregman divergence generated by the negative Shannon entropy.

The same construction extends to probability densities. For densities ρ and η with respect to a fixed reference measure λ, define

$$\Phi(\rho) = \int_E \rho \log \rho \, \mathrm{d}\lambda.$$

Its variational derivative is

$$\frac{\delta\Phi}{\delta\rho} = 1 + \log\rho.$$

For normalized densities, the corresponding Bregman divergence is

$$\begin{aligned} D_\Phi(\rho,\eta) &= \Phi(\rho) - \Phi(\eta) - \int_E (1+\log\eta)(\rho-\eta)\,\mathrm{d}\lambda \\ &= \int_E \rho \log\frac{\rho}{\eta}\,\mathrm{d}\lambda \\ &= R(\rho\lambda\|\eta\lambda). \end{aligned}$$

A distinction from general Bregman divergences

An important point is that a general Bregman divergence is always convex in its first argument, but it is not necessarily convex in its second argument or jointly convex.Indeed, for fixed q, the expression

$$D_\Phi(p,q) = \Phi(p) - \langle \nabla\Phi(q), p\rangle + \text{constant}$$

is convex in p, since Φ is convex and the remaining dependence on p is affine. However, convexity in q does not follow from the convexity of Φ. For example, consider $\Phi(x) = x^4$. The corresponding Bregman divergence is

$$D_\Phi(x,y) = x^4 + 3y^4 - 4xy^3.$$

Fixing $x = 1$, we find

$$\frac{\partial^2}{\partial y^2} D_\Phi(1,y) = 12y(3y-2).$$

This quantity is negative for $0 < y < 2/3$. Therefore, this Bregman divergence is not convex in its second argument.
The KL divergence is special in this respect: it is convex in both arguments and, more strongly, jointly convex.

♦

1.2.4.4 Gibbs tilts and convex duality

Source exercise: Project 11, item 5. *Student Assessment Exercise* [Assessment] (adapted).

Problem

(Gibbs variational identity)
Let $(E,\mathcal{S},\nu)$ be a probability space, and let $U : E \to \mathbb{R}$ be a bounded measurable energy function. Define the normalization constant

$$Z_U := \int_E e^{-U(x)}\,\nu(\mathrm{d}x),$$

and the exponentially tilted probability measure ν_U by

$$\frac{\mathrm{d}\nu_U}{\mathrm{d}\nu}(x) := \frac{e^{-U(x)}}{Z_U}.$$

Show that, for every probability measure $\mu \in \mathcal{P}(E)$,

$$D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \,\mathrm{d}\mu = D_{\mathrm{KL}}(\mu\|\nu_U) - \log Z_U,$$

with the extended-real convention for relative entropy.
Consequently, prove that

$$\inf_{\mu\in\mathcal{P}(E)} \left\{ D_{\mathrm{KL}}(\mu\|\nu) + \int_E U \,\mathrm{d}\mu \right\} = -\log \int_E e^{-U} \,\mathrm{d}\nu,$$

and that the infimum is attained uniquely at

$$\mu^* = \nu_U.$$

Solution.

Let θ be a reference probability measure and define the normalizing constant

$$Z_U := \int e^{-U(x)} \,\theta(\mathrm{d}x).$$

The exponentially tilted probability measure γ_U is defined by

$$\frac{\mathrm{d}\gamma_U}{\mathrm{d}\theta}(x) = \frac{e^{-U(x)}}{Z_U}.$$

Since U is bounded, we have $0 < Z_U < \infty$, and hence γ_U is a well-defined probability measure.

Step 1. Establish the Radon–Nikodym identity. By the definition of γ_U, taking logarithms gives

$$\log \frac{\mathrm{d}\gamma_U}{\mathrm{d}\theta} = -U - \log Z_U.$$

Suppose first that $\gamma \ll \theta$. Since $\mathrm{d}\gamma_U/\mathrm{d}\theta > 0$ θ-almost everywhere, the Radon–Nikodym chain rule yields

$$\frac{\mathrm{d}\gamma}{\mathrm{d}\gamma_U} = \frac{\mathrm{d}\gamma/\mathrm{d}\theta}{\mathrm{d}\gamma_U/\mathrm{d}\theta}.$$

Consequently,

$$\begin{aligned}\log \frac{\mathrm{d}\gamma}{\mathrm{d}\gamma_U} &= \log \frac{\mathrm{d}\gamma}{\mathrm{d}\theta} - \log \frac{\mathrm{d}\gamma_U}{\mathrm{d}\theta} \\ &= \log \frac{\mathrm{d}\gamma}{\mathrm{d}\theta} + U + \log Z_U.\end{aligned}$$

Step 2. Derive the variational identity. Integrating both sides with respect to γ and using the definition of relative entropy, we obtain

$$\begin{aligned} R(\gamma\|\gamma_U) &= \int \log \frac{\mathrm{d}\gamma}{\mathrm{d}\gamma_U}\,\mathrm{d}\gamma \\ &= \int \log \frac{\mathrm{d}\gamma}{\mathrm{d}\theta}\,\mathrm{d}\gamma + \int U\,\mathrm{d}\gamma + \log Z_U \int \mathrm{d}\gamma. \end{aligned}$$

Since γ is a probability measure, $\int \mathrm{d}\gamma = 1$. Therefore,

$$R(\gamma\|\gamma_U) = R(\gamma\|\theta) + \int U\,\mathrm{d}\gamma + \log Z_U.$$

Rearranging gives

$$\boxed{R(\gamma\|\theta) + \int U\,\mathrm{d}\gamma = R(\gamma\|\gamma_U) - \log Z_U.} \tag{1.34}$$

If $\gamma \not\ll \theta$, then $\gamma \not\ll \gamma_U$, because γ_U and θ are equivalent measures. In this case, both relative entropies are infinite, and (1.34) remains valid in the extended-real sense.

Step 3. Establish the lower bound. The non-negativity of relative entropy implies

$$R(\gamma\|\gamma_U) \geq 0.$$

Substituting this inequality into (1.34), we obtain

$$\boxed{R(\gamma\|\theta) + \int U\,\mathrm{d}\gamma \geq -\log Z_U = -\log \int e^{-U}\,\mathrm{d}\theta.}$$

Step 4. Prove existence and uniqueness of the minimizer. Recall that relative entropy satisfies

$$R(\gamma\|\gamma_U) = 0 \quad \Longleftrightarrow \quad \gamma = \gamma_U.$$

Therefore, equality in the variational inequality holds if and only if $\gamma = \gamma_U$. It follows that the infimum is attained uniquely at $\gamma^* = \gamma_U$, and

$$\boxed{\inf_{\gamma \in \mathcal{P}(E)} \left\{ R(\gamma\|\theta) + \int U\,\mathrm{d}\gamma \right\} = -\log \int e^{-U}\,\mathrm{d}\theta.} \tag{1.35}$$

♦

Connection to the Gibbs variational principle

The variational identity (1.35) shows that minimizing the sum of relative entropy and expected potential energy is equivalent to minimizing the relative entropy with respect to the exponentially tilted measure γ_U.
In particular, the two terms in the functional

$$\mathcal{F}(\gamma) = R(\gamma\|\theta) + \int U \, \mathrm{d}\gamma$$

represent competing contributions: the relative entropy penalizes deviations from the reference measure θ, whereas the potential energy favors states with smaller values of U. Their competition determines the unique minimizing distribution.
More generally, introducing an inverse-temperature parameter $\beta > 0$, the same argument gives

$$\inf_{\gamma \in \mathcal{P}(E)} \left\{ R(\gamma\|\theta) + \beta \int U \, \mathrm{d}\gamma \right\} = -\log \int e^{-\beta U} \, \mathrm{d}\theta.$$

The unique minimizer is the Gibbs measure

$$\frac{\mathrm{d}\gamma_{\beta,U}}{\mathrm{d}\theta} = \frac{e^{-\beta U}}{\displaystyle\int e^{-\beta U} \, \mathrm{d}\theta}.$$

This is the Gibbs variational principle, which provides a variational characterization of equilibrium probability distributions and connects relative entropy to thermodynamic free energy.

Theorem 1.32 (Donsker–Varadhan representation of relative entropy)**.**

Source exercise: Project 11, item 6. *Student Assessment Exercise* [Assessment] (adapted).

Let μ, ν be probability measures on a measurable space $(E, \mathcal{S})$, and let $B_b(E)$ denote the space of bounded measurable real-valued functions on E. Then the relative entropy admits the variational representation

$$D_{\mathrm{KL}}(\mu\|\nu) = \sup_{g \in B_b(E)} \left\{ \int_E g \, \mathrm{d}\mu - \log \int_E e^g \, \mathrm{d}\nu \right\}, \tag{1.36}$$

where $D_{\mathrm{KL}}(\mu\|\nu)$ is understood in the extended-real sense, with $D_{\mathrm{KL}}(\mu\|\nu) = +\infty$ whenever $\mu \not\ll \nu$.
The supremum is attained if and only if μ and ν are equivalent measures and

$$\log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} \in L^\infty(\nu).$$

In this case, all maximizers are characterized by

$$g_* = \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} + c, \quad c \in \mathbb{R}, \tag{1.37}$$

In (1.37)*, equality is understood up to ν-almost-everywhere equality.*

Proof. For any $g \in B_b(E)$, define

$$Z_g := \int_E e^g \, \mathrm{d}\nu, \quad \nu_g(\mathrm{d}x) := \frac{e^{g(x)}}{Z_g} \nu(\mathrm{d}x).$$

Since g is bounded, ν_g is a probability measure equivalent to ν.

If $D_{\mathrm{KL}}(\mu\|\nu) < \infty$, then $\mu \ll \nu$ and

$$\begin{aligned} D_{\mathrm{KL}}(\mu\|\nu_g) &= \int_E \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu_g} \, \mathrm{d}\mu \\ &= \int_E \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} \, \mathrm{d}\mu - \int_E g \, \mathrm{d}\mu + \log Z_g \\ &= D_{\mathrm{KL}}(\mu\|\nu) - \int_E g \, \mathrm{d}\mu + \log Z_g. \end{aligned}$$

By the non-negativity of relative entropy,

$$\int_E g \, \mathrm{d}\mu - \log Z_g \le D_{\mathrm{KL}}(\mu\|\nu).$$

The same inequality holds trivially when $D_{\mathrm{KL}}(\mu\|\nu) = +\infty$. Therefore,

$$\sup_{g \in B_b(E)} \left\{ \int_E g \, \mathrm{d}\mu - \log \int_E e^g \, \mathrm{d}\nu \right\} \le D_{\mathrm{KL}}(\mu\|\nu). \tag{1.38}$$

To establish the reverse inequality, suppose first that $\mu \ll \nu$, and write $r := \mathrm{d}\mu / \mathrm{d}\nu$. For each $n \in \mathbb{N}$, define

$$g_n(x) := \log \left(\min \left\{ \max\{r(x), e^{-n}\}, e^n \right\} \right).$$

Then $g_n \in B_b(E)$. Since

$$e^{g_n(x)} = \min \left\{ \max\{r(x), e^{-n}\}, e^n \right\} \longrightarrow r(x),$$

and $e^{g_n} \le r + 1$, the dominated convergence theorem gives

$$\int_E e^{g_n} \, \mathrm{d}\nu \longrightarrow \int_E r \, \mathrm{d}\nu = 1.$$

Consequently,

$$\log \int_E e^{g_n} \, \mathrm{d}\nu \longrightarrow 0.$$

Moreover,

$$\int_E g_n \, \mathrm{d}\mu = \int_E r(x) \log \left(\min \left\{ \max\{r(x), e^{-n}\}, e^n \right\} \right) \, \mathrm{d}\nu.$$

On $\{r > 1\}$, the integrand increases monotonically to $r \log r$. On $\{0 < r \leq 1\}$, it converges pointwise to $r \log r$ and its absolute value is bounded by

$$-r \log r \leq e^{-1}.$$

Thus, by the monotone and dominated convergence theorems, respectively,

$$\int_E g_n \, \mathrm{d}\mu \longrightarrow \int_E r \log r \, \mathrm{d}\nu = D_{\mathrm{KL}}(\mu \| \nu),$$

where the limit is allowed to be $+\infty$. It follows that

$$\sup_{g \in B_b(E)} \left\{ \int_E g \, \mathrm{d}\mu - \log \int_E e^g \, \mathrm{d}\nu \right\} \geq D_{\mathrm{KL}}(\mu \| \nu). \tag{1.39}$$

If $\mu \not\ll \nu$, there exists $A \in \mathcal{S}$ such that $\nu(A) = 0$ and $\mu(A) > 0$. Taking $g_n = n\mathbf{1}_A$, we obtain

$$\begin{aligned} \int_E g_n \, \mathrm{d}\mu - \log \int_E e^{g_n} \, \mathrm{d}\nu &= n\mu(A) - \log 1 \\ &= n\mu(A) \longrightarrow +\infty. \end{aligned}$$

Hence the variational identity also holds in this case. Combining (1.38) and (1.39) establishes (1.36). To characterize the maximizers, define

$$J(g) := \int_E g \, \mathrm{d}\mu - \log \int_E e^g \, \mathrm{d}\nu.$$

Suppose that the supremum is attained at some $g_* \in B_b(E)$. For any $h \in B_b(E)$, consider the perturbation $g_\varepsilon = g_* + \varepsilon h$. Since g_* is a maximizer, the first variation must vanish:

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} J(g_* + \varepsilon h) \bigg|_{\varepsilon = 0} = 0.$$

Differentiating gives

$$\int_E h \, \mathrm{d}\mu = \frac{\int_E h e^{g_*} \, \mathrm{d}\nu}{\int_E e^{g_*} \, \mathrm{d}\nu}.$$

Since this identity holds for every bounded measurable function h, the corresponding probability measures must coincide:

$$\mu(\,\mathrm{d}x) = \frac{e^{g_*(x)}}{Z_{g_*}} \, \nu(\,\mathrm{d}x).$$

Consequently,

$$\frac{\mathrm{d}\mu}{\mathrm{d}\nu} = \frac{e^{g_*}}{Z_{g_*}}.$$

Since g_* is bounded, its exponential is bounded above and bounded away from zero. Therefore, μ and ν are equivalent measures and

$$\log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} = g_* - \log Z_{g_*} \in L^\infty(\nu).$$

Conversely, suppose that μ and ν are equivalent and

$$\log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} \in L^\infty(\nu).$$

Choosing a bounded representative of

$$g_* := \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu},$$

we have

$$\int_E e^{g_*} \, \mathrm{d}\nu = \int_E \frac{\mathrm{d}\mu}{\mathrm{d}\nu} \, \mathrm{d}\nu = 1.$$

Consequently,

$$\begin{aligned} J(g_*) &= \int_E \log \frac{\mathrm{d}\mu}{\mathrm{d}\nu} \, \mathrm{d}\mu - \log \int_E e^{g_*} \, \mathrm{d}\nu \\ &= D_{\mathrm{KL}}(\mu \| \nu). \end{aligned}$$

Thus the supremum is attained.

Finally, the identity

$$\mu(\,\mathrm{d}x) = \frac{e^{g_*(x)}}{Z_{g_*}} \, \nu(\,\mathrm{d}x)$$

shows that every maximizer differs from $\log(\,\mathrm{d}\mu/\,\mathrm{d}\nu)$ only by an additive constant. Conversely, adding any constant to a maximizer leaves the objective unchanged, since

$$\begin{aligned} J(g+c) &= \int_E (g+c) \, \mathrm{d}\mu - \log \int_E e^{g+c} \, \mathrm{d}\nu \\ &= \int_E g \, \mathrm{d}\mu - \log \int_E e^{g} \, \mathrm{d}\nu \\ &= J(g). \end{aligned}$$

This completes the proof. □

Corollary 1.33 (Full Fenchel–Legendre duality for the log-partition functional)**.** *Fix a probability measure Q on $(E, \mathcal{S})$ and define, for $g \in B_b(E)$,*

$$\Lambda_Q(g) := \log \int_E e^g \, \mathrm{d}Q. \tag{1.40}$$

The log-partition functional (1.40) *has the following properties:*

1. ***Convexity.*** *The functional Λ_Q is convex in g, and $P \mapsto D_{\mathrm{KL}}(P\|Q)$ is convex in P.*
2. ***First conjugacy relation.*** *For every $P \in \mathcal{P}(E)$,*

$$D_{\mathrm{KL}}(P\|Q) = \sup_{g\in B_b(E)} \left\{ \int_E g \, \mathrm{d}P - \Lambda_Q(g) \right\}. \tag{1.41}$$

3. ***Second conjugacy relation.*** *For every $g \in B_b(E)$,*

$$\Lambda_Q(g) = \sup_{P\in\mathcal{P}(E)} \left\{ \int_E g \, \mathrm{d}P - D_{\mathrm{KL}}(P\|Q) \right\}. \tag{1.42}$$

The supremum is attained uniquely at the exponentially tilted probability measure

$$P_g(\,\mathrm{d}x) = \frac{e^{g(x)}}{\int_E e^g \,\mathrm{d}Q} \, Q(\,\mathrm{d}x). \tag{1.43}$$

Proof. We first prove that Λ_Q is convex. Let $g_1, g_2 \in B_b(E)$ and $\alpha \in [0,1]$. By the definition of Λ_Q,

$$\begin{aligned} &\Lambda_Q\big(\alpha g_1 + (1-\alpha)g_2\big) \\ &= \log \int_E (e^{g_1})^\alpha (e^{g_2})^{1-\alpha} \,\mathrm{d}Q. \end{aligned}$$

Applying Hölder's inequality gives

$$\int_E (e^{g_1})^\alpha (e^{g_2})^{1-\alpha} \,\mathrm{d}Q \le \left(\int_E e^{g_1} \,\mathrm{d}Q \right)^\alpha \left(\int_E e^{g_2} \,\mathrm{d}Q \right)^{1-\alpha}.$$

Taking logarithms, we obtain

$$\Lambda_Q\big(\alpha g_1 + (1-\alpha)g_2\big) \le \alpha\Lambda_Q(g_1) + (1-\alpha)\Lambda_Q(g_2).$$

Hence Λ_Q is convex. Next, we establish the convexity of the relative entropy. By the Donsker–Varadhan representation,

$$D_{\mathrm{KL}}(P\|Q) = \sup_{g\in B_b(E)} \left\{ \int_E g \, \mathrm{d}P - \Lambda_Q(g) \right\}.$$

For every fixed g, the functional

$$P \longmapsto \int_E g \, \mathrm{d}P - \Lambda_Q(g)$$

is affine in P. Since the pointwise supremum of affine functionals is convex, it follows that

$$P \longmapsto D_{\mathrm{KL}}(P\|Q)$$

is convex on $\mathcal{P}(E)$. The first conjugacy relation,

$$D_{\mathrm{KL}}(P\|Q) = \sup_{g\in B_b(E)} \left\{ \int_E g \, \mathrm{d}P - \Lambda_Q(g) \right\},$$

is precisely the Donsker–Varadhan representation established in Theorem 1.32.

In particular, the KL divergence is the convex conjugate of the log-partition functional Λ_Q with respect to the duality pairing

$$\langle g, P\rangle := \int_E g \, \mathrm{d}P.$$

To prove the second conjugacy relation, fix $g \in B_b(E)$ and write

$$Z_g := \int_E e^g \, \mathrm{d}Q.$$

Define the exponentially tilted probability measure

$$P_g(\,\mathrm{d}x) := \frac{e^{g(x)}}{Z_g} Q(\,\mathrm{d}x).$$

By construction,

$$\int_E P_g(\,\mathrm{d}x) = 1,$$

so $P_g \in \mathcal{P}(E)$. Furthermore, its Radon–Nikodym derivative satisfies

$$\frac{\mathrm{d}P_g}{\mathrm{d}Q} = \frac{e^g}{Z_g},$$

and hence

$$\log \frac{\mathrm{d}P_g}{\mathrm{d}Q} = g - \log Z_g.$$

For any $P \in \mathcal{P}(E)$ with $D_{\mathrm{KL}}(P\|Q) < \infty$, we obtain

$$\begin{aligned} D_{\mathrm{KL}}(P\|P_g) &= \int_E \log \frac{\mathrm{d}P}{\mathrm{d}P_g} \, \mathrm{d}P \\ &= \int_E \log \frac{\mathrm{d}P}{\mathrm{d}Q} \, \mathrm{d}P - \int_E \log \frac{\mathrm{d}P_g}{\mathrm{d}Q} \, \mathrm{d}P \\ &= D_{\mathrm{KL}}(P\|Q) - \int_E g \, \mathrm{d}P + \log Z_g. \end{aligned}$$

Rearranging yields

$$\int_E g \, \mathrm{d}P - D_{\mathrm{KL}}(P\|Q) = \Lambda_Q(g) - D_{\mathrm{KL}}(P\|P_g). \tag{1.44}$$

The identity (1.44) proves the second conjugacy (1.42), with optimizer (1.43); the first con-

jugacy (1.41) is the preceding Donsker–Varadhan representation. Since relative entropy is non-negative, $D_{\mathrm{KL}}(P\|P_g) \geq 0$, we conclude that

$$\int_E g \, \mathrm{d}P - D_{\mathrm{KL}}(P\|Q) \leq \Lambda_Q(g).$$

The same inequality is immediate when $D_{\mathrm{KL}}(P\|Q) = +\infty$. Equality holds if and only if

$$D_{\mathrm{KL}}(P\|P_g) = 0,$$

which is equivalent to $P = P_g$. Moreover, since g is bounded,

$$D_{\mathrm{KL}}(P_g\|Q) = \int_E (g - \log Z_g) \, \mathrm{d}P_g < \infty.$$

Thus P_g is admissible and attains the supremum. Therefore,

$$\Lambda_Q(g) = \sup_{P\in\mathcal{P}(E)} \left\{ \int_E g \, \mathrm{d}P - D_{\mathrm{KL}}(P\|Q) \right\},$$

with the unique maximizer given by $P = P_g$. This completes the proof. □

f-divergences and density-ratio duality. The following definitions support the retained statements of the pending f-divergence and duality problems. Their extended derivations are deferred; the Gibbs and Donsker–Varadhan results above remain fully developed.

Definition 1.34 (f-divergence and convex conjugate)

Let $f : (0,\infty) \to \mathbb{R}$ be convex with $f(1) = 0$. For probability measures $P \ll Q$, with Radon–Nikodym derivative $r = \mathrm{d}P/\mathrm{d}Q > 0$ Q-almost everywhere, define

$$D_f(P\|Q) := \int f(r) \, \mathrm{d}Q. \tag{1.45}$$

The *convex conjugate* (Fenchel–Legendre transform) of f is

$$f^*(u) := \sup_{t>0}\{ut - f(t)\}.$$

The Fenchel–Young inequality reads

$$f(t) + f^*(u) \geq ut, \tag{1.46}$$

with equality when $u \in \partial f(t)$, where $\partial f(t)$ denotes the set of subgradients at t. For differentiable f, this condition becomes $u = f'(t)$.

The pending variational proof below starts from (1.46). The assumption $r > 0$ permits the elementary formulas below without introducing boundary conventions. For general probability measures, the extended f-divergence additionally requires the lower-semicontinuous

extension at $r = 0$ and an appropriate singular-mass term when $P \not\ll Q$; it is *not* mathematically correct to apply (1.45) unchanged in such cases.

Theorem 1.35 (Joint convexity and the variational form of f-divergence)**.**

Source exercise: Project 11, item 9. *Student Assessment Exercise* [Assessment] (adapted).

Let P and Q have strictly positive densities p, q with respect to a common σ-finite measure m. Suppose f is proper, convex, continuously differentiable on $(0, \infty)$ with $f(1) = 0$, and that the displayed integrals exist. Then D_f is jointly convex as a function of (P, Q). Moreover, for every admissible measurable discriminator T whose objective is well defined (no $\infty - \infty$ subtraction),

$$\int T \, \mathrm{d}P - \int f^*(T) \, \mathrm{d}Q \leq D_f(P\|Q).$$

If $T_*(x) = f'(p(x)/q(x))$ is P-integrable and $f^*(T_*)$ is Q-integrable, equality is attained, and therefore

$$\boxed{D_f(P\|Q) = \sup_T \left\{ \mathbb{E}_P[T(X)] - \mathbb{E}_Q[f^*(T(X))] \right\}.}$$

The supremum is over the measurable functions for which the objective is well defined, including T_*. If the candidate fails the stated integrability conditions, a truncation or approximation argument is needed; the attained formula does not cover that case.

Proof. *Proof to be completed.* □

Example 1.36 (KL as an f-divergence)**.**

Problem

Take $f(r) = r \log r$.

To be completed.

Theorem 1.37 (An exponential Fenchel dual for KL)**.**

Source exercise: Project 11, item 10. *Student Assessment Exercise* [Assessment] (adapted).

For probability measures P, Q,

$$\boxed{D_{\mathrm{KL}}(P\|Q) = 1 + \sup_{u \in B_b(E)} \left\{ \mathbb{E}_P[u] - \mathbb{E}_Q[e^u] \right\}.} \tag{1.47}$$

The identity includes infinite entropy. A bounded maximizer exists precisely when $P \sim Q$ and $\log(\mathrm{d}P/\mathrm{d}Q)$ is bounded; it is then $u_* = \log(\mathrm{d}P/\mathrm{d}Q)$, uniquely Q-almost everywhere.

Proof. *Proof to be completed.* □

Problem — Deferred finite-state duality extension

For strictly positive probability vectors, relate the exponential Fenchel dual to the scalar conjugate of the exponential, and justify the associated minimax equality. Distinguish the domains of the log-partition and exponential-integral functionals.

To be completed.

Proposition 1.38 (Optimizing an additive constant recovers Donsker–Varadhan)**.**

Source exercise: Project 11, item 11. *Student Assessment Exercise* [Assessment] (adapted).

For every bounded measurable g and probability law Q, let $Z_g = \int e^g \, \mathrm{d}Q > 0$. Then

$$\sup_{c \in \mathbb{R}} \left\{ 1 + \mathbb{E}_P[g + c] - \mathbb{E}_Q[e^{g+c}] \right\} = \mathbb{E}_P[g] - \log Z_g. \tag{1.48}$$

Consequently, (1.48) *implies that taking the supremum over g in the exponential representation* (1.47) *yields precisely* (1.36).

Proof. *Proof to be completed.* □

1.2.4.5 Density-ratio estimation

Theorem 1.39 (Logistic classification recovers a density ratio)**.**

Source exercise: Project 11, item 12. *Student Assessment Exercise* [Assessment] (adapted).

Let P, Q have strictly positive Lebesgue densities p, q on $V = \mathbb{R}^d$. For a measurable score $g : V \to \mathbb{R}$ define the equally weighted logistic population risk

$$L_{\log}(g) := \mathbb{E}_P[\log(1 + e^{-g(X)})] + \mathbb{E}_Q[\log(1 + e^{g(X)})].$$

If $g_*(x) = \log(p(x)/q(x))$ is in the allowed function class and the risk is finite, it minimizes $L_{\log}$, uniquely up to $(P + Q)$-null sets. For this unrestricted population problem,

$$g_*(x) = \log \frac{p(x)}{q(x)}, \qquad \frac{1}{1 + e^{-g_*(x)}} = \frac{p(x)}{p(x) + q(x)}.$$

The minimum risk equals $2 \log 2 - 2\,\mathrm{JS}(P, Q)$, where

$$\mathrm{JS}(P, Q) = \tfrac{1}{2} D_{\mathrm{KL}}(P \| M) + \tfrac{1}{2} D_{\mathrm{KL}}(Q \| M), \quad M = \tfrac{1}{2}(P + Q).$$

Proof. *Proof to be completed.* □

What density-ratio estimation does and does not prove

The logistic score $g_* = \log(p/q)$ is a *population* solution when an unrestricted measurable discriminator is available. A neural network of restricted capacity or a finite training sample need not recover this function exactly. On regions where only one density is positive, the optimal log ratio is extended-valued, so a finite everywhere optimizer need not exist. The identity is therefore a mathematical target for learning, not a finite-data guarantee.

1.2.4.6 Entropy-regularized measure optimization

The final item of Project 11 is retained as a pending extension from Gibbs tilting to optimization over probability laws.

Theorem 1.40 (Reward-guided mean-field exponential tilt)**.**

Source exercise: Project 11, item 13. *Student Assessment Exercise* [Assessment] (adapted).

Let E be a Polish space and ν a fixed Borel probability law. Suppose $\mathcal{R} : \mathcal{P}(E) \to \mathbb{R}$ is concave, bounded above, and upper semicontinuous for weak convergence. Then

$$\sup_{\mu \in \mathcal{P}(E)} \{\mathcal{R}(\mu) - D_{\mathrm{KL}}(\mu \| \nu)\}$$

admits a unique maximizer $\mu_* \ll \nu$. Suppose additionally μ_* has a strictly positive density $r_* = \mathrm{d}\mu_*/\mathrm{d}\nu$, the differentiability conditions below hold, and $\mathcal{R}$ has a functional derivative $\delta\mathcal{R}/\delta\mu(x;\mu_*)$ characterized by

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\bigg|_{\varepsilon=0} \mathcal{R}((1+\varepsilon\eta)\mu_*) = \int \frac{\delta\mathcal{R}}{\delta\mu}(x;\mu_*)\eta(x)\,\mu_*(\mathrm{d}x)$$

for every bounded η with $\int \eta \,\mathrm{d}\mu_* = 0$ and sufficiently small $|\varepsilon|$. Assume the derivative and $\log r_*$ are integrable against μ_*. Then the maximizer satisfies

$$\begin{aligned} \mu_*(\mathrm{d}x) &= \frac{\exp\left(\frac{\delta\mathcal{R}}{\delta\mu}(x;\mu_*)\right)}{Z(\mu_*)}\,\nu(\mathrm{d}x), \\ Z(\mu_*) &:= \int \exp\left(\frac{\delta\mathcal{R}}{\delta\mu}(x;\mu_*)\right)\nu(\mathrm{d}x). \end{aligned} \tag{1.49}$$

In (1.49), the functional derivative is only defined up to an additive constant, which is absorbed by $Z(\mu_*)$.

Proof. *Proof to be completed.* □

1.3 Conditional Probability and Effective Dynamics

The previous section transported a law by a prescribed state flow. Coarse-graining instead asks what remains predictable after only selected observables are retained. Before conditioning, fix the microscopic stochastic model used in the projection and learned-flow arguments below. For inverse temperature $\beta > 0$, a C^2 potential $\mathcal{V} : \mathbb{R}^d \to \mathbb{R}$, and a standard d-dimensional Brownian motion independent of X_0, take

$$\mathrm{d}X_t = -\nabla\mathcal{V}(X_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t. \tag{1.50}$$

Assume local Lipschitz drift, non-explosion, and the integrability stipulated in each result below. On test functions $\varphi \in C_c^2(\mathbb{R}^d)$ its generator is $\mathcal{L}\varphi = -\nabla\mathcal{V}\cdot\nabla\varphi + \beta^{-1}\Delta\varphi$. If $\mu_t = \mathrm{Law}(X_t)$ has density ρ_t, its forward equation is

$$\partial_t\rho_t = \nabla\cdot(\rho_t\nabla\mathcal{V}) + \beta^{-1}\Delta\rho_t, \qquad \rho_{t=0} = \rho_0. \tag{1.51}$$

The equation is distributional for integrable densities, and classical under sufficient parabolic regularity; integrations over $\mathbb{R}^d$ require vanishing flux at infinity. It follows by Itô's formula and duality, as developed with explicit periodic domains in Section 2.1. The static Gibbs laws used below are reference measures when their partition functions are finite; time-dependent conditional kernels refer to the actual law of (1.50). Thus a stationary conditional average and a time-dependent projected coefficient need not agree.

Here the ambient Euclidean state space is $V = \mathbb{R}^d$, and the reduced space is $\mathcal{Z} = \mathbb{R}^m$, with $d \geq m \geq 1$; both carry their Borel sigma-algebras. We first apply measurable pushforward and conditioning, then impose the smoothness needed for level-set geometry.

Collective variables and the limits of marginal information. A physical description often retains a small number of quantities rather than the entire microscopic configuration. A measurable map $\xi : V \to \mathcal{Z}$, with $\mathcal{Z}$ typically lower-dimensional, is called a *collective variable* or *reaction coordinate* when its values represent the chosen coarse description. If $Z_t = \xi(X_t)$ and $\mu_t = \mathrm{Law}(X_t)$, then

$$\nu_t := \mathrm{Law}(Z_t) = \xi_{\#}\mu_t,$$
$$\mathbb{E}[g(Z_t)] = \int_V g(\xi(x))\,\mu_t(\,\mathrm{d}x) = \int_{\mathcal{Z}} g(z)\,\nu_t(\,\mathrm{d}z).$$

If ν_t has a Lebesgue density, denote it by $p_{\xi,t}$. This is an exact statement about one-time distributions. It does not imply that (Z_t) admits a closed Markov dynamics. Indeed, two models may have identical one-time marginals while differing in correlations or transition statistics. Finite-dimensional agreement, meaning equality of all joint laws at finitely many times, is stronger. Pathwise agreement under a specified coupling, meaning equality of both sample paths almost surely, is stronger still. For continuous-path processes, equality of all finite-dimensional distributions does determine equality of laws on the canonical continuous-path space, but it does not imply sample-path equality under an arbitrary coupling.

The missing microscopic information can be encoded by conditional laws. For suitable integrable observables A, the conditional expectation $\mathbb{E}[A(X_t) \mid \xi(X_t)]$ records what remains predictable from the coarse variable alone. Because Euclidean state spaces are standard Borel spaces, such conditional expectations admit regular conditional probability representations, unique up to null sets. The geometric construction on level sets, effective free energies, and the problem of dynamical closure will be developed only after their additional assumptions have been stated.

1.3.1 Conditional laws and geometric disintegration

For geometric calculations, strengthen the measurable map to $\xi : \mathbb{R}^d \to \mathbb{R}^m$ of class C^2 with rank $D\xi(x) = m$ everywhere in the region of interest. For a probability measure μ, write $\nu = \xi_{\#}\mu$. The disintegration theorem on standard Borel spaces gives a family of probability kernels μ_z, defined for ν-almost every z, such that

$$\int_{\mathbb{R}^d} f(x)g(\xi(x))\mu(\,\mathrm{d}x) = \int_{\mathbb{R}^m} g(z)\left[\int_{\mathbb{R}^d} f(x)\mu_z(\,\mathrm{d}x)\right]\nu(\,\mathrm{d}z). \tag{1.52}$$

The measure μ_z is supported on the level set $\mathcal{M}_z = \{x : \xi(x) = z\}$ for almost every z. The conditioning symbol $\mu(\,\mathrm{d}x \mid \xi(x) = z)$ represents this measure kernel, not an ordinary Lebesgue density on $\mathbb{R}^d$.

Example 1.41 (Gaussian conditional law for a coordinate average)**.**

Source exercise: Project 12, item 1. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let $X \sim \mathcal{N}(0, I_d)$ and $\xi(X) = d^{-1}\sum_{j=1}^{d} X^j =: Z$.

Solution.

$$\begin{aligned} Z &\sim \mathcal{N}(0, d^{-1}), \\ \mathrm{Law}(X \mid Z = z) &= \mathcal{N}_{\mathcal{M}_z}\left(z\mathbf{1}, I_d - \tfrac{1}{d}\mathbf{1}\mathbf{1}^\top\right). \end{aligned} \tag{1.53}$$

The second Gaussian is *degenerate* and supported on $\{x : \mathbf{1}^\top x = nz\}$. Decompose X into its orthogonal projections on span$\{\mathbf{1}\}$ and its orthogonal complement: $X = Z\mathbf{1} + (I - d^{-1}\mathbf{1}\mathbf{1}^\top)X$. The two components are independent jointly Gaussian vectors. Their means and covariances yield (1.53). For this linear coordinate, $J_\xi = \|\nabla\xi\| = 1/\sqrt{d}$, so the delta measure in the source is

$$\delta(\xi(x) - z)\,\mathrm{d}x = \sqrt{d}\;\mathrm{d}\mathcal{H}^{d-1}(x)\big|_{\mathcal{M}_z}.$$

This is an identity under integration, not a probability law: weighting by the Gaussian density and normalizing produces the conditional law. The general justification is the coarea formula below. ♦

1.3.1.1 Noisy conditioning and the sharp-level limit

Source exercise: Project 12, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Problem — Noisy conditioning

Let $X \sim \mathcal{N}(0, I_d)$, $Z = d^{-1}\mathbf{1}^\top X$, and $Y_\sigma = Z + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ is independent of X. Determine the conditional law of X given $Y_\sigma = z$ and study its sharp-level limit. For a general collective variable ξ, formulate the analogous noisy posterior and the conditions needed for its limit.

To be completed.

1.3.1.2 Coarea representation of the marginal density

For $x \in \mathbb{R}^d$ define the positive-definite Gram matrix $G_\xi(x) = D\xi(x)D\xi(x)^\top \in \mathbb{R}^{m\times m}$ and $J_\xi(x) = \sqrt{\det G_\xi(x)}$. Write $\mathcal{H}^{d-m}\restriction_{\mathcal{M}_z}$ for the $(d-m)$-dimensional Hausdorff surface measure on $\mathcal{M}_z$.

Theorem 1.42 (Coarea disintegration and potential of mean force)**.**

Source exercise: Project 12, item 3. *Student Assessment Exercise* [Assessment] (adapted).

Suppose $\mu(\mathrm{d}x) = Z^{-1}e^{-\beta\mathcal{V}(x)}\,\mathrm{d}x$, where $\beta > 0$, $0 < Z < \infty$, and the full-rank condition holds on the support under consideration. Then $\nu = \xi_\#\mu$ has (almost-everywhere defined) density

$$p_\xi(z) = \frac{1}{Z}\int_{\mathcal{M}_z} \frac{e^{-\beta\mathcal{V}(x)}}{J_\xi(x)}\,\mathrm{d}\mathcal{H}^{d-m}(x),$$

For almost every z with $0 < p_\xi(z) < \infty$, the conditional kernel is the normalized surface measure

$$\mu_z(\mathrm{d}x) = \frac{e^{-\beta\mathcal{V}(x)}J_\xi(x)^{-1}}{Zp_\xi(z)}\,\mathrm{d}\mathcal{H}^{d-m}(x), \quad x \in \mathcal{M}_z. \tag{1.54}$$

At points with $0 < p_\xi(z) < \infty$, the free energy of the coordinate is

$$F_\xi(z) = -\beta^{-1}\log p_\xi(z). \tag{1.55}$$

Proof. The coarea formula applied to the integrable function $e^{-\beta\mathcal{V}(x)}f(x)g(\xi(x))/Z$ gives

$$\begin{aligned}&\int_{\mathbb{R}^d} f(x)g(\xi(x))\mu(\mathrm{d}x)\\ &\quad= \int_{\mathbb{R}^m} g(z)\left[\frac{1}{Z}\int_{\mathcal{M}_z}\frac{f(x)e^{-\beta\mathcal{V}(x)}}{J_\xi(x)}\,\mathrm{d}\mathcal{H}^{d-m}(x)\right]\mathrm{d}z.\end{aligned}$$

Set $f = 1$ to identify the marginal density, then compare the expression with (1.52) to identify

the conditional kernel. Equation (1.55) is a definition; its additive constant depends on the density convention and normalization. □

Physical meaning of free energy

The potential F_ξ is associated with a *probability measure on the chosen coordinate space*. It is not simply the microscopic potential $\mathcal{V}$ evaluated on a representative configuration. Entropic contributions arise from the number and geometry of microscopic states compatible with $\xi(x) = z$. Changing coordinates changes the reference volume element, and hence changes the density-based free-energy function.

1.3.1.3 The mean force from conditional expectation

Let G_ξ^{-1} denote the inverse Gram matrix and define vector fields

$$B_i^\xi(x) = \sum_{j=1}^{m} [G_\xi(x)^{-1}]_{ij} \nabla \xi^j(x), \quad D\xi(x) B_i^\xi(x) = e_i.$$

Theorem 1.43 (Mean-force identity)**.**

Source exercise: Project 12, item 4. *Student Assessment Exercise* [Assessment] (adapted).

Assume that $\mathcal{V}, \xi$ are sufficiently smooth, the density p_ξ is positive, finite, and differentiable almost everywhere on the region considered. Assume $B_i^\xi \in C^1$, $\int (\|B_i^\xi\| + |B_i^\xi \cdot \nabla \mathcal{V}| + |\nabla \cdot B_i^\xi|)\, \mathrm{d}\mu < \infty$, and vanishing boundary or cutoff terms in the following integrations by parts. Then for almost every such z,

$$\partial_{z^i} F_\xi(z) = \mathbb{E}_\mu \left[B_i^\xi \cdot \nabla \mathcal{V} - \beta^{-1} \nabla \cdot B_i^\xi \,\middle|\, \xi(X) = z \right]. \tag{1.56}$$

Proof. For a compactly supported smooth φ on $\mathbb{R}^m$, $B_i^\xi \cdot \nabla(\varphi \circ \xi) = \partial_{z^i} \varphi(\xi)$. Integrating against $Z^{-1} e^{-\beta \mathcal{V}(x)}\, \mathrm{d}x$ and using integration by parts yields

$$\begin{aligned} \int \partial_{z^i} \varphi(z) p_\xi(z)\, \mathrm{d}z &= \int B_i^\xi \cdot \nabla(\varphi \circ \xi) \mu(\mathrm{d}x) \\ &= \int \varphi(\xi(x)) [\beta B_i^\xi \cdot \nabla \mathcal{V} - \nabla \cdot B_i^\xi] \mu(\mathrm{d}x). \end{aligned}$$

On the other hand, this equals $-\int \varphi(z) \partial_{z^i} p_\xi(z)\, \mathrm{d}z$. Disintegration identifies $-\partial_{z^i} \log p_\xi$ as the conditional expectation in brackets. Multiplication by β^{-1} gives the result. □

Later thought — Geometry and conditional drift

To revisit. Connect Projects 12.3–12.4 with Project 13.1: for a nonlinear coordinate, track the normalization and geometric terms from conditional laws to the mean force.
Idea. Start with a regular two-dimensional coordinate and compare (1.54) with (1.56).
Status: pending; no worked solution here.

1.3.2 Effective dynamics and the limits of Markovian closure

For a time-dependent law, write $\mu_{t,z} := \mathrm{Law}(X_t \mid \xi(X_t) = z)$ for a selected regular conditional kernel; μ_z below denotes the static reference kernel. The marginal-projection viewpoint is closely related to the mimicking theorem of Gyöngy [Gyö86].

1.3.2.1 Conditional least squares and total covariance

The conditional expectation $\mathbb{E}[Y \mid \mathcal{G}]$ is the orthogonal projection of any $Y \in L^2(\Omega)$ onto the closed subspace $L^2(\mathcal{G})$. This interpretation is the basis for learning effective coefficients by regression.

Proposition 1.44 (Law of total covariance)**.**

Source exercise: Project 13, item 1. *Student Assessment Exercise* [Assessment] (adapted).

For a square-integrable random vector X and $Z = \xi(X)$,

$$\mathrm{Cov}(X) = \mathrm{Cov}(\mathbb{E}[X \mid Z]) + \mathbb{E}[\mathrm{Cov}(X \mid Z)].$$

Proof. Write $X - \mathbb{E}[X] = [\mathbb{E}(X \mid Z) - \mathbb{E}X] + [X - \mathbb{E}(X \mid Z)]$. The mixed terms vanish after conditioning on Z. Expand the outer product and take expectations. □

Remark 1.45 (The two coordinate-choice illustrations in Project 13.1)**.** For a centered non-degenerate Gaussian X with covariance Σ, take $Z = u^\top X$ with $\|u\| = 1$. The conditional reconstruction error is

$$\mathbb{E}[X \mid Z] = \frac{\Sigma u}{u^\top \Sigma u} Z,$$
$$\mathbb{E}\|X - \mathbb{E}[X \mid Z]\|^2 = \mathrm{tr}\,\Sigma - \frac{u^\top \Sigma^2 u}{u^\top \Sigma u}.$$

Since $u^\top \Sigma^2 u \le \lambda_{\max}(\Sigma) u^\top \Sigma u$, a top eigenvector minimizes this error. For the additional double-well illustration, consider $\mathcal{V}(x^1, x^2) = ((x^1)^2 - 1)^2/4 + (x^2)^2/2$ and compare retaining x^1 or x^2 under reconstruction-error and inter-well-transition criteria.
To be completed.

Conditional regression supplies the instantaneous predictor in Project 13.1. Project 13.2 now identifies the dynamics of the chosen coordinate through Itô calculus.

1.3.2.2 The exact projected semimartingale equation

For the scalar projection, retain the ambient dimension d and take the reduced dimension to be one. For $\xi : \mathbb{R}^d \to \mathbb{R}$ of class C^2, apply Itô calculus to $Z_t^0 = \xi(X_t)$ where X_t solves (1.50).

Theorem 1.46 (Itô formula for a collective variable)**.**

Source exercise: Project 13, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Assume sufficient regularity and non-explosion. Then

$$\begin{aligned} \mathrm{d}Z_t^0 &= A_\xi(X_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\nabla\xi(X_t)\cdot\,\mathrm{d}W_t, \\ A_\xi(x) &= -\nabla\mathcal{V}(x)\cdot\nabla\xi(x) + \beta^{-1}\Delta\xi(x). \end{aligned}$$

The instantaneous quadratic variation is $\mathrm{d}[Z^0]_t = 2\beta^{-1}\|\nabla\xi(X_t)\|^2\,\mathrm{d}t$.

Proof. Write the microscopic equation in coordinates: $\mathrm{d}X_t^i = -\partial_i\mathcal{V}(X_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t^i$. The multidimensional Itô formula says

$$\mathrm{d}\xi(X_t) = \sum_i \partial_i\xi(X_t)\,\mathrm{d}X_t^i + \frac{1}{2}\sum_{i,j}\partial_{ij}\xi(X_t)\,\mathrm{d}[X^i,X^j]_t.$$

Because the Brownian components are independent, $\mathrm{d}[X^i,X^j]_t = 2\beta^{-1}\delta_{ij}\,\mathrm{d}t$. Substitution gives

$$\begin{aligned} \mathrm{d}\xi(X_t) &= -\sum_i \partial_i\xi\,\partial_i\mathcal{V}\;\mathrm{d}t + \sqrt{2\beta^{-1}}\sum_i\partial_i\xi\;\mathrm{d}W_t^i + \beta^{-1}\sum_i\partial_{ii}\xi\;\mathrm{d}t \\ &= \big(-\nabla\mathcal{V}\cdot\nabla\xi + \beta^{-1}\Delta\xi\big)(X_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}}\nabla\xi(X_t)\cdot\,\mathrm{d}W_t. \end{aligned}$$

The martingale part has quadratic variation $2\beta^{-1}\int_0^t\|\nabla\xi(X_s)\|^2\,\mathrm{d}s$, so its instantaneous variance coefficient is the *squared magnitude* $\|\nabla\xi\|^2$. Averaging the amplitude $\|\nabla\xi\|$ instead would not preserve this quadratic-variation identity. □

To represent the projected noise by a scalar Brownian motion, choose a measurable unit vector $e(x) = \nabla\xi(x)/\|\nabla\xi(x)\|$ where the gradient is nonzero, and any fixed unit vector where it vanishes. Then $W_t^\xi = \int_0^t e(X_s)\cdot\,\mathrm{d}W_s$ is a continuous martingale with quadratic variation t, hence a Brownian motion. The noise term can be written $\sqrt{2\beta^{-1}}\|\nabla\xi(X_t)\|\,\mathrm{d}W_t^\xi$. This representation does not close the drift or diffusion in $\xi(X_t)$.

1.3.2.3 A Markov model with the same one-time marginals

Assume on each finite time horizon that

$$\mathbb{E}\int_0^T\big(|A_\xi(X_t)| + \|\nabla\xi(X_t)\|^2\big)\,\mathrm{d}t < \infty.$$

This makes the projected drift integrable and the martingale term square integrable for the expectation calculation below. Write $\nu_t = \mathrm{Law}(Z_t^0)$ and choose jointly Borel versions of the conditional moments (initially specified $\mathrm{d}t\,\nu_t(\,\mathrm{d}z)$-almost everywhere):

$$\begin{aligned} b_t(z) &= \mathbb{E}[A_\xi(X_t)\mid\xi(X_t)=z], \\ a_t(z) &= \mathbb{E}[\|\nabla\xi(X_t)\|^2\mid\xi(X_t)=z]. \end{aligned} \tag{1.57}$$

The two coefficients in (1.57) track the first moment and the quadratic variation: averaging the noise *amplitude* is generally not equivalent to averaging its square.

Theorem 1.47 (Projected time marginals)**.**

Source exercise: Project 13, item 3. *Student Assessment Exercise* [Assessment] (adapted).

Extend the selected families b_t, a_t to jointly Borel coefficients, with $a_t \geq 0$, and suppose they admit a non-explosive Markov weak solution of

$$\mathrm{d}Y_t = b_t(Y_t)\,\mathrm{d}t + \sqrt{2\beta^{-1}a_t(Y_t)}\,\mathrm{d}W_t^{\mathrm{red}}, \quad \mathrm{Law}(Y_0) = \mathrm{Law}(\xi(X_0)), \tag{1.58}$$

with a scalar standard Brownian motion W^{red} and the corresponding drift and variance integrability. Assume uniqueness of the forward Fokker–Planck equation in a class containing both the law of this solution and ν_t. Then, for all times covered by these assumptions,

$$\mathrm{Law}(Y_t) = \mathrm{Law}(\xi(X_t)).$$

The Markov property of the selected weak solution is part of the assumption; for example, it follows from a well-posed martingale problem, but it is not a consequence of the marginal calculation alone. This does not assert equality of path laws.

Proof. For each $\varphi \in C_c^\infty(\mathbb{R})$, Theorem 1.46 and conditional expectation show that the following identity holds for almost every t, with an absolutely continuous left-hand side:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\nu_t(\mathrm{d}z) = \int \left[b_t(z)\varphi'(z) + \beta^{-1}a_t(z)\varphi''(z)\right]\nu_t(\mathrm{d}z).$$

The law of (1.58) satisfies exactly the same weak equation and initial condition. By the stipulated uniqueness of the forward problem their laws coincide at each t. □

1.4 Learned Probability Flows and Structure-Preserving Approximation

1.4.1 Probability-flow velocities and conditional learning

We begin with velocity fields that represent a prescribed marginal evolution. Probability current and stochastic interpolation provide two constructions; conditional regression then supplies a learning objective for such coefficients.

1.4.1.1 The probability-current velocity

Assume $\rho_t(x) > 0$ is a sufficiently smooth density solving the Fokker–Planck equation (1.51). Define the probability current $\boldsymbol{j}_t(x) = -\rho_t(x)\nabla\mathcal{V}(x) - \beta^{-1}\nabla\rho_t(x)$ and its current

velocity, defined only on this positive-density region,

$$v_t^{\mathrm{cur}}(x) := \frac{\boldsymbol{j}_t(x)}{\rho_t(x)} = -\nabla \mathcal{V}(x) - \beta^{-1} \nabla \log \rho_t(x).$$

Then $\partial_t \rho_t + \nabla \cdot (\rho_t v_t^{\mathrm{cur}}) = 0$. This identity links stochastic evolution to the deterministic continuity equation in Section 1.2. If the velocity generates a well-defined flow with suitable regularity, it can transport the same *one-time densities* as the diffusion. It does not reproduce Brownian sample paths, quadratic variation, or their multi-time laws.

1.4.1.2 Stochastic interpolation and conditional velocity

Let (X_0, X_1) have a joint law γ on $\mathbb{R}^d \times \mathbb{R}^d$, with prescribed marginal laws μ_0, μ_1. Suppose α, λ are C^1 functions on $[0,1]$ with $\alpha_0 = 1, \alpha_1 = 0, \lambda_0 = 0, \lambda_1 = 1$, and set

$$I_t = \alpha_t X_0 + \lambda_t X_1.$$

The joint law γ is a *coupling*; its choice affects the path $t \mapsto \mathrm{Law}(I_t)$ and the velocity required to transport it.

Example 1.48 (Exact Gaussian interpolation: single and two-time laws)**.**

Source exercise: Project 14, item 1. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Assume X_0 and X_1 are independent standard Gaussian vectors in $\mathbb{R}^d$.

Solution.

Then $I_t = \alpha_t X_0 + \lambda_t X_1$ is centered Gaussian, and independence makes the covariance calculation explicit:

$$\mathrm{Cov}(I_t) = (\alpha_t^2 + \lambda_t^2) I_d,$$
$$\mathrm{Cov}(I_s, I_t) = (\alpha_s \alpha_t + \lambda_s \lambda_t) I_d.$$

Indeed, expand $\mathbb{E}[I_s I_t^\top]$ into four terms. The cross terms vanish because the endpoint Gaussians are independent and centered, while $\mathbb{E}[X_i X_i^\top] = I_d$. For the linear interpolation $\alpha_t = 1-t$, $\lambda_t = t$, we find $I_t \sim \mathcal{N}(0, [(1-t)^2 + t^2] I_d)$, which is narrower at $t = 1/2$ than at either endpoint. Thus even an interpolation between identical endpoint marginals need not have constant intermediate marginals, and the coupling is mathematically consequential. ♦

Theorem 1.49 (Conditional interpolation velocity)**.**

Source exercise: Project 14, item 2. *Student Assessment Exercise* [Assessment] (adapted).

Assume $\mathbb{E}(\|X_0\| + \|X_1\|) < \infty$*, and let* $\mu_t = \mathrm{Law}(I_t)$*. Choose a jointly Borel conditional velocity, defined* $\mathrm{d}t\,\mu_t(\mathrm{d}z)$*-almost everywhere, by*

$$v_t^{\mathrm{int}}(z) = \mathbb{E}[\dot{\alpha}_t X_0 + \dot{\lambda}_t X_1 \mid I_t = z]. \tag{1.59}$$

Then the measure-valued continuity equation

$$\partial_t \mu_t + \nabla \cdot (v_t^{\mathrm{int}} \mu_t) = 0$$

holds in distributions on $(0,1) \times \mathbb{R}^d$*. Only if* μ_t *admits a Lebesgue density do we write* $\mu_t(\mathrm{d}z) = \rho_t(z)\,\mathrm{d}z$ *and obtain* $\partial_t \rho_t + \nabla \cdot (\rho_t v_t^{\mathrm{int}}) = 0$ *as a density equation.*

Proof. The C^1 coefficient functions and the first-moment hypothesis give $\mathbb{E}\int_0^1 \|\dot{I}_t\|\,\mathrm{d}t < \infty$. For $\varphi \in C_c^1(\mathbb{R}^d)$, boundedness of its gradient permits differentiation under expectation, yielding

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}[\varphi(I_t)] = \mathbb{E}[\nabla\varphi(I_t)\cdot \dot{I}_t] = \int \nabla\varphi(z)\cdot v_t^{\mathrm{int}}(z)\mu_t(\mathrm{d}z).$$

Moreover $\int \|v_t^{\mathrm{int}}\|\,\mathrm{d}\mu_t \le \mathbb{E}\|\dot{I}_t\|$ by conditional Jensen. Applying the same chain rule to a compactly supported space–time test ψ and integrating in t gives

$$\int_0^1 \int \big(\partial_t\psi + \nabla\psi \cdot v_t^{\mathrm{int}}\big)\mu_t(\mathrm{d}z)\,\mathrm{d}t = 0,$$

which proves the claimed measure-valued equation without assuming existence of a density. □

For example, $X_0 = X_1 = 0$ almost surely gives $\mu_t = \delta_0$: the theorem applies although no Lebesgue density exists. A conditional velocity specified only μ_t-almost everywhere does not, without additional regularity and well-posedness assumptions, automatically define a global deterministic ODE flow.

Remark 1.50 (Nonuniqueness of transporting velocity)**.** For a density path $\mu_t(\mathrm{d}z) = \rho_t(z)\,\mathrm{d}z$ on a domain with suitable boundary conditions, if v_t^{int} transports ρ_t, all other transporting velocities are precisely $v_t^{\mathrm{int}} + w_t$ with $\nabla\cdot(\rho_t w_t) = 0$ in distributions, subject to the same boundary flux and integrability conditions. Necessity follows by subtracting the two continuity equations; sufficiency follows by adding the divergence-free current. For general laws the corresponding condition is $\nabla\cdot(w_t\mu_t) = 0$, with the required local integrability. Thus a marginal-law path alone does not uniquely determine an ambient velocity field. The conditional construction in (1.59) supplies a *particular* velocity; its uniqueness requires further criteria such as an appropriate minimal-action principle or a one-dimensional zero-flux condition.

Proposition 1.51 (Conditional score identity for Gaussian interpolation)**.**

Source exercise: Project 14, item 3. *Student Assessment Exercise* [Assessment] (adapted).

Suppose $X_0 \sim \mathcal{N}(0, I_d)$ is independent of an arbitrary random vector X_1, and $I_t = \alpha_t X_0 + \lambda_t X_1$ with $\alpha_t \neq 0$. Then I_t has a smooth strictly positive density ρ_t and

$$\nabla \log \rho_t(x) = -\frac{1}{\alpha_t}\mathbb{E}[X_0 \mid I_t = x].$$

If both endpoints are independent standard Gaussians, the right side reduces to $-x/(\alpha_t^2 + \lambda_t^2)$.

Proof. Write $\mu_1 = \mathrm{Law}(X_1)$. Conditioning on $X_1 = y$ shows that I_t has Gaussian density $\varphi_{|\alpha_t|}(x - \lambda_t y)$. Hence

$$\rho_t(x) = \int \varphi_{|\alpha_t|}(x - \lambda_t y)\mu_1(\,\mathrm{d}y).$$

Every derivative of the fixed-variance Gaussian kernel is bounded as a function of its argument. Since μ_1 is a probability measure, dominated differentiation on compact x-sets proves smoothness and permits differentiation under the integral:

$$\nabla\rho_t(x) = -\frac{1}{\alpha_t^2}\int (x - \lambda_t y)\varphi_{|\alpha_t|}(x - \lambda_t y)\mu_1(\,\mathrm{d}y).$$

Divide by $\rho_t(x)$ and recognize the conditional expectation $\mathbb{E}[x - \lambda_t X_1 \mid I_t = x] = \alpha_t\mathbb{E}[X_0 \mid I_t = x]$. The formula is therefore a Gaussian integration identity, not a general claim for arbitrary endpoint coupling. For standard independent Gaussian endpoints, elementary joint-Gaussian conditioning gives $\mathbb{E}[X_0 \mid I_t = x] = \alpha_t x/(\alpha_t^2 + \lambda_t^2)$. □

1.4.1.3 Conditional regression as a learning objective

For a square-integrable observable $G_t = G(X_t)$, the variational identity

$$g_t^* = \arg\min_{g \in L^2(\mathrm{Law}(Z_t))} \mathbb{E}\|g(Z_t) - G_t\|^2 = \mathbb{E}[G_t \mid Z_t = \cdot]$$

is understood as equality of equivalence classes almost everywhere. This conditional regression is also the population objective used in flow matching and stochastic-interpolant methods [Lip+23; ABV25]. A neural network can parameterize $g_{\theta,t}(z)$ to approximate an effective drift or conditional velocity. The learning problem is not mathematically identical to the population optimum: finite samples, representation restrictions, and optimization errors introduce additional approximation layers.

1.4.2 Variational construction of transport maps

A learned probability flow can be chosen by minimizing a mismatch between an initial law transported by a map and a prescribed target law. The dependence of a density on the

transporting map is nonlinear; therefore a small perturbation of the map must be propagated through its pushforward, rather than differentiated as if density and map were independent.

Theorem 1.52 (First variation of a pushforward KL objective)**.**

Source exercise: Project 9, item 13. *Student Assessment Exercise* [Assessment] (adapted).

Let ρ_0 and q be smooth positive probability densities on $\mathbb{R}^d$, and let F be an orientation-preserving C^2 diffeomorphism. Define its pushforward density by $\rho(y)\,\mathrm{d}y = F_\#(\rho_0(x)\,\mathrm{d}x)$, and put $\mathcal{J}(F) = D_{\mathrm{KL}}(q\|\rho)$. For a compactly supported smooth vector field w, perturb F through $F_\varepsilon = (\mathrm{id} + \varepsilon w) \circ F$. If differentiation under the integral is justified, then

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\mathcal{J}(F_\varepsilon)\Big|_{\varepsilon=0} = \int_{\mathbb{R}^d} q(y)[\nabla \log \rho(y) - \nabla \log q(y)] \cdot w(y)\,\mathrm{d}y. \tag{1.60}$$

Equivalently, for a Lagrangian map perturbation $\delta F(x) = w(F(x))$,

$$\delta\mathcal{J} = \int_{\mathbb{R}^d} q(F(x)) \det DF(x)[\nabla \log \rho - \nabla \log q](F(x)) \cdot \delta F(x)\,\mathrm{d}x. \tag{1.61}$$

Proof. For any compactly supported test function φ,

$$\int \varphi(y)\rho_\varepsilon(y)\,\mathrm{d}y = \int \varphi(y + \varepsilon w(y))\rho(y)\,\mathrm{d}y,$$

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_0 \int \varphi\rho_\varepsilon\,\mathrm{d}y = \int \nabla\varphi(y) \cdot w(y)\rho(y)\,\mathrm{d}y = -\int \varphi\nabla \cdot (\rho w)\,\mathrm{d}y.$$

Therefore $\delta\rho = -\nabla \cdot (\rho w)$ in the sense of distributions. Since q is fixed,

$$\delta\mathcal{J} = -\int q(y)\frac{\delta\rho(y)}{\rho(y)}\,\mathrm{d}y = \int q(y)\,[\nabla \cdot w(y) + w(y) \cdot \nabla \log \rho(y)]\,\mathrm{d}y.$$

Integrate the divergence term by parts to obtain $\int q\nabla \cdot w = -\int q\, w \cdot \nabla \log q$. This proves (1.60). Finally, set $y = F(x)$ so that $\mathrm{d}y = \det DF(x)\,\mathrm{d}x$ and $w(y) = \delta F(F^{-1}(y))$, giving (1.61). The factor $\det DF$ is necessary when expressing the variation relative to the initial-coordinate Lebesgue measure; omitting it mixes Eulerian and Lagrangian representations. □

Example 1.53 (Constructing a transport toward a tilted Gaussian law)**.**

Source exercise: Project 9, item 14. *Student Assessment Exercise* [Assessment] (adapted).

Problem

Let ρ_0 denote the standard Gaussian density in one dimension, let $w \geq 0$ be measurable, and assume $0 < Z_w = \int w(x)\rho_0(x)\,dx < \infty$. The desired density is $\rho_w(x) = w(x)\rho_0(x)/Z_w$.

To be completed.

Proposition 1.54 (Formal optimality system for a kinetic-action flow)**.**

Source exercise: Project 9, item 15. *Student Assessment Exercise* [Assessment] (adapted).

Let ρ_0 be prescribed and let $\mathcal{E}(\rho_1)$ be a specified, differentiable terminal functional of the final probability law. Consider, formally under sufficient smoothness,

$$\min_{\rho,v}\left\{\int_0^1\int_{\mathbb{R}^d}\tfrac{1}{2}\rho_t(x)\|v_t(x)\|^2\,dx\,dt+\mathcal{E}(\rho_1)\right\},$$
$$\partial_t\rho+\nabla\cdot(\rho v)=0.$$

For a smooth interior stationary point with positive density that admits a smooth Lagrange multiplier ψ, the necessary stationarity conditions, in a choice of additive gauge for ψ, are

$$v_t=\nabla\psi_t,$$
$$\partial_t\psi_t+\tfrac{1}{2}\|\nabla\psi_t\|^2=0,$$
$$\psi_1=-\frac{\delta\mathcal{E}}{\delta\rho}(\rho_1),$$

along with the continuity equation and the initial condition. The original assessment leaves the terminal functional unspecified; therefore it does not determine a unique optimal flow without an additional choice of $\mathcal{E}$.

Proof to be completed.

1.4.3 Structure-preserving numerical probability evolution

The Markov generator structure is central to numerical approximations of Fokker–Planck dynamics [Pav14].

A finite-volume Markov-chain discretization Consider on the bounded interval $[0,1]$ the Fokker–Planck equation

$$\partial_t\rho_t=\partial_x(\mathcal{V}'\rho_t)+D_{\text{diff}}\partial_{xx}\rho_t,\quad D_{\text{diff}}=\tfrac{1}{2},$$

The assessment specifies $\partial_x \rho_t = 0$ at both endpoints but also asks for a mass-conserving Markov discretization. These demands are incompatible for a general potential: integration of the PDE gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_0^1 \rho_t(x)\,\mathrm{d}x = [\mathcal{V}'\rho_t + D_{\text{diff}}\partial_x\rho_t]_0^1 = [\mathcal{V}'\rho_t]_0^1 \quad \text{when } \partial_x\rho_t = 0.$$

The construction below therefore uses the explicitly corrected *zero-flux* condition $\boldsymbol{j}_t = -\mathcal{V}'\rho_t - D_{\text{diff}}\partial_x\rho_t = 0$ at both endpoints. It agrees with the stated Neumann condition when $\mathcal{V}'(0) = \mathcal{V}'(1) = 0$. For other potentials it solves the reflecting problem, rather than silently asserting conservation for the original boundary data. Divide $[0,1]$ into cells I_i of width $\Delta x = 1/N_x$ with centers x_i. For this fixed mesh, write $m_i(t) := m_i^{\Delta x}(t) \approx \int_{I_i} \rho_t(x)\,\mathrm{d}x$ for the semidiscrete *cell masses*, not pointwise densities. Abbreviate $q_{i\to j} := q_{i\to j}^{\Delta x}$ and $\pi_i := \pi_i^{\Delta x}$ on this mesh. Set $\mathcal{V}_i = \mathcal{V}(x_i)$ and nearest-neighbor rates

$$\begin{aligned} q_{i\to i+1} &= \frac{D_{\text{diff}}}{(\Delta x)^2}\exp\left[-\frac{\mathcal{V}_{i+1}-\mathcal{V}_i}{2D_{\text{diff}}}\right], \\ q_{i+1\to i} &= \frac{D_{\text{diff}}}{(\Delta x)^2}\exp\left[\frac{\mathcal{V}_{i+1}-\mathcal{V}_i}{2D_{\text{diff}}}\right]. \end{aligned} \tag{1.62}$$

No transitions are permitted beyond the two boundary cells. The edge flux is $j_{i+1/2} = q_{i\to i+1}m_i - q_{i+1\to i}m_{i+1}$; the cell balance $\dot m_i = j_{i-1/2} - j_{i+1/2}$ gives the system below, with $j_{1/2} = j_{N_x+1/2} = 0$. Its interior backward operator is consistent: for $\mathcal{V}, f \in C^4$ on a neighborhood of the interval, Taylor expansion gives

$$\begin{aligned} &q_{i\to i+1}(f(x_i+\Delta x) - f(x_i)) + q_{i\to i-1}(f(x_i-\Delta x) - f(x_i)) \\ &\quad = -\mathcal{V}'(x_i)f'(x_i) + D_{\text{diff}}f''(x_i) + O((\Delta x)^2). \end{aligned}$$

This is an interior consistency calculation, not a boundary error estimate or a proof of convergence of the semidiscrete density.

Theorem 1.55 (Positivity, conservation, and detailed balance)**.**

Source exercise: Project 8, item Main task. *Student Assessment Exercise* [Assessment] (adapted).

Let the semidiscrete cell-mass vector satisfy

$$\dot m_i = \sum_{j\neq i}(q_{j\to i}m_j - q_{i\to j}m_i), \quad m_i(0) \geq 0, \quad \sum_i m_i(0) = 1, \tag{1.63}$$

with the rates in (1.62)*. Then* $m_i(t) \geq 0$ *and* $\sum_i m_i(t) = 1$ *for all* $t \geq 0$*. Moreover,* $\pi_i = Z_{\Delta x}^{-1}e^{-\mathcal{V}_i/D_{\text{diff}}}$ *is stationary and satisfies detailed balance:* $\pi_i q_{i\to j} = \pi_j q_{j\to i}$*.*

Proof. The matrix of (1.63) has nonnegative off-diagonal entries and columns summing to zero. Calling it $Q^{\Delta x}$, choose $\lambda \geq \max_i(-Q_{ii}^{\Delta x})$; then $Q^{\Delta x} + \lambda I$ has nonnegative entries, so $e^{tQ^{\Delta x}} = e^{-\lambda t}\sum_{k\geq 0} t^k(Q^{\Delta x} + \lambda I)^k/k!$ is entrywise nonnegative. The identity $\mathbf{1}^\top Q^{\Delta x} = 0$ implies $\mathbf{1}^\top e^{tQ^{\Delta x}} = \mathbf{1}^\top$, so this exponential is column-stochastic and pre-

serves probability cell masses. Summing (1.63) over i cancels every internal flux. Finally, $q_{i\to i+1}/q_{i+1\to i} = \exp[-(\mathcal{V}_{i+1} - \mathcal{V}_i)/D_{\text{diff}}] = \pi_{i+1}/\pi_i$. □

Proposition 1.56 (Discrete relative-entropy dissipation)**.** *Suppose* $m_i(t) > 0$ *and write* $r_i = m_i/\pi_i$. *Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_i m_i \log\frac{m_i}{\pi_i} = -\sum_{i=1}^{N_x-1} \pi_i q_{i\to i+1}(r_{i+1} - r_i)(\log r_{i+1} - \log r_i) \le 0. \tag{1.64}$$

Proof. Mass conservation removes the constant term from the entropy derivative. For each edge let $c_i = \pi_i q_{i\to i+1} = \pi_{i+1} q_{i+1\to i}$. The contribution to $\dot{m}_i$ from that edge is $c_i(r_{i+1} - r_i)$, and its contribution to $\dot{m}_{i+1}$ is the negative. Multiplying by $\log r_i$ and $\log r_{i+1}$ and adding gives precisely $-c_i(r_{i+1} - r_i)(\log r_{i+1} - \log r_i)$. Summing the edges proves the displayed identity and its sign. All rates on this connected chain are positive, so a nonzero initial probability vector becomes strictly positive for $t > 0$; the identity then applies even when some initial cells have zero mass. □

Exercise 1.57 (The central-difference positivity test)**.**

Source exercise: Project 8, item 1. *Student Assessment Exercise* [Assessment] (adapted).

The assessment proposes a centered approximation for the drift–diffusion PDE with ghost-cell boundary values. Rewrite it as an ODE for cell masses; determine whether the off-diagonal entries are nonnegative and whether the boundary closure conserves total mass.

To be completed.

Verification protocol The numerical study should monitor (i) total mass and sign of each cell mass, (ii) convergence of the semidiscrete solution under mesh refinement, (iii) equilibrium detailed balance and entropy dissipation, and (iv) discrepancy against an analytic solution or an independent finer-grid reference. A low loss on finitely many collocation points, by itself, is not a proof of a global density approximation.

Executed Markov semidiscretization check.

On $[0, 1]$, use $D_{\text{diff}} = 1/2$, $\mathcal{V}(x) = 2(x - 0.3)^2 + 0.7x$, and 60 finite-volume cells. The continuous-time generator from (1.62) is propagated by a matrix-exponential action; time is not discretized. The initial cell-mass vector is proportional to $\exp[-((x_i - .75)/.11)^2]$. Total mass, minimum cell mass, and discrete relative entropy provide numerical checks of the analytic invariants: The largest absolute mass discrepancy in these reported time points is below 2×10^{-13}, consistent with roundoff. Entropy decreases, as required by the exact semidiscrete dissipation identity (1.64). This fixed-grid check does not establish spatial convergence to the PDE; mesh refinement remains to be carried out.

Table 1.3: Computed conservation, positivity, and equilibrium relative entropy.

t	$\min_i m_i(t)$	$D_{\mathrm{KL}}(m_t\|\pi)$
0.000	1.5453e-21	2.144403
0.015	2.6201e-07	1.472250
0.050	7.0263e-04	0.826780
0.200	3.4791e-03	0.077412
0.500	1.6399e-03	0.000610

1.5 A Coupled OU Benchmark: Exact, Numerical, and AI Results

1.5.1 Exact OU dynamics and closure diagnostics

The analytic benchmark connects the microscopic equilibrium law, time-dependent conditional coefficients, and the correlation diagnostic for a reduced closure. These exact identities provide the reference for the numerical and regression experiments in the next subsection.

1.5.1.1 The microscopic OU model and its Gibbs equilibrium

For $a, c, \beta > 0$ and $\kappa \in \mathbb{R}$, consider

$$\mathcal{V}(x,y) = \frac{a}{2}x^2 + \frac{c}{2}(y-\kappa x)^2, \quad \xi(x,y) = x.$$

Writing $R_t = (X_t, Y_t)^\top$, the microscopic SDE becomes

$$\mathrm{d}R_t = -A_{\mathrm{OU}} R_t \,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t, \quad A_{\mathrm{OU}} = \begin{pmatrix} a + c\kappa^2 & -c\kappa \\ -c\kappa & c \end{pmatrix}. \tag{1.65}$$

For every $(u,v) \neq (0,0)$, $(u,v)A_{\mathrm{OU}}(u,v)^\top = au^2 + c(v-\kappa u)^2 > 0$. Thus A_{OU} is symmetric positive definite, and its determinant is $ac > 0$.

Proposition 1.58 (Gaussian equilibrium and projected free energy)**.** *The unique invariant law of* (1.65) *is* $\mathcal{N}(0, \Sigma_\infty)$*, where*

$$\Sigma_\infty = \beta^{-1} A_{\mathrm{OU}}^{-1} = \beta^{-1} \begin{pmatrix} 1/a & \kappa/a \\ \kappa/a & 1/c + \kappa^2/a \end{pmatrix}.$$

At equilibrium, $X \sim \mathcal{N}(0, (\beta a)^{-1})$ *and* $Y \mid X = z \sim \mathcal{N}(\kappa z, (\beta c)^{-1})$*. Consequently the free energy of* $\xi(x,y) = x$ *is* $F_\xi(z) = az^2/2 + \text{const}$.

Proof. The exact solution for an initial state independent of the Brownian increments is

$$R_t = e^{-A_{\mathrm{OU}}t}R_0 + \sqrt{2\beta^{-1}}\int_0^t e^{-A_{\mathrm{OU}}(t-s)}\,\mathrm{d}W_s.$$

The stochastic integral is centered Gaussian with covariance $2\beta^{-1}\int_0^t e^{-A_{\mathrm{OU}}s}e^{-A_{\mathrm{OU}}s}\,\mathrm{d}s$. Because A_{OU} is symmetric positive definite, $e^{-A_{\mathrm{OU}}t} \to 0$ and that covariance tends to $\beta^{-1}A_{\mathrm{OU}}^{-1}$. The stationary covariance therefore solves $A_{\mathrm{OU}}\Sigma+\Sigma A_{\mathrm{OU}} = 2\beta^{-1}I$ and equals $\Sigma = \beta^{-1}A_{\mathrm{OU}}^{-1}$. Integrating the factorized Gibbs density $\exp[-\beta a z^2/2]\exp[-\beta c(y-\kappa z)^2/2]$ over y identifies the marginal and conditional laws. Positivity of A_{OU} makes the deterministic propagator exponentially stable. The explicit solution converges in law to the stated Gaussian for any initial probability law independent of future noise; an invariant law must equal this limit. This proves uniqueness, not just uniqueness of the covariance matrix. □

1.5.1.2 Exact time-dependent conditional projection

If $R_0 \sim \mathcal{N}(m_0^{\mathrm{OU}}, \Sigma_0)$, independent of future Brownian increments, with Σ_0 symmetric positive semidefinite, then

$$\begin{aligned} m_t^{\mathrm{OU}} &= e^{-A_{\mathrm{OU}}t}m_0^{\mathrm{OU}}, \\ \Sigma_t &= \Sigma_\infty + e^{-A_{\mathrm{OU}}t}(\Sigma_0 - \Sigma_\infty)e^{-A_{\mathrm{OU}}t}. \end{aligned} \tag{1.66}$$

For $(m_t^{\mathrm{OU}})_x$ and $(m_t^{\mathrm{OU}})_y$ denoting coordinate means, the exact conditional expectation is

$$g_t(z) = \mathbb{E}[Y_t \mid X_t = z] = (m_t^{\mathrm{OU}})_y + \frac{(\Sigma_t)_{yx}}{(\Sigma_t)_{xx}}(z - (m_t^{\mathrm{OU}})_x), \tag{1.67}$$

whenever $(\Sigma_t)_{xx} > 0$. The exact projected drift is

$$b_t(z) = -(a + c\kappa^2)z + c\kappa g_t(z), \tag{1.68}$$

and the diffusion coefficient remains $\sqrt{2\beta^{-1}}$. Equations (1.67)–(1.68) make conditional regression a testable learning problem with a known target.

1.5.1.3 Exact marginals and incorrect time correlations

Theorem 1.59 (Failure of stationary free-energy closure for correlations)**.** *Suppose the full model starts in its invariant law and $\kappa \neq 0$. The equilibrium conditional drift is $-az$, so the stationary free-energy closure, driven by a scalar standard Brownian motion W^{red}, is*

$$\mathrm{d}\bar{X}_t = -a\bar{X}_t\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t^{\mathrm{red}}.$$

Both X_t and $\bar{X}_t$ have the same stationary one-time marginal $\mathcal{N}(0, (\beta a)^{-1})$, but their

time autocovariances differ:

$$C_{\text{full}}(t) = e_1^\top e^{-A_{\text{OU}}t}\Sigma_\infty e_1,$$
$$C_{\text{closure}}(t) = (\beta a)^{-1}e^{-at}.$$

In particular, their right second derivatives at zero satisfy

$$C''_{\text{full}}(0+) = \beta^{-1}(a + c\kappa^2), \quad C''_{\text{closure}}(0+) = \beta^{-1}a.$$

Proof. The equilibrium conditional mean is $\mathbb{E}[Y \mid X = z] = \kappa z$, giving $b(z) = -az$. The OU semigroup yields the correlation formula. Twice differentiate the matrix exponential: $C''_{\text{full}}(0+) = e_1^\top A_{\text{OU}}^2 \Sigma_\infty e_1 = \beta^{-1} e_1^\top A_{\text{OU}} e_1$. The scalar expression follows by differentiating e^{-at}. For $\kappa \neq 0$, the results differ, so the correlations cannot agree for all t despite identical one-time stationary laws. □

1.5.2 Numerical validation and drift learning

The accompanying experiments implement exact Gaussian propagation, conditional projected coefficients, affine regression, and neural regression on sampled microscopic states. Their population objective is

$$\min_\theta \mathbb{E}\left|g_{\theta,t}(X_t) - Y_t\right|^2.$$

By orthogonality of conditional expectation, the unrestricted population minimizer is (1.67). The numerical experiment below reports test errors from a trained two-hidden-layer neural model and a Gaussian affine baseline.

Algorithm 1: Reproducible conditional-projection benchmark

1: Choose A_{OU}, β, an initial Gaussian law, and evaluation times
2: Compute $m_t^{\text{OU}}, \Sigma_t$ from (1.66)
3: Sample pairs (X_t, Y_t) from the exact joint Gaussian
4: Fit $g_{\theta,t}(x)$ to Y_t under squared loss, obtaining $\widehat{\theta}$
5: Form $\widehat{b}_t(x) = -(a + c\kappa^2)x + c\kappa g_{\widehat{\theta},t}(x)$
6: Evaluate learned conditional mean and drift on an independent test set
7: Compute exact stationary correlations for microscopic and reduced models

Later thought — A nonlinear extension of the OU benchmark

To revisit. Replace the quadratic x-potential by a double well and compare conditional drift learning with temporal observables. **Idea.** Separate analytic equilibrium checks from transition statistics that require independent microscopic simulations and time-step refinement. **Status: planned extension; not solved or simulated here.**

Executed numerical experiment: identical marginals do not fix time correlations.

For the coupled OU model take $(a, c, \kappa, \beta) = (0.8, 1.7, 1.1, 1)$. The equilibrium X-marginal has variance $(\beta a)^{-1} = 1.25$ in both the microscopic model and the Markov closure. The two time correlations at lag one, however, are

$$C_{\text{full}}(1) = 0.798008, \quad C_{\text{closure}}(1) = 0.561661.$$

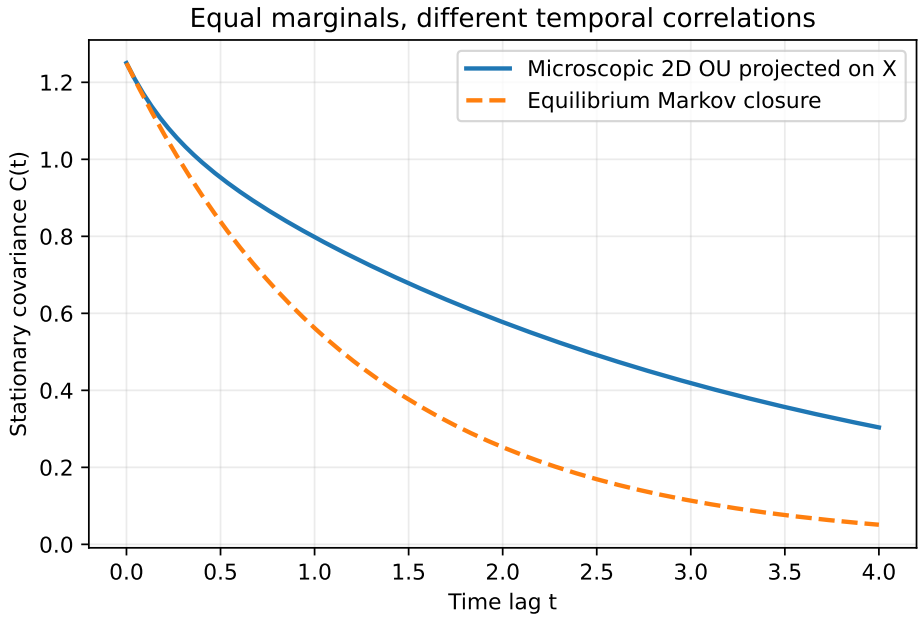


Figure 1.9: Exact stationary two-time covariances; the curves share the same value at zero lag but separate as soon as their temporal dynamics matters.

Executed AI experiment: learning a projected drift.

With the same model and nonstationary Gaussian initial data $m_0^{\text{OU}} = (1, -0.8)^\top$, $\Sigma_0 = \text{diag}(0.12, 0.25)$, we sampled the exact Gaussian law at times $t \in \{0.25, 0.5, 1, 2\}$. The regression input is (t, X_t), its noisy label is Y_t, and the population target is the conditional mean $g_t(x) = \mathbb{E}[Y_t \mid X_t = x]$ from (1.67). A two-hidden-layer Tanh network with 48 units per layer is fit by Adam with fixed seed `20261102`, learning rate 0.008, 260 epochs, and batches of at most 256 samples. Inputs and labels are standardized using training data only; the four time slices are sampled uniformly. We also fit a separate affine least-squares model at each training time; the latter incorporates the Gaussian conditional structure and is a meaningful physics-informed baseline. All errors are measured on 12,000 independently generated test samples against the *exact conditional mean*, rather than against the noisy response. The affine and neural fits share the test set but use separately sampled training sets. Thus their single-run difference combines model and training-sample effects; it is not a paired comparison on identical data.

The model is a regression surrogate for a drift coefficient; it does not by itself demonstrate correct long-time trajectories. In this one-seed experiment the larger training set improves the structure-aware affine baseline, while the unconstrained neural network does not improve

Table 1.4: Held-out RMSE in the learned conditional mean $g_t(x)$. The projected-drift RMSE equals $c|\kappa|$ times these values.

Training pairs	Neural network	Affine Gaussian baseline
512	0.07427	0.07414
4096	0.09911	0.03552

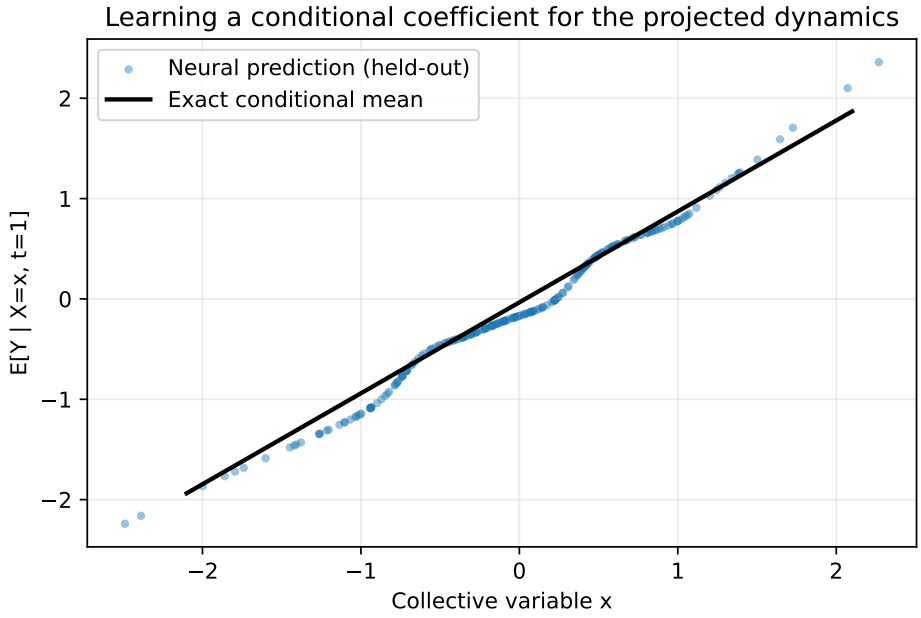


Figure 1.10: Neural conditional-mean predictions on held-out samples at $t = 1$ compared with the exact Gaussian regression function.

monotonically. This distinction is part of the result, not a convergence claim from two sample sizes. The exact Gaussian formulas provide an unambiguous ground truth for later training-size and architecture studies.

Saved results and the historical reproducibility audit. Table 1.4 retains the supplied historical results. The isolated audit rerun of 4 November 2026, with NumPy 2.4.4, SciPy 1.17.1, and PyTorch 2.11.0 on CPU reproduced the affine errors to rounding precision. Its neural conditional-mean RMSEs were 0.07427 and 0.09316 for 512 and 4096 training pairs, respectively. Thus the saved neural numbers are not reproduced bit for bit in that recorded environment, although the nonmonotone training-size observation remains. The original fully pinned training environment is unavailable, so no unique cause of the discrepancy is inferred. The comparison and environment record are saved separately in `results/audit-2026-11-04/`; the original data and figure are unchanged.

Reproducible experiments

Run `python experiments/run_benchmarks.py`, `python experiments/hamiltonian_benchmark.py`, and `python experiments/project11_kl_benchmarks.py` after installing `numpy`, `scipy`, `matplotlib`, and `torch`. These scripts write the PDF figures and CSV/JSON tables used in the local numerical validations and the OU benchmark. The seed, matrix parameters, initial laws, optimization settings, and reference formulas are contained in the corresponding source files.

Scope of the assessment answers. The retained Projects 8–14 distinguish 17 Core, 14 Concise, and 14 Pending entries; the provisional extension adds Projects 2–3. Core questions develop flows, probability transport, relative-entropy variational structure, conditional reduction, and learned velocities. Concise entries retain supporting formulas, short arguments, and selected existing numerical checks. Pending entries retain their statements and source anchors; their original solutions and proofs are preserved as line comments, and the visible completion notices do not represent completed proofs.

Project 8 retains the corrected no-flux Markov structure, while its independent ghost-cell exercise is pending. The score equation is retained, with its Hessian and long-time extensions deferred. The same distinction applies to the total-covariance identity and its deferred double-well illustration. The core Markovian projection result still assumes existence and forward-equation uniqueness; matching one-time marginals does not identify the full process law. The conditional-score identity still requires independent Gaussian initial noise and $\alpha_t \neq 0$. The existing numerical results are unchanged; further learning studies and spatial refinement remain outside their scope. The new PDE module reports its own integration checks separately in Section 2.6.

Chapter 2

Neural PDE Solvers: Fokker–Planck Dynamics, Committor Equations, and Operator Learning

Chapter 1 distinguishes a microscopic state, an observable on states, and a probability law, then follows their evolution through conditioning, closure, and learning. A related computational question is how to approximate the differential equations induced by a specified stochastic model. A density solver advances a law; a committor solver computes a state-dependent hitting probability; an operator learner approximates a family of such solutions. These are different outputs, even when they share the same generator.

This chapter is a *provisional research module*. Mathematical statements with proofs below are established within their stated assumptions. The sparse discretization and neural architectures are existing numerical approaches, implemented here as a low-dimensional baseline. The executed integration check establishes that the components run together; the accuracy, generalization, and high-dimensional studies are *Planned Experiments*. Projects 2 and 3 of the assessment motivate the extension. The latter's printed signs, repeated target center, and periodic-potential convention require explicit treatment rather than silent corrections.

2.1 Fokker–Planck dynamics and PDE representation

To isolate the relation between an observable generator and a density equation, take the state space to be the flat torus $\mathcal{D} = \mathbb{T}_L^d = (\mathbb{R}/L\mathbb{Z})^d$, represented by $[-L/2, L/2)^d$, where $L > 0$ is its side length. Its Lebesgue volume is $|\mathcal{D}| = L^d$. This $\mathcal{D}$ is a physical domain; it is distinct from the probability sample space Ω in the notation tables. Fix $\varepsilon > 0$, a periodic C^1 drift $b : \mathcal{D} \to \mathbb{R}^d$, and a standard d-dimensional Brownian motion W_t, independent of the initial state. Consider

$$\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + \sqrt{2\varepsilon}\,\mathrm{d}W_t, \qquad \mu_t = \mathrm{Law}(X_t). \tag{2.1}$$

Here X_t is a random state, $(X_t)_{t\geq 0}$ is a stochastic process, and μ_t is a probability measure. When $\mu_t(\,\mathrm{d}x) = p_t(x)\,\mathrm{d}x$, p_t is its density. In the previous Langevin model, $\varepsilon = \beta^{-1}$ and $b = -\nabla\mathcal{V}$.

For a scalar test observable $\varphi \in C^2(\mathcal{D})$, Itô's formula gives the backward differential realization $\mathcal{L} : C^2(\mathcal{D}) \to C(\mathcal{D})$. Its formal Lebesgue adjoint has differential realization $\mathcal{L}^* : C^2(\mathcal{D}) \to C(\mathcal{D})$:

$$\mathcal{L}\varphi = b \cdot \nabla\varphi + \varepsilon\Delta\varphi, \qquad \mathcal{L}^* p = -\nabla \cdot (bp) + \varepsilon\Delta p. \tag{2.2}$$

All derivatives and functions here are periodic. These smooth realizations suffice for the following calculations; an unbounded operator on $L^2(\mathcal{D})$ has the specified Sobolev realization $\mathcal{L} : H^2(\mathcal{D}) \subset L^2(\mathcal{D}) \to L^2(\mathcal{D})$, and likewise for $\mathcal{L}^*$. The space H^2 consists of square-integrable functions with square-integrable weak derivatives through order two. The star means adjoint under Lebesgue pairing, not a backward generator acting on densities.

Proposition 2.1 (Duality, forward evolution, and probability current)**.** *Under the preceding assumptions, for smooth periodic φ, p,*

$$\int_{\mathcal{D}} (\mathcal{L}\varphi)p \,\mathrm{d}x = \int_{\mathcal{D}} \varphi\mathcal{L}^* p \,\mathrm{d}x.$$

If μ_0 has nonnegative density $p_0 \in L^1(\mathcal{D})$ with integral one, the diffusion law solves

$$\partial_t p = \mathcal{L}^* p, \qquad p(0,\cdot) = p_0, \qquad \partial_t p + \nabla \cdot \boldsymbol{j} = 0, \quad \boldsymbol{j} = bp - \varepsilon\nabla p, \tag{2.3}$$

in the weak sense, classically for positive times under the usual parabolic regularity. It remains nonnegative and has unit integral. A stationary probability density defines an invariant probability measure, and conversely, in this class.

Proof. Integration by parts on a torus has no boundary term; applying it once to the drift and twice to the diffusion proves the duality. Taking expectations in Itô's formula yields $\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi \,\mathrm{d}\mu_t = \int \mathcal{L}\varphi \,\mathrm{d}\mu_t$, which identifies the weak forward equation. Substitution of the current gives its continuity form. Integrating that equation proves mass conservation; nonnegativity also follows directly because μ_t is the law of a Markov process (or from the parabolic maximum principle). Uniqueness of the forward problem identifies a time-independent solution with a law invariant under every transition operator. The converse follows by testing the invariant law against smooth observables. □

The current in (2.3) includes diffusion. Only where $p > 0$ may one define the current velocity $\boldsymbol{j}/p = b - \varepsilon\nabla \log p$ as in Section 1.4. This rewrites a marginal-density equation; it does not replace the SDE path law by an ODE law. On $\mathbb{R}^d$ the same integrations require decay of the flux at infinity; on a reflecting bounded domain they require $\boldsymbol{j} \cdot n = 0$, with n the outward unit normal. Arbitrary absorbing boundaries do not preserve total probability in the unabsorbed domain.

For a smooth periodic potential $\mathcal{V}$, $b = -\nabla\mathcal{V}$ gives $p_\infty = Z_\varepsilon^{-1} e^{-\mathcal{V}/\varepsilon}$, where $Z_\varepsilon = \int_{\mathcal{D}} e^{-\mathcal{V}/\varepsilon}\,\mathrm{d}x$ is finite and positive. Indeed $bp_\infty - \varepsilon\nabla p_\infty = 0$. This checks invariance using

(2.2); it is not an assumption that an arbitrary stationary current must vanish.

2.2 Committor PDE and transition path theory

Density evolution describes an ensemble at a time. A transition question instead asks which of two sets a trajectory will reach first. Let $A, B \subset \mathcal{D}$ be nonempty open sets with disjoint closures and $C^{2,\alpha}$ boundaries, where $0 < \alpha < 1$ specifies a Hölder exponent, and let $D = \mathcal{D} \setminus (\overline{A} \cup \overline{B})$ be connected. Strengthen the drift to $b \in C^{1,\alpha}$. Define stopping times and the forward committor observable by

$$\tau_A = \inf\{t \geq 0 : X_t \in \overline{A}\}, \qquad \tau_B = \inf\{t \geq 0 : X_t \in \overline{B}\}, \qquad q(x) = \mathbb{P}_x(\tau_B < \tau_A). \tag{2.4}$$

We use closed hitting sets for the boundary-value formulation. For this nondegenerate diffusion and regular boundaries, the hitting times defined with the open sets have the same infima almost surely. The subscript x specifies the initial state $X_0 = x$. Extend q by zero in A and one in B; it is an observable taking values in $[0, 1]$, not a density.

Proposition 2.2 (Backward boundary-value representation)**.** *Assume also that $\tau_A \wedge \tau_B < \infty$ almost surely for every starting state in D. The unique classical Dirichlet solution is the hitting probability in (2.4):*

$$\mathcal{L}q = \varepsilon \Delta q + b \cdot \nabla q = 0 \quad \text{in } D, \qquad q|_{\partial A} = 0, \quad q|_{\partial B} = 1. \tag{2.5}$$

The outer identifications of $\mathcal{D}$ are periodic. This stationary elliptic problem has no initial condition. For the homogeneous difference of two solutions the operator acts as $\mathcal{L} : \{u \in H^2(D) : \mathrm{Tr}_{\partial A \cup \partial B}\, u = 0\} \subset L^2(D) \to L^2(D)$, with periodic identifications inherited from the torus.

Proof. Uniform ellipticity, the smooth compact geometry, and the stated Hölder assumptions give the classical Dirichlet solution. The elliptic maximum principle gives $0 \leq q \leq 1$ and uniqueness. Apply Itô's formula to this solution up to $t \wedge \tau$, where $\tau = \tau_A \wedge \tau_B$, first localizing away from the boundary. Since $\mathcal{L}q = 0$, the stopped bounded process is a martingale. Bounded convergence as the localization is removed and $t \to \infty$ gives $q(x) = \mathbb{E}_x[q(X_\tau)] = \mathbb{P}_x(\tau_B < \tau_A)$. This also explains why the backward operator appears: the argument conditions a future event on a present state, rather than evolving a density. □

The committor is a central ingredient of transition path theory, which studies the portions of trajectories going from A to B [EV06]. It distinguishes transition likelihood from an arbitrary geometric distance or a coordinate chosen only to minimize static reconstruction error. The full theory also uses invariant statistics and, in general, backward-in-time committors; the single forward function is not by itself every reactive-path statistic. Classical grid methods for (2.5) use N^d nodes at N points per coordinate. This cost motivates coordinate networks and operator surrogates, without establishing their accuracy in high dimension.

The assessment sign discrepancy. The source prints $-\varepsilon\Delta q + b_f \cdot \nabla q = 0$ and $b_f = -\nabla\mathcal{V}_f$. Multiplication by -1 gives $\varepsilon\Delta q - b_f \cdot \nabla q = 0$, the backward equation for SDE drift $-b_f = +\nabla\mathcal{V}_f$. In contrast, the SDE (2.1) with drift b_f gives (2.5), or equivalently $-\varepsilon\Delta q - b_f \cdot \nabla q = 0$. The implementation consistently chooses the standard SDE with $b_f = -\nabla\mathcal{V}_f$ and the *positive* diffusion backward residual. It does not claim to solve the literal printed equation for the same drift. Which sign the source intended remains an editorial question.

2.3 Physics-informed neural networks and their limitations

A physics-informed neural network (PINN) represents an unknown solution by a coordinate-dependent network and fits its differential equation and data through a loss [RPK17]. For a density approximation $p_\theta(t, x)$, the residual is $r_\theta = \partial_t p_\theta - \mathcal{L}^* p_\theta$; for a committor $q_\theta(x)$ it is $r_\theta = \mathcal{L} q_\theta$. With nonnegative weights $\lambda_{\text{PDE}}, \lambda_{\text{BC}}, \lambda_{\text{IC}}$, the time-dependent objective takes the form

$$\mathcal{J}_{\text{total}}(\theta) = \lambda_{\text{PDE}}\mathcal{J}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{J}_{\text{BC}} + \lambda_{\text{IC}}\mathcal{J}_{\text{IC}}.$$

Here θ denotes network parameters; $\mathcal{J}$ denotes a scalar loss, avoiding confusion with $\mathcal{L}$. For example, $\mathcal{J}_{\text{PDE}} = M^{-1}\sum_{i=1}^{M} |r_\theta(x_i)|^2$ for interior committor samples $x_i \in D$, and $\mathcal{J}_{\text{BC}}$ averages $|q_\theta|^2$ on ∂A and $|q_\theta - 1|^2$ on ∂B. There is no IC term for the stationary committor. Smooth activations and automatic differentiation supply gradients and Hessian traces; a piecewise linear network's almost-everywhere second derivative does not automatically represent the required classical Laplacian. In dimension d the exact Hessian trace uses d directional second derivatives.

Uniform interior sampling, dedicated surface sampling, and extra points near transition layers serve different purposes. Adaptive sampling should be accompanied by independent validation points; if a specific continuum norm is the target, nonuniform samples need the corresponding quadrature or importance weights. A sigmoid enforces $0 < q_\theta < 1$ but cannot impose exact endpoint values at finite logits. Hard boundary constructions or explicit absorbing-set values are additional operations. For a density, positivity and integral-one normalization are separate constraints.

Proposition 2.3 (A finite-time-sampling failure for the periodic heat equation)**.**

Source exercise: Project 2, item Main task. *Student Assessment Exercise* [Assessment] (adapted).

Fix $T > 0$, $\varepsilon > 0$, and $0 < \zeta < 1$. On the torus $[0, 2\pi)$, the density

$$p(t, x) = \frac{1 + \zeta e^{-\varepsilon t}\cos x}{2\pi}$$

solves $\partial_t p = \varepsilon\partial_{xx} p$ with periodic boundary conditions and initial density $p_0(x) = (1 +$

$\zeta \cos x)/(2\pi)$. *Its stationary density is* $p_\infty = (2\pi)^{-1}$. *Define* $H(s) = 1 - 10s^3 + 15s^4 - 6s^5$ *on* $[0,1]$ *and*

$$\eta_k(t) = \begin{cases} 1, & 0 \le t \le k^{-1}, \\ H(kt-1), & k^{-1} < t < 2k^{-1}, \\ 0, & t \ge 2k^{-1}, \end{cases} \qquad u_k = p_\infty + \eta_k(p - p_\infty), \quad k \in \mathbb{N}.$$

Then u_k *is a nonnegative, normalized, periodic density,* C^2 *in time, with the exact initial value* p_0. *For any fixed finite set of nonnegative time samples, and any spatial samples, the empirical squared PDE residual eventually equals zero. Nevertheless, for each* $t > 0$,

$$u_k(t,\cdot) \longrightarrow p_\infty, \qquad \|u_k(t,\cdot) - p(t,\cdot)\|_{L^2(0,2\pi)} \longrightarrow \frac{\zeta e^{-\varepsilon t}}{2\sqrt{\pi}} > 0.$$

Proof. The function H decreases from one to zero because $H'(s) = -30s^2(1-s)^2$, and its first two derivatives vanish at both endpoints. Thus $0 \le \eta_k \le 1$ and its joins are C^2. Convex mixing of two probability densities preserves nonnegativity and mass. The IC is exact because $\eta_k = 1$ near zero. Direct differentiation gives

$$r_k := \partial_t u_k - \varepsilon \partial_{xx} u_k = \eta_k'(t) \frac{\zeta e^{-\varepsilon t} \cos x}{2\pi}. \tag{2.6}$$

This residual vanishes outside $(1/k, 2/k)$, including at zero. For a fixed finite sample set with smallest strictly positive time δ, choose $k > 2/\delta$; every sampled residual is then zero. If all times are zero the assertion is immediate. The sequence itself does not depend on any samples. For each fixed $t > 0$, eventually $u_k = p_\infty$; integrating $\cos^2 x$ gives the asserted nonzero error. The pointwise discrepancy is nonzero where $\cos x \neq 0$; it is not nonzero at every spatial point. □

This does not contradict uniqueness or a continuum stability estimate. Indeed, for $2/k < T$, the *true* residual norm is

$$\|r_k\|^2_{L^2((0,T)\times(0,2\pi))} = \frac{\zeta^2 k}{4\pi} \int_0^1 |H'(s)|^2 e^{-2\varepsilon(s+1)/k} \, \mathrm{d}s,$$

which grows linearly in k. Equation (2.6) separates this continuum residual from the zero sampled loss and the nonzero solution error. The result fixes the time sample set before taking $k \to \infty$; it does not establish failure for indefinitely refreshed samples, a resolving sequence of time grids, or a particular trained neural architecture. For the heat equation, an error e satisfying $e(0) = 0$ and forced residual r obeys $\|e(t)\|_2 \le \int_0^t \|r(s)\|_2 \, \mathrm{d}s$ by the L^2 contraction of the heat semigroup. Continuum residual control with the correct IC is consequently a stronger condition than the empirical loss here.

2.4 Neural operator learning and Fourier neural operator

A coordinate network approximates one function. An operator learner instead targets a map $\mathcal{G} : \mathcal{U} \to \mathcal{Q}$ between declared function spaces. Here one may take a bounded, finite-mode subset $\mathcal{U} \subset C^\infty(\mathcal{D})$ of potential perturbations, and $\mathcal{Q} = \{q \in H^1(D) : \mathrm{Tr}_{\partial A}\, q = 0, \mathrm{Tr}_{\partial B}\, q = 1,\ 0 \le q \le 1\}$ with periodic identifications. With fixed geometry and ε, $\mathcal{G}(f) = q_f$ is defined by (2.5) with $b_f = -\nabla \mathcal{V}_f$. It is nonlinear in f, although the PDE is linear in q_f. Rough inputs in a full L^2 ball do not in general define a classical gradient drift; the finite Fourier expansion is a substantive regularity restriction.

The Fourier neural operator (FNO) lifts input features to a channel field v_0, iterates spectral and local layers, then projects to output values [Li+20]. A layer has the form

$$v_{\ell+1}(x) = \sigma\left(W_\ell v_\ell(x) + \mathcal{F}^{-1}\big[R_\ell(k)\mathcal{F}v_\ell(k)\big](x) + c_\ell(x)\right).$$

Here ℓ indexes layers, σ is a pointwise activation, W_ℓ mixes channels locally, c_ℓ is a bias, $\mathcal{F}$ is the componentwise Fourier transform on the periodic domain, and the learned complex matrices $R_\ell(k)$ act only on retained modes k. Fourier transformation is an isometry from L^2 to square-summable Fourier sequences under consistent normalization; the retained multiplier defines a finite-rank convolution operator on channel fields. A grid implementation uses FFTs, truncation, nonlinear activation, and an output projection. Learning the multiplier approximates nonlocal dependence while channel mixing and nonlinearity build a nonlinear map.

The two-dimensional implementation adds periodic sine/cosine position channels to the input field f. Otherwise a translation-equivariant network receiving constant f cannot locate the fixed sets A, B. A sigmoid projection bounds the output, and absorbing-set values are imposed separately. Grid resolution transfer is an experiment to validate, not a consequence of using an FFT. In Section 1.4, a learned probability flow targets velocities or distribution evolution; here $f \mapsto q_f$ targets a stationary observable. A network predicting a time-dependent density solution would be another neural PDE task, rather than automatically a learned flow of states.

2.5 Research problem: high-dimensional committor operator learning

Source exercise: Project 3, item Main task. *Student Assessment Exercise* [Assessment] (adapted).

Problem — Provisional formulation of Project 3

On $\mathcal{D} = \mathbb{T}^d_{20}$, learn $\boxed{\mathcal{G} : f \mapsto q_f}$ with the potential structure

$$\mathcal{V}_f(x) = 2e^{-\|x-\boldsymbol{a}\|^2/2} + 2e^{-\|x-\boldsymbol{b}\|^2/2} + e^{-\|x-\boldsymbol{c}\|^2/2} + f(x), \qquad b_f = -\nabla \mathcal{V}_f, \tag{2.7}$$

where $\boldsymbol{a} = (-1, \ldots, -1)$, $\boldsymbol{b} = (1, \ldots, 1)$, and $\boldsymbol{c} = (0, \ldots, 0)$ are centers, distinct from the vector field b_f. The notation $\mathcal{V}_f$ denotes the source's potential V, avoiding conflict with the state vector space V. The source specifies radius 0.5, A centered at $\boldsymbol{a}$, and B as a union of two balls both centered at $\boldsymbol{b}$. Keep both entries as configurable centers $(\boldsymbol{z}_{B,1}, \boldsymbol{z}_{B,2})$, with literal default $(\boldsymbol{b}, \boldsymbol{b})$. Their union then equals a single ball. No second center is inferred to be $\boldsymbol{c}$.

Periodic model qualification. The Gaussian terms in (2.7) on a finite box are not exactly periodic, despite their small tails near the faces when $L = 20$. The code's default `potential` retains that Euclidean formula. For the *explicitly periodized benchmark*, replace each Gaussian by

$$g_{\boldsymbol{z}}^{\mathrm{per}}(x) = \sum_{m \in \mathbb{Z}^d} e^{-\|x - \boldsymbol{z} + Lm\|^2/2} = \prod_{j=1}^{d} \frac{\sqrt{2\pi}}{L} \left[1 + 2 \sum_{n=1}^{\infty} e^{-(2\pi n/L)^2/2} \cos\left(\frac{2\pi n (x^j - z^j)}{L} \right) \right].$$

The equality follows by Fourier coefficients of the periodized one-dimensional Gaussian and factorization. Both potential and drift are then smooth and periodic. The code truncates the one-dimensional series at $K_{\mathrm{per}} = \lceil L\sqrt{2\log(10^{15})}/(2\pi) \rceil$; the first omitted Gaussian coefficient is below 10^{-15}, which is a coefficient threshold rather than a dimension-independent total-error bound. All reference labels and FNO residuals use this same truncation. This is a documented modeling completion of the torus specification, not exact equality to the original nonperiodic formula.

Input sampling and the source norm. Use a finite canonical integer-mode set $\mathcal{K} \subset \mathbb{Z}^d$, at most one member of each pair $\{k, -k\}$, with the first nonzero entry positive. Then

$$f(x) = \sum_{k \in \mathcal{K}} \left[a_k \cos(2\pi k \cdot x/L) + b_k \sin(2\pi k \cdot x/L) \right], \qquad \frac{\|f\|_{L^2(\mathcal{D})}}{|\mathcal{D}|} \le 0.1. \tag{2.8}$$

The coefficients a_k, b_k are real; set $b_0 = 0$ if the zero mode is included. The $2\pi/L$ factor specializes the expression $\cos(k \cdot x)$ to a torus of side L. Orthogonality gives the exact normalization formula

$$\|f\|_{L^2(\mathcal{D})}^2 = |\mathcal{D}| \left[a_0^2 + \frac{1}{2} \sum_{k \neq 0} (a_k^2 + b_k^2) \right],$$

with $a_0 = 0$ if absent. Sample independent Gaussian coefficients, then rescale their vector to the desired radius. The sampler returns a direction on the boundary of a chosen finite-dimensional ball; varying the radius samples its interior but requires an explicitly chosen distribution. The source constraint is $\|f\|_2/|\mathcal{D}| \le 0.1$, whereas $\mathrm{RMS}(f) = \|f\|_2/\sqrt{|\mathcal{D}|}$. They are different. At $L = 20$ the permitted RMS is $0.1\,20^{d/2}$, so blindly saturating the source bound across dimensions can greatly change the drift scale. Cross-dimensional studies must declare amplitude and bandwidth as well as the source bound.

Reference labels and joint training. For $d = 2$, an endpoint-excluded $N \times N$ grid has spacing $h = L/N$. Sample the periodized drift at each node and use the backward jump generator

$$(\mathcal{L}_h u)_i = \sum_{j=1}^{2} \sum_{s\in\{-1,1\}} \left(\frac{\varepsilon}{h^2} + \frac{\max(s b_f^j(x_i), 0)}{h} \right) (u_{i+se_j} - u_i). \tag{2.9}$$

Here i is a two-dimensional grid index, e_j a coordinate index step, and neighbors wrap periodically. Thus $\mathcal{L}_h : \mathbb{R}^{N^2} \to \mathbb{R}^{N^2}$ acts on grid observables; its transpose would act on mass columns. Set $q_h = 0$ on all nodes inside $\overline{A}$ and $q_h = 1$ inside $\overline{B}$, and solve the remaining sparse equations $\mathcal{L}_h q_h = 0$. No clipping is used to conceal residuals. Positive rates and connectivity to absorbing nodes give the discrete maximum principle: an interior strict maximum cannot exceed all connected neighbors; following the connected graph reaches a prescribed boundary value. The same argument gives uniqueness and $0 \le q_h \le 1$. Taylor expansion of (2.9) gives $b_f \cdot \nabla u + \varepsilon \Delta u + O(h)$ for smooth u, b_f away from the absorbing boundary. The diffusion stencil alone is second order. Staircase ball geometry introduces a separate error; a small sparse-system residual proves neither spatial convergence nor a continuum error rate.

Generate training pairs by sampling f first, then solving this reference problem. Split data by entire coefficient vectors before training, rather than treating nodes of the same function as independent test functions. The FNO objective combines supervised grid mean-square error, the squared interior residual of the *same* $\mathcal{L}_h$, soft boundary error on raw sigmoid outputs, and a range penalty (identically zero for exact sigmoid outputs). Hard grid values are used inside the residual stencil so that labels and training solve the same absorbing problem. A separate pointwise PINN residual uses automatic differentiation for the continuous (2.5). Differentiating a coupled grid FNO with respect to its input field does not compute coordinate derivatives of q. Residuals on masked absorbing nodes are excluded; dedicated boundary and independent validation samples remain necessary. A smooth hard boundary construction for off-grid inference is not implemented.

High-dimensional scope and open questions. A dense-grid FNO still stores N^d values per channel and performs FFT work proportional to $dN^d \log N$; at fixed channel widths its retained spectral weights also grow with the number of retained mode tuples. Even $N = 32, d = 10$ requires $32^{10} = 2^{50}$ grid entries before channels or batches. Factorized Fourier layers on the same full tensor do not by themselves remove this storage cost. Sparse Fourier inputs, low-rank function representations, non-grid coordinate decoders, stochastic trace estimators, and trajectory-based reference labels are possible extensions, not implemented results. The `Field` callable interface separates field evaluation from a tensor grid, and the Fourier sampler and AD residual support general d; the reference solver and FNO intentionally remain two-dimensional.

The open research tasks are to control gradient scales under (2.8), resolve rare-event sampling costs at small ε, quantify boundary and residual errors on independent samples, establish reference-label accuracy, and validate generalization across coefficient amplitudes

and bandwidths. Variable ε could be added as a constant input channel with corresponding reference labels; varying geometry likewise requires geometry features. Neither extension is trained here. The repeated center and intended drift sign remain source ambiguities.

2.6 Computational research roadmap

The implementation is `src/probflow/neural_pde_solver.py`; mathematical checks are in `tests/test_neural_pde_solver.py`, and the executable integration check is `experiments/neural_pde_smoke.py`. Its report is `results/neural_pde/smoke.json`. The research stages and acceptance evidence are:

1. *Low-dimensional reference validation.* Check positive rates, periodic wrap, prescribed values, range, linear-system residual, an independent analytic strip committor, and manufactured-function consistency. Compare finer grids while separating ball-geometry error from interior discretization error.

2. *Fourier input generation.* Verify periodicity, exact Parseval normalization, reproducible seeds, coefficient coverage, and numerical drift gradients; save modes, amplitudes, and train/test function identifiers.

3. *FNO implementation.* Check retained spectral bands, lifting, position features, projection, output shape, and boundedness. Record memory, parameter count, per-epoch cost, and sensitivity to resolution and mode cutoff.

4. *Supervised and residual training.* Report each loss separately, independent continuum residuals, boundary error, relative L^2 and maximum error, range violations, seed variability, and held-out function errors. Multi-seed accuracy, ablations, and out-of-distribution tests are *Planned Experiments*.

5. *PINN versus FNO.* At fixed geometry, noise, coefficient distribution, and reference tolerance, compare per-instance PINN training with FNO data-generation and amortized training/inference cost. Include initial-layer diagnostics motivated by the sampling counterexample (Proposition 2.3). This comparison is a *Planned Experiment*.

6. *High-dimensional algorithms.* Before $d \geq 10$, select a representation that avoids full-grid storage, develop independently validated reference observables, and measure dimension-dependent memory, sampling, and error. This stage is a *Planned Experiment*, not an extrapolation of the two-dimensional smoke test.

The executed smoke configuration uses $d = 2$, $L = 20$, $\varepsilon = 0.5$, radius 0.5, the literal repeated B center $(1, 1)$, $N = 32$, seed 20261115, four training functions and two held-out functions. Input modes are $(1, 0), (0, 1), (1, 1), (1, -1), (2, 0), (0, 2)$ at the boundary of (2.8). The FNO has width eight, two spectral layers, and four retained modes per band. Adam takes 12 full-batch steps at learning rate 0.003 with loss weights $(1, 0.01, 1, 1)$ in supervised/PDE/boundary/range order. The report records the actual losses, held-out errors,

and grid comparisons at $N = 32, 64, 128$. These tiny datasets and coarse voxelized balls test integration and reproducibility; they establish no learning rate, spatial convergence order, or reliable generalization estimate. The heat-equation diagnostic uses $T = 1$, $\varepsilon = 1$, $\zeta = 0.5$, $k = 64$, and eleven time samples spaced by 0.1.

These PDE tools extend Chapter 1's microscopic-to-effective framework without changing its dependencies: the stochastic model specifies the generator, the generator specifies a density or hitting-probability problem, and numerical or learned solvers approximate that declared problem. A solver's empirical accuracy is evidence to measure separately from the mathematical identities that define its target.

Appendix A

ODE Foundations and Linear Evolution

This appendix collects the supporting ODE arguments used throughout Project 9. The source problems and their links remain in the main text; in particular, Example 1.10 contains the variation-of-constants calculation itself.

A.1 Existence, Uniqueness, and Continuation

We prove Theorem 1.9.

Notice

The proof is divided into three parts.

- First, we establish the local existence of a solution near the initial time.
- Second, we establish uniqueness, showing that the same initial condition determines at most one solution.
- Finally, we extend the local solution to the entire prescribed time interval, establishing global existence.

Together, these results establish the existence and uniqueness of a global solution.

Proof.

1. Local existence. Fix an initial condition $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^d$. We seek a local solution by reformulating the initial-value problem as a fixed-point equation for an integral operator. Since f is continuous, choose $a, \rho > 0$ and consider the compact cylinder

$$Q = [t_0 - a, t_0 + a] \times \overline{B}_\rho(x_0).$$

By continuity and compactness, f is bounded on Q. Define

$$M := \sup_{(t,y)\in Q} \|f(t, y)\| < \infty.$$

By the local Lipschitz assumption, there exists $L < \infty$ such that

$$\|f(t, y) - f(t, z)\| \leq L\|y - z\|$$

for all $t \in [t_0 - a, t_0 + a]$ and $y, z \in \overline{B}_\rho(x_0)$. Choose $0 < \delta \leq a$ sufficiently small such that

$$M\delta \leq \rho, \quad L\delta < 1.$$

Let $J_\delta = [t_0 - \delta, t_0 + \delta]$. We now construct the space of candidate trajectories:

$$\mathcal{C}_\delta := \left\{\gamma \in C(J_\delta, \mathbb{R}^d) : \|\gamma_t - x_0\| \leq \rho, \quad t \in J_\delta\right\},$$

equipped with the supremum norm

$$\|\gamma\|_\infty := \sup_{t \in J_\delta} \|\gamma_t\|.$$

Since $C(J_\delta, \mathbb{R}^d)$ is a Banach space and $\mathcal{C}_\delta$ is closed, $\mathcal{C}_\delta$ is complete under the supremum norm.

For the fixed initial data, define the Picard operator $\mathsf{Q} : \mathcal{C}_\delta \longrightarrow C(J_\delta, \mathbb{R}^d)$ by

$$(\mathsf{Q}\gamma)_t := x_0 + \int_{t_0}^{t} f(r, \gamma_r)\, \mathrm{d}r.$$

- We first show that Q maps $\mathcal{C}_\delta$ into itself. For any $\gamma \in \mathcal{C}_\delta$,

$$\|(\mathsf{Q}\gamma)_t - x_0\| = \left\| \int_{t_0}^{t} f(r, \gamma_r)\, \mathrm{d}r \right\| \leq M|t - t_0| \leq M\delta \leq \rho.$$

  Therefore, $\mathsf{Q}(\mathcal{C}_\delta) \subseteq \mathcal{C}_\delta$.

- Next, we establish the contraction property. For any $\gamma, \eta \in \mathcal{C}_\delta$, the Lipschitz condition gives

$$\begin{aligned} \|(\mathsf{Q}\gamma)_t - (\mathsf{Q}\eta)_t\| &= \left\| \int_{t_0}^{t} \big(f(r, \gamma_r) - f(r, \eta_r)\big)\, \mathrm{d}r \right\| \\ &\leq L|t - t_0| \|\gamma - \eta\|_\infty \\ &\leq L\delta \|\gamma - \eta\|_\infty. \end{aligned}$$

  Taking the supremum over $t \in J_\delta$, we obtain

$$\|\mathsf{Q}\gamma - \mathsf{Q}\eta\|_\infty \leq L\delta \|\gamma - \eta\|_\infty.$$

  Since $L\delta < 1$, the Picard operator is a contraction on the complete metric space $\mathcal{C}_\delta$.

By the Banach fixed-point theorem, there exists a unique fixed point $\gamma^* \in \mathcal{C}_\delta$ satisfying

$$\mathsf{Q}\gamma^* = \gamma^*.$$

Equivalently,

$$\gamma_t^* = x_0 + \int_{t_0}^{t} f(r, \gamma_r^*)\, \mathrm{d}r.$$

Since f and γ^* are continuous, the fundamental theorem of calculus implies that γ^* is continuously differentiable and satisfies

$$\dot{\gamma}_t^* = f(t, \gamma_t^*), \quad \gamma_{t_0}^* = x_0.$$

Thus, a local C^1 solution exists.

2. Uniqueness. Let $x(t)$ and $y(t)$ be any two solutions with the same initial condition, defined on a common interval containing t_0. We show that they agree throughout their common interval.

Fix an arbitrary time t in this interval. On the compact time interval between t_0 and t, both trajectories are continuous and therefore bounded. Hence, there exists $R > 0$ such that

$$x(\tau), y(\tau) \in \overline{B}_R(0)$$

throughout this interval. By the local Lipschitz assumption, there exists a constant $L < \infty$ such that

$$\|f(\tau, x) - f(\tau, y)\| \leq L\|x - y\|.$$

Choose $\delta > 0$ sufficiently small so that $L\delta < 1$.

Consider any subinterval $[a, b]$ of length at most δ on which the two solutions agree at one endpoint. Their integral representations then give

$$\|x(t) - y(t)\| \leq L \left| \int_a^t \|x(r) - y(r)\|\, \mathrm{d}r \right| \leq L\delta \|x - y\|_{C([a,b])}.$$

The same estimate holds when the common endpoint is b rather than a. Taking the supremum over $t \in [a, b]$ yields

$$\|x - y\|_{C([a,b])} \leq L\delta \|x - y\|_{C([a,b])}.$$

Since $L\delta < 1$, we conclude that

$$x(t) = y(t), \quad t \in [a, b].$$

We can divide the compact interval between t_0 and the arbitrary time t into finitely many subintervals of length at most δ. Starting from the common initial condition, the preceding argument can be applied successively on each subinterval, in either time direction. Consequently,

$$x(t) = y(t)$$

for every t in their common interval of definition.This establishes uniqueness and completes the proof. □

A.2 Composition, Reversibility, and Differentiable Dependence

Proof. It suffices to prove the claims on an arbitrary compact time window, which we write as $[0, T]$ after translating the time coordinate. Uniqueness makes the constructions agree on overlapping windows, giving the flow throughout I. By Theorem 1.9, each initial condition determines a unique solution on $[0, T]$, including evolution to times earlier than the initial time. The existence, uniqueness, and continuation arguments are given in Appendix A.1. We use this result to construct the flow and establish its properties.

1. Construction of the flow and its algebraic properties. We first formulate the flow in terms of the solution set. Let

$$\mathcal{S}_v := \left\{\eta \in C^1([0,T], V) : \dot{\eta}_t = v_t(\eta_t)\right\}.$$

For each $r \in [0, T]$, define the evaluation map

$$E_r : \mathcal{S}_v \to V, \quad E_r(\eta) := \eta_r.$$

By global existence, every initial state $x \in V$ determines a solution on $[0, T]$, so E_r is surjective. By uniqueness, two solutions agreeing at time r must coincide throughout the interval, so E_r is injective. Hence E_r is a bijection. We can therefore define the two-time flow by

$$\Phi_{s,t} := E_t \circ E_s^{-1}.$$

In other words, E_s^{-1} assigns to an initial state its unique solution trajectory, while E_t evaluates that trajectory at time t. Thus, writing $x_\cdot = E_s^{-1}(x)$, we recover $\Phi_{s,t}(x) = x_t$, whose terminal-time evolution and initial condition are given by (1.10).

The identity, composition, and reversibility properties now follow directly from the algebra of maps. First,

$$\Phi_{s,s} = E_s \circ E_s^{-1} = \mathrm{id}_V .$$

For any $r, s, t \in [0, T]$,

$$\Phi_{r,t} \circ \Phi_{s,r} = (E_t \circ E_r^{-1}) \circ (E_r \circ E_s^{-1}) = E_t \circ E_s^{-1} = \Phi_{s,t}.$$

Similarly, taking the inverse of the defining composition,

$$\Phi_{s,t}^{-1} = (E_t \circ E_s^{-1})^{-1} = E_s \circ E_t^{-1} = \Phi_{t,s}.$$

Thus every $\Phi_{s,t}$ is a bijection, with

$$\Phi_{s,s} = \mathrm{id}_V, \quad \Phi_{r,t} \circ \Phi_{s,r} = \Phi_{s,t}, \quad \Phi_{s,t}^{-1} = \Phi_{t,s}.$$

2. Differentiability with respect to the initial state. The preceding construction establishes the algebraic properties of the flow. To show that it is a C^1 diffeomorphism, we must additionally establish its differentiability with respect to the initial state.

Fix $s \in [0,T]$ and $x \in V$, and write

$$x_t := \Phi_{s,t}(x).$$

The candidate derivative is determined by the linear variational equation

$$\begin{cases} \partial_t M_{s,t}(x) = D_x v_t(x_t) M_{s,t}(x), \\ M_{s,s}(x) = I_V. \end{cases}$$

Since $D_x v_t(x_t)$ is continuous in time, this linear matrix equation has a unique solution on $[0,T]$, including both forward and backward times.

We now verify that $M_{s,t}(x)$ is indeed the derivative of the flow. For $\delta x \in V$, define

$$\Delta_t(\delta x) := \Phi_{s,t}(x+\delta x) - \Phi_{s,t}(x).$$

The integral equations for the two trajectories and the global Lipschitz condition, with uniform constant L, imply, by Gronwall's inequality,

$$\sup_{t\in[0,T]} \|\Delta_t(\delta x)\| \le e^{LT}\|\delta x\|.$$

Set

$$A_{x,t} := D_x v_t(x_t).$$

The continuity of $D_x v$ implies uniform continuity on every compact neighborhood of the reference trajectory $\{(t,x_t) : t \in [0,T]\}$. Consequently, the first-order Taylor remainder

$$R_t^{\text{lin}}(\delta x) := v_t(x_t + \Delta_t(\delta x)) - v_t(x_t) - A_{x,t}\Delta_t(\delta x)$$

satisfies

$$\sup_{t\in[0,T]} \|R_t^{\text{lin}}(\delta x)\| = o(\|\delta x\|).$$

Subtracting the integral equation for the reference trajectory from that of the perturbed trajectory gives

$$\Delta_t(\delta x) = \delta x + \int_s^t \left[A_{x,r}\Delta_r(\delta x) + R_r^{\text{lin}}(\delta x)\right] \, \mathrm{d}r.$$

On the other hand, the variational equation gives

$$M_{s,t}(x)\delta x = \delta x + \int_s^t A_{x,r} M_{s,r}(x)\delta x \, \mathrm{d}r.$$

Subtracting these two relations yields

$$\Delta_t(\delta x) - M_{s,t}(x)\delta x = \int_s^t A_{x,r}\big[\Delta_r(\delta x) - M_{s,r}(x)\delta x\big]\ \mathrm{d}r + \int_s^t R_r^{\mathrm{lin}}(\delta x)\ \mathrm{d}r.$$

Since A_x is bounded on $[0,T]$, another application of Gronwall's inequality, in both time directions, gives

$$\sup_{t\in[0,T]} \|\Delta_t(\delta x) - M_{s,t}(x)\delta x\| = o(\|\delta x\|).$$

Therefore,

$$D_x\Phi_{s,t}(x) = M_{s,t}(x).$$

Moreover, continuous dependence of solutions on their coefficients, together with the continuous dependence of $\Phi_{s,t}(x)$ on x and the joint continuity of $D_x v$, implies that $M_{s,t}(x)$ varies continuously with x. Hence $\Phi_{s,t}$ is continuously differentiable with respect to the initial state.

Since the same argument applies with s and t interchanged, both $\Phi_{s,t}$ and its inverse $\Phi_{t,s}$ are C^1. Thus

$$\Phi_{s,t} \in \mathrm{Diff}^1(V).$$

3. Autonomous flows and the one-parameter group. Suppose now that $v_t = v$ is autonomous and that solutions exist for all real times. Define

$$\phi_t := \Phi_{0,t}.$$

For any initial time s, the trajectory $\tau \mapsto \Phi_{s,s+\tau}(x)$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Phi_{s,s+\tau}(x) = v\big(\Phi_{s,s+\tau}(x)\big), \quad \Phi_{s,s}(x) = x.$$

By autonomy and uniqueness, it coincides with $\phi_\tau(x)$. Therefore,

$$\Phi_{s,t} = \phi_{t-s}.$$

The composition law then becomes

$$\phi_{t+r} = \Phi_{0,t+r} = \Phi_{r,t+r} \circ \Phi_{0,r} = \phi_t \circ \phi_r.$$

In particular, setting $r = -t$ gives $\phi_{-t} = \phi_t^{-1}$. Thus the autonomous flow defines a one-parameter group of C^1 diffeomorphisms of V.

4. Time reversal. Finally, let x_t be any solution on $[0,T]$ and define the reversed trajectory

$$y_t := x_{T-t}.$$

By the chain rule,

$$\dot{y}_t = -\dot{x}_{T-t} = -v_{T-t}(x_{T-t}) = -v_{T-t}(y_t).$$

Moreover, $y_0 = x_T$. Hence the reversed trajectory satisfies

$$\begin{cases} \dot{y}_t = -v_{T-t}(y_t), \\ y_0 = x_T. \end{cases}$$

Equivalently, if $\widetilde{\Phi}_{s,t}$ denotes the flow of the reversed vector field $\widetilde{v}_t = -v_{T-t}$, then

$$\widetilde{\Phi}_{s,t} = \Phi_{T-s,T-t}.$$

This completes the proof. □

A.3 Construction and Completeness of the Fundamental Matrix

Let $A : I \to \mathbb{R}^{d\times d}$ be continuous on a time interval I. Consider the homogeneous system $\dot{y}_t = A_t y_t$. Use the Euclidean vector norm and its induced matrix norm. Let $e_1, \ldots, e_d$ be the standard basis of $\mathbb{R}^d$. Theorem 1.9, applied on compact time intervals, provides the column solutions u_j on I such that

$$\dot{u}_j(t) = A_t u_j(t), \quad u_j(s) = e_j.$$

Define the normalized fundamental matrix by its columns:

$$U(t,s) = \begin{pmatrix} u_1(t) & \cdots & u_d(t) \end{pmatrix}.$$

Its initial columns and their differential equations give

$$\partial_t U(t,s) = A_t U(t,s), \quad U(s,s) = I_d. \tag{A.1}$$

Columnwise uniqueness also proves uniqueness of this matrix initial-value problem.

Theorem A.1 (Fundamental matrix)**.** *The normalized fundamental matrix $U(t,s)$ is invertible for all $s,t \in I$. Every homogeneous solution is $y_t = U(t,s)y_s$, and $U(t,r)U(r,s) = U(t,s)$ with $U(t,s)^{-1} = U(s,t)$.*

Proof. We prove completeness and invertibility here; the algebraic identities are established in the next section. For constant scalars $c_1, \ldots, c_d$, linearity implies

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{j=1}^{d} c_j u_j(t) = A_t \sum_{j=1}^{d} c_j u_j(t).$$

This is the superposition principle for homogeneous solutions. Given any homogeneous

solution y_t, linear algebra writes its initial vector uniquely as $y_s = \sum_j c_j e_j$. The curve $\sum_j c_j u_j(t) = U(t,s)y_s$ solves the same equation and has the same initial value. ODE uniqueness, rather than linear algebra alone, now forces it to equal y_t at every time. In particular, evaluation at s is a linear bijection between the vector space of homogeneous solutions and $\mathbb{R}^d$. Its inverse sends c to the solution $U(\cdot, s)c$; thus the solution space has dimension d and the d column solutions form a basis.

To prove independence at an arbitrary time $t_0 \in I$, suppose $U(t_0, s)c = 0$. The solution $z_t = U(t,s)c$ then has zero value at t_0. Uniqueness with initial time t_0, in both time directions, makes it the zero solution throughout I. At s this gives $c = 0$. Hence $U(t_0, s)$ has trivial kernel, and a square matrix with trivial kernel is invertible. This proves that the columns remain independent at every time, without assuming an inverse or a composition identity in advance. The remaining assertions follow from Appendix A.4. □

A.4 Algebraic Identities for the Evolution Matrix

Fix any $r, s \in I$. As functions of t, the matrices $F(t) = U(t,r)U(r,s)$ and $G(t) = U(t,s)$ satisfy $F' = A_t F$ and $G' = A_t G$. At $t = r$ both equal $U(r,s)$. Columnwise uniqueness with initial time r therefore yields

$$U(t,r)U(r,s) = U(t,s).$$

No ordering of the three times is needed, since the solutions exist on all of I. Taking the terminal time equal to s, and then interchanging s and t, gives

$$U(s,t)U(t,s) = I_d, \quad U(t,s)U(s,t) = I_d.$$

Consequently $U(t,s)^{-1} = U(s,t)$, consistently with the independent kernel argument above. In particular,

$$U(t,s)U(\tau,s)^{-1} = U(t,s)U(s,\tau) = U(t,\tau).$$

These are identities for the linear homogeneous evolution matrix. They do not rely on properties of a general state flow.

For later differentiation under the integral, joint continuity can be checked directly: for a fixed $a \in I$, $U(t,\tau) = U(t,a)U(\tau,a)^{-1}$. Both factors are continuous in their respective variables, since inversion is continuous on invertible matrices. Thus $U(t,\tau)$ and $\partial_t U(t,\tau) = A_t U(t,\tau)$ are jointly continuous.

A.5 Matrix Exponentials and Time Ordering

The fundamental matrix packages all homogeneous initial vectors into one matrix: it avoids solving a new system for each y_s. Its definition as an initial-value problem always applies; an ordinary matrix exponential is an additional representation requiring suitable

coefficients.

Constant matrix. For $A_t = A$, the power series

$$E(t) = e^{A(t-s)} = \sum_{n=0}^{\infty} \frac{(t-s)^n A^n}{n!}$$

and its derivative series converge uniformly on compact time intervals by comparison with scalar exponential series in $\|A\|$. Termwise differentiation gives $E' = AE$ and $E(s) = I_d$. Uniqueness of (A.1) proves $U(t,s) = e^{A(t-s)}$.

Pairwise commuting coefficients. Assume $A_t A_r = A_r A_t$ for all $r, t \in I$ and put $B_A(t,s) = \int_s^t A_\tau \, \mathrm{d}\tau$. Then $\partial_t B_A(t,s) = A_t$ and $B_A(t,s)A_t = A_t B_A(t,s)$, by integrating the commutation relation. The product rule, before using commutativity, reads

$$\frac{\mathrm{d}}{\mathrm{d}t} B_A(t,s)^n = \sum_{k=0}^{n-1} B_A(t,s)^k A_t B_A(t,s)^{n-1-k}.$$

Commutativity makes this $nA_t B_A(t,s)^{n-1}$. On a compact interval, $\|B_A\|$ and $\|A\|$ are bounded, so both the exponential series and its derivative series converge uniformly. It follows that

$$\frac{\mathrm{d}}{\mathrm{d}t} e^{B_A(t,s)} = A_t e^{B_A(t,s)}, \quad e^{B_A(s,s)} = I_d, \quad U(t,s) = \exp\left(\int_s^t A_\tau \, \mathrm{d}\tau\right).$$

For noncommuting coefficients the terms in the product-rule sum cannot in general be combined. Thus differentiating $e^{B_A(t,s)}$ does not in general give $A_t e^{B_A(t,s)}$; time dependence alone does not justify the ordinary exponential formula.

Peano–Baker series from the integral equation. Integrating (A.1) gives

$$U(t,s) = I_d + \int_s^t A_r U(r,s) \, \mathrm{d}r. \tag{A.2}$$

Set $P_0(t,s) = I_d$ and recursively define

$$P_n(t,s) = \int_s^t A_{t_1} P_{n-1}(t_1,s) \, \mathrm{d}t_1, \quad n \geq 1.$$

For $t \geq s$, repeated substitution yields

$$\begin{aligned} U(t,s) = I_d + & \int_s^t A_{t_1}\,\mathrm{d}t_1 \\ & + \int_s^t \int_s^{t_1} A_{t_1} A_{t_2}\,\mathrm{d}t_2\,\mathrm{d}t_1 \\ & + \sum_{n=3}^{\infty} \int_{s \leq t_n \leq \cdots \leq t_1 \leq t} A_{t_1} \cdots A_{t_n}\,\mathrm{d}t_n \cdots \mathrm{d}t_1. \end{aligned}$$

To justify the series, take a compact $K \subset I$ containing s and put $M = \max_K \|A_t\|$. Induction in the recursive integral gives, in either time direction,

$$\|P_n(t,s)\| \leq \frac{M^n |t-s|^n}{n!}, \quad t \in K.$$

In forward time, the factor $|t-s|^n/n!$ is the volume of the ordered integration simplex. In backward time the same bound follows by taking absolute values in each oriented integral. The scalar exponential series therefore proves absolute uniform convergence of $S(t,s) = \sum_{n=0}^{\infty} P_n(t,s)$ on K. Passing its uniformly convergent sum through the integral gives $S = I_d + \int_s^t A_r S(r,s)\,\mathrm{d}r$. The fundamental theorem of calculus gives $\partial_t S = A_t S$ and $S(s,s) = I_d$, so uniqueness identifies $S = U$. Equivalently, $\partial_t P_n = A_t P_{n-1}$ and

$$\|\partial_t P_n(t,s)\| \leq \frac{M^n |t-s|^{n-1}}{(n-1)!}, \quad n \geq 1,$$

which also justifies termwise differentiation on K.

This construction is exactly Picard iteration of (A.2): starting with $U^{[0]} = I_d$, $U^{[m+1]} = I_d + \int_s^t A_r U^{[m]}(r,s)\,\mathrm{d}r$ gives $U^{[m]} = \sum_{n=0}^{m} P_n$. Thus the ordered series records the successive matrix Picard iterates, rather than a new existence principle.

Meaning of the time-ordered exponential. For forward time, the notation

$$U(t,s) = \mathcal{T} \exp\left(\int_s^t A_\tau\,\mathrm{d}\tau \right), \quad t \geq s,$$

denotes precisely the Peano–Baker series. The symbol $\mathcal{T}$ places later-time matrices on the left: at second order the product is $A_{t_1} A_{t_2}$ for $t_1 \geq t_2$, so the earlier contribution acts first on a column vector. Equivalently, the nth term is

$$P_n(t,s) = \frac{1}{n!} \int_{[s,t]^n} \mathcal{T}\big[A_{t_1} \cdots A_{t_n}\big]\,\mathrm{d}t_1 \cdots \mathrm{d}t_n.$$

The cube splits into $n!$ ordered simplices, apart from boundaries of zero volume. Sorting the factors makes all these integrals equal, which explains the factor $1/n!$. With commuting coefficients the ordering can be removed, and the cube integral is $B_A(t,s)^n$; without commutativity this simplification is unavailable.

For $t < s$, use the same recursive oriented integrals for P_n or the already proved identity $U(t,s) = U(s,t)^{-1}$. In that direction the nested products place earlier physical times on the left (reverse chronological ordering), with a sign $(-1)^n$ in the nth term after reorienting the integrals. The forward $\mathcal{T}$ convention above should therefore not be read as unchanged chronological ordering for a backward interval.

The solution required by the source question remains (1.7); the ordered exponential specifies its homogeneous matrix when the coefficients do not commute. Return to Example 1.10 for the concise worked solution.

A.6 Uniform Single-Step Jacobian Estimates

This section supplies the technical estimates used in parts (a) and (c) of the numerical flow-Jacobian analysis linked at the discrete Jacobian problem. Retain its assumptions: v is C^2 on a neighborhood of $[0,1] \times B$, where $B \subset V = \mathbb{R}^d$ is compact and convex and contains the exact and sufficiently fine Euler trajectories. For $h = 1/K$, $t_k = kh$, and $V_k(x) = x + hv_{t_k}(x)$, the single-step derivative is $DV_k(x) = I_d + hDv_{t_k}(x)$.

To expand this determinant, consider a matrix $A \in \mathrm{End}(V)$ with columns $a_1, \dots, a_d$. By multilinearity in the columns $e_j + ha_j$, $\det(I_d + hA)$ is a polynomial in h. Its constant term is $\det I_d = 1$. Choosing ha_j from exactly one column contributes $h \det(e_1, \dots, a_j, \dots, e_d) = hA_{jj}$, so the linear coefficient is $\mathrm{tr}(A)$. Define the determinant remainder by

$$R_A^{\det}(h) := \det(I_d + hA) - 1 - h\,\mathrm{tr}(A) = \sum_{j=2}^{d} c_j(A) h^j.$$

Each remaining term selects at least two columns from hA, and hence contains at least h^2. The coefficients $c_j(A)$ are polynomials in the entries of A. We use the Euclidean vector norm and its induced matrix norm, as in the main analysis. For a fixed bound $\|A\| \le L_v$, continuity on the compact matrix ball gives the finite constant

$$C_{d,L_v} := \sum_{j=2}^{d} \sup_{\|A\| \le L_v} |c_j(A)|, \quad |R_A^{\det}(h)| \le C_{d,L_v} h^2 \quad (|h| \le 1).$$

The sum is zero when $d = 1$. This constant depends only on d and L_v, and is independent of A and h.

By the stated regularity on the compact set $[0,1] \times B$, fix L_v bounding $\|Dv_t(x)\|$ there. Taking $A = Dv_{t_k}(x)$ and using $\mathrm{tr}(Dv_{t_k}(x)) = \nabla \cdot v_{t_k}(x)$ gives

$$\begin{aligned} R_k^{\det}(h,x) &:= R^{\det}_{Dv_{t_k}(x)}(h), \\ \det DV_k(x) &= 1 + h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h,x), \quad |R_k^{\det}(h,x)| \le C_{d,L_v} h^2. \end{aligned}$$

Thus the determinant expansion has an $O(h^2)$ remainder uniform in k and $x \in B$. The same derivative bound gives $|\nabla \cdot v_t(x)| \le dL_v$; write $B_{\mathrm{div}} := dL_v$. Set

$$u_k(h,x) := h\nabla \cdot v_{t_k}(x) + R_k^{\det}(h,x), \quad h_0 := \min\left\{1, \frac{1}{2(1 + B_{\mathrm{div}} + C_{d,L_v})}\right\}.$$

For $0 < h < h_0$, uniformly in k and $x \in B$,

$$|u_k(h,x)| \le (B_{\mathrm{div}} + C_{d,L_v})h < \frac{1}{2}, \quad \det DV_k(x) = 1 + u_k(h,x) \ge \frac{1}{2} > 0.$$

Consequently all these determinants have well-defined real logarithms for the same range of step sizes.

For the logarithmic expansion, define the scalar remainder

$$r_{\log}(u) := \log(1+u) - u.$$

On $|u| \le 1/2$, the second derivative of $\log(1+u)$ has absolute value at most 4. Taylor's theorem therefore gives $|r_{\log}(u)| \le C_{\log} u^2$ with the universal constant $C_{\log} = 2$. Define the combined logarithmic remainder separately by

$$r_k^{\log}(h,x) := R_k^{\det}(h,x) + r_{\log}(u_k(h,x)).$$

Then

$$\log\det DV_k(x) = h\nabla \cdot v_{t_k}(x) + r_k^{\log}(h,x),$$
$$|r_k^{\log}(h,x)| \le \big[C_{d,L_v} + C_{\log}(B_{\mathrm{div}} + C_{d,L_v})^2\big]h^2 =: Ch^2.$$

Here C depends only on d and the fixed derivative bound L_v, and is independent of k, $x \in B$, and $0 < h < h_0$. This proves the uniform $O(h^2)$ logarithmic expansion without identifying the determinant and logarithmic remainders.

Notation and Conventions

The tables distinguish mathematical objects by type and scope. Global entries apply throughout unless a local definition explicitly qualifies them; topic-specific entries belong to the indicated construction. They are an index to the definitions and hypotheses in the text, not additional existence, regularity, or convergence assumptions.

In these notes, hats usually indicate numerical approximations or statistical estimates, bars usually identify defined reduced models, and tildes usually indicate transformations or modified objects. These are common conventions within this manuscript, not universal rules. Every exception is defined locally. A change of model, a sample from an exact law, and a time-discretization error are distinct operations.

Operators use upright symbols such as D_A, M_A, and Q. Scalar functionals use ordinary letters: S for differential entropy, F for free energy, $H_{m_{\mathrm{ref}}}$ for negative entropy, and $L_{\log}$ for logistic risk. Calligraphic letters continue to denote the explicitly defined spaces and collections.

Global Notation

States, time, and approximation

Symbol	Mathematical object	Definition and convention
$V = \mathbb{R}^d$	State space	The full Euclidean state space; d is its dimension. A general measurable state space is written $(E, \mathcal{S})$.
$\mathcal{Z} = \mathbb{R}^m$	Reduced space	The range of a collective variable $\xi : V \to \mathcal{Z}$, with $m \leq d$ when reduction is intended.
$x,\ x_0$	States	A generic deterministic state and a prescribed initial state. The fixed initial time is stated locally.
$x_t,\ x_{t_k}$	Exact states	The exact trajectory and its value at a grid point: $x_t = \Phi_{0,t}(x_0)$ when the initial time is zero.
$X_t,\ Z_t$	Random states	A full random state and the projected process $Z_t = \xi(X_t)$. Random initial conditions do not turn an ODE into an SDE.

States, time, and approximation (continued)

Symbol	Mathematical object	Definition and convention
$x^i,\ X_t^i$	Components	Coordinate indices are superscripts; vectors are columns. Parenthesized superscripts $x^{(\ell)}$ instead label sample realizations.
s, t, r, τ	Continuous times	Usually s is the initial time and t the terminal time; integration variables are local dummy variables.
$T,\ K,\ h,\ t_k$	Time mesh	On $[0,T]$, $h = T/K$ and $t_k = kh$. From an initial time s to T, use $h = (T-s)/K$ and $t_k = s + kh$.
$N_x,\ \Delta x,\ x_i$	Spatial mesh	The cell count, cell width, and spatial cell centers. Spatial and temporal refinement parameters are distinct.
N_{samp}	Sample count	The number of observations or samples; it is neither a time-step count nor a spatial cell count.
$\widehat{x}_k$	ODE approximation	A numerical state at time t_k, with $\widehat{x}_0 = x_0$ when the initial datum is represented exactly.
$\widehat{X}_k$	SDE approximation	A time-discrete approximation to a random state. A pathwise error comparison with X_{t_k} requires a specified common coupling.
e_k	State error	$e_k := \widehat{x}_k - x_{t_k}$. Its norm is a scalar error; the vector itself is not an error magnitude.
$\widehat{\cdot},\ \bar{\cdot},\ \widetilde{\cdot}$	Common decorations	In these notes, usually an approximation/estimate, a defined reduced model, and a transformation/modification, respectively. These are local conventions, not universal rules; exceptions are defined where used.

Flows, derivatives, and evolution

Symbol	Mathematical object	Definition and convention
$v_t(x)$	Velocity field	$v_t(x) := v(t,x)$, with $\dot{x}_t = v_t(x_t)$. It is an instantaneous field, not a finite-time state map.
$\Phi_{s,t},\ \phi_t$	Exact flow maps	$\Phi_{s,t}$ maps a state at s to its value at t. In a specified autonomous complete system, $\phi_t := \Phi_{0,t}$.
$A_t,\ U(t,s)$	Linear evolution	$\partial_t U(t,s) = A_t U(t,s)$ and $U(s,s) = I_d$. The traditional matrix notation puts terminal time first: $\Phi_{s,t}(x) = U(t,s)x$ in the homogeneous linear case.
$D_x F(x)$	State derivative	The derivative matrix, with entries $\partial F^i/\partial x^j$. The subscript x may be omitted when the differentiated variable is unambiguous.
$M_{s,t}(x)$	Flow derivative	$M_{s,t}(x) := D_x \Phi_{s,t}(x)$ is a square matrix. Write $M_t := M_{0,t}$ only after fixing the initial time and state.

Flows, derivatives, and evolution (continued)

Symbol	Mathematical object	Definition and convention
$J_{s,t}(x)$	Flow determinant	$J_{s,t}(x) := \det M_{s,t}(x)$ is a scalar. It is positive under the stated smooth-flow assumptions; $J_t := J_{0,t}$ is an explicit abbreviation.
$V_k(x)$	Euler map	$V_k(x) = x + hv_{t_k}(x)$ and $\widehat{x}_{k+1} = V_k(\widehat{x}_k)$. Capital V_k and lowercase v_t have different meanings.
$\widehat{\Phi}_{h,k}$	Numerical flow	For Euler, $V_{k-1} \circ \cdots \circ V_0$, from the initial state to t_k. Other methods carry a method label when compared.
$\widehat{M}_{h,k},\ \widehat{J}_{h,k}$	Numerical derivatives	The derivative of the numerical flow and its determinant, with the initial-state argument retained where needed. Integrating a variational system gives this derivative only when the updates are consistent.
$\mathrm{id}_V,\ I_d$	Identities	The identity state map and the $d \times d$ identity matrix. An omitted dimension must be unambiguous.
$\mathcal{A} = B_b(V)$	Observable algebra	Bounded Borel measurable real functions, with pointwise algebra operations and supremum norm; unbounded observables require their own integrability conditions.
$\mathcal{K}_F,\ \mathcal{K}_{s,t}$	Koopman operators	$\mathcal{K}_F f = f \circ F$ and $\mathcal{K}_{s,t} f = f \circ \Phi_{s,t}$. These act on observables and reverse the order of state-map composition.
$\mathsf{T}_F,\ \mathsf{T}_{s,t}$	Law transport	$\mathsf{T}_F \mu = F_{\#}\mu$ and $\mathsf{T}_{s,t}\mu = (\Phi_{s,t})_{\#}\mu$. These act on measures, not on state vectors.
$\mathsf{K}_{s,t}$	Transition kernel	$\mathsf{K}_{s,t}(x, \mathrm{d}y)$ is a Markov probability kernel. The typeface distinguishes it from the time-step count K.
$\mathsf{P}_{s,t}$	Markov operator	$\mathsf{P}_{s,t} f(x) = \int f(y) \mathsf{K}_{s,t}(x, \mathrm{d}y)$. For a deterministic flow it equals $\mathcal{K}_{s,t}$.
$\mathcal{L}_t$	Generator	An infinitesimal operator on observables. Its domain and limit topology are specified locally; its differential expression depends on the underlying dynamics.
X_{obs}	Observable space	A selected invariant Banach space for a norm-generator; it is not the trajectory space $\mathcal{X}$ used in the linear ODE example.

Probability, analysis, and errors

Symbol	Mathematical object	Definition and convention
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space	The sample space, its sigma-algebra, and its probability measure. A moving physical region is instead written $\mathcal{D}_t$.
$\mu_t,\ \nu_t$	Probability laws	When full and reduced states are compared, $\mu_t = \mathrm{Law}(X_t)$ and $\nu_t = \mathrm{Law}(Z_t)$.
$\rho_t,\ p_{\xi,t}$	Densities	Only when the laws are absolutely continuous: $\mu_t(\,\mathrm{d}x) = \rho_t(x)\,\mathrm{d}x$ and $\nu_t(\,\mathrm{d}z) = p_{\xi,t}(z)\,\mathrm{d}z$. Static marginals may be written p_ξ.
$\pi,\ \rho_{\mathrm{eq}}$	Equilibrium law	π denotes an invariant probability law where established; if it has a density, $\pi(\,\mathrm{d}x) = \rho_{\mathrm{eq}}(x)\,\mathrm{d}x$. Invariance is a model-dependent conclusion.
$\mathbb{E}_\mu,\ \mathrm{Law}$	Expectations and laws	Expectation under a specified law and the law of a random variable. The expectation subscript may be omitted on a fixed probability space.
$\langle f, \mu\rangle$	Integration pairing	$\int f\ \mathrm{d}\mu$, distinct from the Euclidean inner product $x^\top y = x \cdot y$.
$\|x\|,\ \|A\|$	Default norms	Euclidean vector norm and its induced matrix norm. Function-space norms carry an explicit subscript, such as $\|f\|_\infty$.
$\|A\|_{\mathrm{F}},\ \|a\|$	Other magnitudes	Frobenius matrix norm and scalar absolute value. For a measurable region, $\|\mathcal{D}\|$ denotes its Lebesgue volume.
$\nabla,\ \nabla^2,\ \Delta,\ \nabla\cdot$	Spatial derivatives	The column gradient, Hessian matrix, Laplacian, and divergence. For a vector field v, its divergence is $\nabla \cdot v$.
$\mathrm{d},\ \partial_t,\ \delta$	Differential symbols	Integration/SDE differential, time partial derivative, and a defined variation. A dot denotes an ordinary time derivative.
$\mu_n \xrightarrow{\mathrm{w}} \mu$	Weak convergence of laws	On the declared topological space: convergence of integrals of all bounded continuous test functions. It does not mean function-space weak convergence.
$f_n \rightharpoonup f$	Function-space weak limit	Weak convergence in an explicitly named Banach or Hilbert space, tested against its continuous dual. No such convergence is inferred from law convergence alone.
$\Longrightarrow$	Logical implication	Reserved here for logical implication; it is not the notation for either of the preceding limits.
$\ll,\ \sim$	Measure relations	$\mu \ll \nu$ means absolute continuity; $P \sim Q$ means mutual absolute continuity, whereas $X \sim \mu$ specifies a distribution. Scalar asymptotic equivalence is defined in its own context.
$\delta_x,\ \mathbf{1}_A,\ \#$	Measure operations	A Dirac probability law, an indicator, and pushforward. A formal Dirac-delta density on a level set is interpreted under integration.

Probability, analysis, and errors (continued)

Symbol	Mathematical object	Definition and convention
$O(h^p),\ o(h^p)$	Asymptotic orders	The limit parameter and fixed data are stated. A uniform bound identifies the variables excluded from its constant; order notation is not the name of a particular remainder.
$C,\ L_x,\ L_t$	Estimate constants	Constants are local unless identified explicitly. Their sources, parameter dependence, and uniformity are stated at first use; a reused letter does not assert an unchanged numerical value.

Topic-specific Notation

Linear ODEs, geometry, and mechanics

Symbol	Mathematical object	Definition and convention
B_t	Affine forcing	The continuous vector-valued external term in $\dot{x}_t = A_t x_t + B_t$, with $B : I \to V$. Brownian motions use W with a stated superscript when needed.
$\mathcal{X},\ \mathcal{Y}$	Trajectory spaces	$C^1(I;V)$ and $C(I;V)$ in the affine ODE example. Generic path evaluation $\gamma(t)$ is the same value as γ_t.
$\mathsf{M}_A,\ \mathsf{D}_A$	Trajectory operators	$(\mathsf{M}_A\gamma)(t) = A_t\gamma(t)$ and $\mathsf{D}_A = \partial_t - \mathsf{M}_A$. Neither is the observable algebra or generator.
$\mathcal{H},\ E_t,\ S_t$	Homogeneous solutions	$\mathcal{H} = \ker \mathsf{D}_A$; E_t evaluates a homogeneous trajectory, and $S_t = E_t^{-1}$ supplies its complete solution. S_t is not a Hessian.
$\gamma_{\mathrm{p}},\ \mathcal{K}_s$	Affine geometry	The chosen zero-initial-value particular solution and the space of paths vanishing at s. The affine solution set and initial-value slice are defined in the example.
T_t	Temperature	Temperature in the cooling example, with environment value T_{env} and relaxation rate λ. It is not the terminal time T.
$\mathcal{D}_t,\ \Theta_w(t)$	Volume and weighted action	$\mathcal{D}_t = \Phi_{0,t}(\mathcal{D}_0)$ is a moving region. For the weighted-path problem only, $\Theta_w(t) = \int_0^t w_r^{-1}\,\mathrm{d}r$.
$\mathcal{V},\ H(q,p)$	Potential and Hamiltonian	The potential is model-specific. The Hamiltonian and physical hypotheses are stated in each model; $\frac{1}{2}\|p\|^2 + \mathcal{V}(q)$ is used only for the unit-mass mechanical model in these notes.
$q, p,\ z = (q,p)$	Canonical variables	In mechanics $q, p \in \mathbb{R}^n$ and $d = 2n$. In the particle introduction, subscripts on $q_i, p_i \in \mathbb{R}^3$ label particles.

Linear ODEs, geometry, and mechanics (continued)

Symbol	Mathematical object	Definition and convention
$\mathbb{J}_n,\ \nabla^2 H(z_t)$	Symplectic geometry	$\mathbb{J}_n$ is the standard $2n \times 2n$ block matrix with upper-right block I_n and lower-left block $-I_n$. The Hamiltonian Hessian is written directly.
$\widehat{q}_k,\ \widehat{p}_k$	Numerical mechanics	Computed position and momentum, including a hatted half-step momentum in Verlet. $\widehat{H}_k = H(\widehat{q}_k, \widehat{p}_k)$ is the computed true energy.
$V_h,\ \widetilde{H}_h$	Verlet quantities	The locally defined Verlet update matrix and its modified energy; the latter is not a numerical approximation to the state.
$\delta x,\ \Delta_t(\delta x)$	Initial-data sensitivity	The vector perturbation and difference of two exact flow trajectories in the differentiability proof; distinct from numerical error e_k.
$\mathsf{Q}_{s,x},\ \mathsf{Q}_\delta$	Picard operators	The integral map for initial data (s, x) and its restriction to the local trajectory set on J_δ. With the interval fixed, the latter is abbreviated Q; it is distinct from the Markov operator $\mathsf{P}_{s,t}$.
$P_n(t,s),\ U^{[n]}(t,s)$	Ordered series	A Peano–Baker term and a Picard partial sum. $B_A(t,s) = \int_s^t A_r \,\mathrm{d}r$ is an integrated coefficient matrix, not a Hamiltonian.
$\mathcal{T}$	Time ordering	The forward ordering places later-time matrices on the left; backward intervals use the explicitly stated reversed ordering.

Entropy, convexity, and information

Symbol	Mathematical object	Definition and convention
$S(\rho),\ \ell_t$	Entropy and log density	$S(\rho) = -\int \rho \log \rho \,\mathrm{d}x$ when finite; $\ell_t = \log \rho_t$ requires positive density.
$s_t^\rho,\ H_t^\rho$	Score and log Hessian	$s_t^\rho = \nabla \log \rho_t$ and $H_t^\rho = \nabla^2 \ell_t$. Neither is a solution map or the Hamiltonian.
$F(\rho),\ \beta$	Free energy	The diffusion free-energy functional and inverse temperature, under the local potential, integrability, and boundary assumptions. This functional is not the sigma-algebra $\mathcal{F}$.
$P, Q,\ p, q$	Information-theory laws	Locally, P, Q are probability measures and p, q their densities under a declared reference measure. These density symbols are not mechanical position and momentum.

Entropy, convexity, and information (continued)

Symbol	Mathematical object	Definition and convention
$D_{\mathrm{KL}},\ D_f$	Divergences	The directed relative entropy and the specified f-divergence. Density arguments are explicitly declared shorthand for measure arguments.
$r,\ f^*,\ \partial f$	Convex duality	$r = \mathrm{d}P/\,\mathrm{d}Q$ is a density ratio; f^* is the scalar convex conjugate and ∂f its subdifferential. A star on an optimizer has a separate, locally stated role.
$\vartheta,\ m_{\mathrm{ref}}$	Convex mixtures and base measure	The scalar mixing weight and the fixed reference measure. θ is instead used for model parameters.
$H_{m_{\mathrm{ref}}}$	Negative entropy	$\int p \log p \ \mathrm{d}m_{\mathrm{ref}}$ on its declared density space; its Bregman representation is used only under the stated directional differentiability conditions.
$c,\ Z_c,\ \nu_c$	Gibbs tilt	$Z_c = \int e^{-c}\,\mathrm{d}\nu$ and $\nu_c = Z_c^{-1} e^{-c}\nu$. An abstract tilt cost equals $\beta\mathcal{V}$ only when that identification is specified.
$\Lambda_Q,\ \Psi_Q$	Partition functionals	The log-exponential integral and exponential integral, respectively; their natural conjugate domains differ.
$\mathcal{J}_\alpha,\ m(b)$	Truncated Gaussian example	The constrained objective and inverse Mills ratio; b is a threshold, φ the standard Gaussian density and $\bar{\Phi}$ its upper tail. The bar here does not denote a reduced dynamical model.
$L_{\log},\ \mathcal{R}$	Risk and reward	The logistic population risk and the concave measure-valued reward functional. Neither is the generator $\mathcal{L}_t$.

Conditioning, learned transport, and numerics

Symbol	Mathematical object	Definition and convention
$\mu_z,\ \mu_{t,z}$	Conditional kernels	A static regular conditional law and a selected version of $\mathrm{Law}(X_t \mid \xi(X_t) = z)$. Null-set and geometric regularity qualifications are retained.
$\mathcal{M}_z,\ G_\xi$	Level geometry	$\mathcal{M}_z = \{x : \xi(x) = z\}$ and $G_\xi = D\xi D\xi^\top$. Surface measure is $\mathcal{H}^{d-m}$ on this level set.
J_ξ	Coarea factor	$\sqrt{\det G_\xi}$, positive under the stated full-rank assumption. It is not the square-flow derivative determinant $J_{s,t}$.
$F_\xi,\ B_i^\xi$	Mean-force objects	$F_\xi = -\beta^{-1} \log p_\xi$ for the specified static Gibbs marginal. The lifting field satisfies $D\xi B_i^\xi = e_i$.
$Y_\sigma,\ \varphi_{\sigma,m}$	Noisy observations	A noisy collective-variable observation and its Gaussian smoothing kernel. The observation noise parameter σ is not a time index.

Conditioning, learned transport, and numerics (continued)

Symbol	Mathematical object	Definition and convention
$W_t,\ W_t^{\mathrm{red}},\ d_W$	Brownian motions	The full noise, a scalar reduced noise, and full noise dimension. The scalar noise W_t^{ξ} is constructed in the projection argument. Coupling or independence is specified locally.
$b_t,\ \sigma_t,\ a_t$	SDE coefficients	Full drift and noise matrix, or in the scalar projected model a conditional drift and conditional squared-gradient factor. Averaging a noise amplitude is not averaging its square.
$A_\xi,\ Z_t^0,\ \bar{X}_t$	Projected models	The scalar projected drift before conditioning, the exact projected process, and a specified closure model. Matching marginals does not assert matching path laws.
$\boldsymbol{j}_t,\ v_t^{\mathrm{cur}}$	Probability current	$v_t^{\mathrm{cur}} = \boldsymbol{j}_t/\rho_t$ only where $\rho_t > 0$; no velocity is defined by this quotient at a zero of the density.
$\gamma,\ I_t,\ v_t^{\mathrm{int}}$	Stochastic interpolation	In this subsection only, γ is an endpoint coupling and $I_t = \alpha_t X_0 + \lambda_t X_1$. The conditional interpolation velocity is defined almost everywhere under its time marginals.
$g_{\theta,t},\ \widehat{\theta}$	Learning notation	The time-dependent parameterized family (equivalently the network interface $g_\theta(t,z)$) and fitted parameters. Write $\widehat{g}_t = g_{\widehat{\theta},t}$ and $\widehat{b}_t$ for learned functions.
$F_{\mathrm{tr}},\ F_t^{\mathrm{tr}}$	Transport maps	A quantile/triangular transport and its time interpolation. Coordinate components are F_{tr}^i; CDFs F_0, F_w and the weight function w are defined in that example.
$\widehat{\mu}_{w,N_{\mathrm{samp}}}$	Weighted empirical law	The normalized weighted sample measure; finite-sample approximation is not a smooth diffeomorphic image of a Gaussian.
$\psi_t,\ \mathcal{E},\ \mathfrak{L}$	Kinetic control	The scalar multiplier, terminal functional and Lagrangian of the formal stationarity system. ψ_t is not the autonomous flow ϕ_t.
$R_A^{\mathrm{det}},\ r_{\log}$	Explicit remainders	The determinant-polynomial and scalar logarithmic remainders; their definitions and uniform estimates are in Appendix A.6.
$r_k^{\log},\ r_k^{\mathrm{quad}}$	Stepwise remainders	The combined logarithmic remainder and interval quadrature remainder. State dependence is retained until evaluation at $\widehat{x}_k$.
$R_t^{\mathrm{lin}},\ B_{\mathrm{div}}$	Differentiability and bounds	The initial-data Taylor remainder, distinct from determinant remainders; B_{div} bounds the divergence in the Euler argument.
$D_{\mathrm{diff}},\ q_{i\to j}^{\Delta x}$	Spatial discretization	The scalar diffusion constant and outgoing transition rate from cell i to cell j. Fixed-mesh formulas abbreviate the rate as $q_{i\to j}$.
$Q^{\Delta x},\ \pi_i^{\Delta x}$	Finite-state evolution	The mass-column generator (columns sum to zero) and stationary cell masses. The backward matrix on observables is its transpose; fixed-mesh formulas may write π_i.

Conditioning, learned transport, and numerics (continued)

Symbol	Mathematical object	Definition and convention
$m_i^{\Delta x}(t)$	Semidiscrete mass	The exact continuous-time solution of the spatially discrete system, approximating a PDE cell mass. The local abbreviation is $m_i(t)$; a further time approximation would be $\widehat{m}_{k,i}^{\Delta x}$.
$A_{\mathrm{OU}},\ R_t$	OU benchmark	The positive-definite matrix and full two-dimensional state in $\mathrm{d}R_t = -A_{\mathrm{OU}} R_t\,\mathrm{d}t + \sqrt{2\beta^{-1}}\,\mathrm{d}W_t$. The drift sign is part of the model.
$m_t^{\mathrm{OU}},\ \Sigma_t$	OU Gaussian statistics	The mean and covariance; Σ_∞ is the established equilibrium covariance. The mean is distinct from the finite-volume mass vector.
$C_{\mathrm{full}},\ C_{\mathrm{closure}}$	Time correlations	The locally defined stationary autocovariances; equality of one-time laws does not imply equality of these functions.

The principal definitions are developed in Sections 1.1 and 1.2; the ODE constructions are detailed in Appendix A. Sections 1.3–1.5 develop conditional and effective dynamics, learned flows, and the OU benchmark. The provisional extension is Chapter 2. A new symbol should be defined at first use, assigned an object type and scope, and checked against these tables. Standard local notations such as mechanical p, q and information-theory densities p, q are retained only with their scope made explicit.

Neural PDE module (local conventions). In Chapter 2, $\mathcal{D} = \mathbb{T}_L^d$ is the periodic physical domain, distinct from the probability sample space Ω; D is its region outside the absorbing sets. The parameter $\varepsilon > 0$ is diffusivity, equal to β^{-1} when the Langevin conventions are identified. The observable generator $\mathcal{L} = b_f \cdot \nabla + \varepsilon\Delta$ and its Lebesgue adjoint $\mathcal{L}^*$ act on observables and densities, respectively. The center $\boldsymbol{b}$ is distinct from the drift $b_f = -\nabla\mathcal{V}_f$. The committor q_f is a scalar hitting probability, not a probability density; $\mathcal{G}$ maps input fields f to these observables. Grid versions are $\mathcal{L}_h, q_h$. The symbols η_k, H, ζ denote the time cutoff, its polynomial profile, and the amplitude in the heat-equation counterexample only. Losses use $\mathcal{J}$; they are scalar functionals, not generators.

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